Probability: an open-problem working book

A twenty-hour research campaign, one worker

Checkpoint: 2026-09-06T22:55:18Z

Work began 6 September 2026 at15:13:26 UTC. The twenty-hour deadline is7 September 2026 at11:13:26 UTC. This is an evolving checkpoint, not a claim that the mandate is finished.

232 distinct questions admitted; 232 with a substantive recorded attempt; 0 original full solutions promoted. Target:1,000 distinct sourced questions. Shortfall at this checkpoint:768.

Primary sources supply the questions and historical open-status evidence. An old open-problem statement is not proof that the question remains open today. Current-status searches, known literature resolutions, author-checked partial arguments, and candidate proofs are recorded separately.

P0174 has a scoped independent analytic check that found no gap in the reviewed calculations; broader semantic review, novelty and registration remain pending. A successful ledger check, private repository push, PDF build, or math-brain retrieval test does not validate a proof. No routine exercise or cosmetic parameter variant is counted toward the target.

Current result limits

Seven full candidates are preserved: the Brownian prefix construction, the stationary nonergodic Galton-Watson flow, the sharp free-Brownian diameter constant, the rotation-algebra Brown-measure counterexample, the compact-initial-law multiplicative counterexample, the wreath-graph parameter counterexample, and the uniform-CLT counterexample. Their exact assumptions and review limits are in the proof appendices. None is promoted as an original solved open problem. Known theorem rederivations remain separate.

Persistent math brain

Sixteen private draft entries and 18 links have been ingested in two accepted bundles. Nine positive paraphrase searches retrieved the intended entries at rank 1; four negative controls were clean. A rejected intermediate bundle with invalid license metadata was corrected and is not counted separately. These are retrieval checks, not independent proof review or public promotion.

Operational record

The live wake for this thread reported a latest dream age of30.5 hours against a30.0-hour ceiling; the degraded finding was reported. The signed-project-contract cache warning remains unresolved. Work is confined to private research artifacts; no public publication, deployment, GPU training, or trading-box work is part of this campaign.

How to read each entry

Each question has a source and location, a paraphrased mathematical target, scope caveats, and its own attempted route. A failed route is reported as such; an example against a proof method is not relabeled as a counterexample to the source conjecture. Closely overlapping source formulations are grouped, while excluded literature resolutions appear separately.

source_says_open

The primary source presents this as unresolved at its writing date.

unverified

The current literature status is not established by this campaign.

no_resolution_found_in_documented_search

A dated search found no matching resolution; absence of a search hit is not proof of openness.

attempted_unsolved

An individual mathematical attempt is recorded, but the full target is not settled.

candidate proof

A full argument has been drafted, with assumptions and novelty/current-status checks still pending.

proved_here_author_checked

Reserved for a complete source-matched argument that survives repeated author checks; this is weaker than independent review.

resolved_in_literature

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P0001 Negative correlation in uniform forests

Source: Selected problems in probability theory | Conjecture 4.2, PDF page 8

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: no_resolution_found_in_documented_search | Result: partial_result

Question

For every finite simple graph, does its uniform acyclic edge subset have pairwise nonpositive edge covariance?

Dated literature checks

{"date": "2026-09-06", "queries": ["uniform forests negative correlation site:arxiv.org"], "evidence_url": "https://arxiv.org/html/2603.10738v1", "evidence_locator": "Conjecture 1.3", "scope": "March 2026 primary paper still states general conjecture; later complete-graph results do not settle arbitrary graphs"}

P0001-A001 | Condition independent weighted Bernoulli edges on acyclicity; factor disjoint cycle constraints in cactus graphs

Argument or result

Complete author-checked proof of negative association for all finite cactus graphs and finite positive edge weights; target on arbitrary graphs remains unsolved here

Check

Direct subset counts for uniform cycles give covariance -2^(m-2)/(2^m-1)^2; triangle -2/49 and tree zero agree

Remaining obstruction

Overlapping cycle constraints do not factor; generic conditioning on a decreasing event fails. Need an inequality using graphic-matroid circuit elimination.

Verification: author_checked | Independent review: none | Full question solved: False

P0002 Self-avoiding-walk counting exponents

Source: Selected problems in probability theory | Conjecture 2.1, PDF page 3 and footnote

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the asymptotic correction to c_n=mu^n for lattice self-avoiding walks in dimensions two, three, and four.

Scope

The source predicts gamma_2=43/32 and explicitly requires logarithmic corrections in dimension four. High-dimensional cases already proved are excluded from the unresolved target. This remains one source problem, not separate parameter rows.

P0002-A001 | Submultiplicativity and its limits

Argument or result

On Z^d, splitting a walk gives c_(m+n)<=c_m c_n. Positive-coordinate walks give d^n<=c_n, and the nonbacktracking bound gives c_n<=2d(2d-1)^(n-1). Fekete determines the exponential growth rate. However a_n=mu^n exp(sqrt(n)) is submultiplicative and has no fixed polynomial prefactor.

Check

The example is a real sequence, not a self-avoiding-walk counterexample. The square-root subadditivity checks the claimed property directly.

Remaining obstruction

Submultiplicativity cannot determine the critical exponent or the dimension-four logarithmic correction.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0003 Hexagonal self-avoiding-walk scaling limit

Source: Selected problems in probability theory | Question 2.2, PDF page 3

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does the rescaled uniform n-step hexagonal-lattice self-avoiding walk converge to SLE_(8/3)?

Scope

The source leaves the precise rescaling, curve topology, parametrization and limiting SLE geometry implicit; these are proof obligations, not permissions to change the question.

P0003-A001 | Exact finite-domain conditional law

Argument or result

For a finite-graph a-to-b self-avoiding path with weight z^length, condition on an initial simple prefix. The remaining path has the same fugacity law on the graph with earlier prefix vertices removed. The factor z^prefix_length cancels between numerator and partition function.

Check

The endpoint stays in the graph; deleting it would invalidate the argument. Fixed-length conditioning also fixes the remaining length and is a different ensemble.

Remaining obstruction

This domain Markov property supplies neither tightness nor conformal invariance and does not identify an SLE scaling limit.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0004 Existence of the quenched connective limit

Source: Selected problems in probability theory | Question 2.3, PDF pages 3-4

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: partial_result

Question

On the infinite supercritical bond-percolation cluster of Z^2, does c_n(v)^(1/n) converge almost surely, independently of v?

P0004-A001 | Adjacent-root walk decomposition and a Markov/Borel-Cantelli bound

Argument or result

Root-independent limsup on every infinite connected bounded-degree graph; annealed upper bound p mu

Check

Walks avoiding the neighboring root inject by prepending; walks visiting it inject into reversed-prefix/suffix pairs. These are disjoint cases and preserve lengths.

Remaining obstruction

The cut-walk inequality has an environment-dependent endpoint maximum. No stationary scalar subadditive process or liminf lower bound was obtained.

Verification: author_checked | Independent review: none | Full question solved: False

P0005 Ehrenfest mirror escape

Source: Selected problems in probability theory | Question 3.1, PDF pages 4-5

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Which mirror densities p give positive escape probability in the square-lattice model with equally likely NE/NW mirrors?

P0005-A001 | Invertible routing and finite-volume diagnostics

Argument or result

The local mirror pairings define an invertible map on oriented light states. Its orbits are cycles or bi-infinite trajectories. On a finite periodic torus every orbit is a cycle. A mirror-free initial straight run of N sites has probability (1-p)^N.

Check

The finite-state bijection proves the torus assertion independently of sampling. The straight-run bound tends to zero for each fixed p>0.

Remaining obstruction

Finite periodic loop counts cannot decide infinite-volume escape; the N-to-infinity and p-to-zero limits cannot be interchanged from this bound.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0006 Poisson mirror escape threshold

Source: Selected problems in probability theory | Question 3.2(a-c), PDF pages 5-6

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For nondegenerate Poisson-mirror orientation laws, characterize escape below the vacant-set threshold, including uniform orientations.

Scope

All three source subquestions are grouped to avoid counting nested parameter variants separately. The direction quantifier for theta_mu must be resolved from the model before a theorem claim.

P0006-A001 | Exact first-collision distribution for Poisson mirrors

Argument or result

For a straight test ray segment of length L in direction alpha and mirrors of total length epsilon with orientation law mu, the centers whose mirror crosses the segment form a parallelogram of area epsilon L |sin(alpha-beta)| for angle beta. Unit-intensity marked Poisson counting gives no-collision probability exp(-epsilon L integral |sin(alpha-beta)| mu(dbeta)). Thus the first free flight is exponential whenever this rate is positive.

Check

The factor is total mirror length, not half-length. Parallel mirrors contribute zero area and endpoint coincidences have zero intensity. The rate is positive for every alpha if mu is not concentrated on a single unoriented direction.

Remaining obstruction

After reflection, previously examined empty regions and previously hit mirrors make the flight history dependent. The first-flight computation gives no independent-flight construction or escape threshold.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0007 Unbounded mirror component trapping every ray

Source: Selected problems in probability theory | Question 3.3, PDF page 6

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Can an unbounded component of a mirror complement contain an origin from which every reflected ray stays bounded?

P0007-A001 | Attempted reduction to straight visibility

Argument or result

An unbounded unobstructed straight ray from the origin immediately rules out the proposed all-directions trapped configuration. However an unbounded connected planar open set need not contain any unbounded straight ray: a sufficiently narrow unbounded spiral corridor is a topological counterexample to that inference.

Check

The corridor is only a counterexample to the topological inference. It is not asserted to be a configuration of the source mirror model.

Remaining obstruction

Unboundedness of the vacant component gives a continuous route, not a realizable billiard route. No mirror configuration trapping all angles, or theorem preventing one, was constructed.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0008 Diffusive Poisson mirror trajectories

Source: Selected problems in probability theory | Question 3.4, PDF page 7

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For rationally supported nondegenerate mirror orientations, do escaping trajectories have finite positive diffusivity and a Gaussian limit?

P0008-A001 | Green-Kubo variance and a failed CLT shortcut

Argument or result

For a stationary mean-zero scalar velocity with integrable covariance C(s), Var(integral_0^t V_s ds)/t = 2 integral_0^t (1-s/t) C(s) ds tends to 2 integral_0^infinity C(s) ds. Covariance control alone does not imply a Gaussian limit: let Y_i=A epsilon_i with independent fair signs and independent A equal to 1 or 2 with probability 1/2. Distinct increments are uncorrelated, but n^(-1/2) sum Y_i converges to the equally weighted mixture of N(0,1) and N(0,4).

Check

Conditioning on A proves the mixture limit; its fourth moment differs from that of a Gaussian with the same variance. This is not a unit-speed mirror-model counterexample.

Remaining obstruction

Need mixing or a martingale approximation for the billiard environment, including the effect of conditioning on an unbounded trajectory.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0009 Manhattan pinball localization

Source: Selected problems in probability theory | Question 3.5, PDF page 7; model Figure 3.2

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does every positive mirror density localize the Manhattan-oriented pinball trajectory?

P0009-A001 | Small-density straight paths and lack of an order argument

Argument or result

At q=0 a ray moves indefinitely along its Manhattan street. The event that its first N potential collisions contain no mirror has probability (1-q)^N. This only controls a finite initial flight and gives no positive lower bound as N increases at fixed q.

Check

The finite-run probability agrees with independence and has the correct q=0 endpoint. Adding a mirror changes routing rather than merely deleting traversable states, so an edgewise percolation monotonicity proof has not been supplied.

Remaining obstruction

The known localization interval near and above one-half cannot be extended to every q>0 using this finite-run bound or an unproved monotonicity assumption.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0010 Biased random square-lattice orientations

Source: Selected problems in probability theory | Question 5.1, PDF page 9

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does every nonzero orientation bias give positive probability of an infinite directed path from the origin?

P0010-A001 | Directed-percolation subgraph comparison

Argument or result

Reflecting both coordinate axes maps p to 1-p, so the infinite-path probability is symmetric about one-half. North-east paths use independent edges pointing north/east with probability p; survival of the usual north-east directed percolation therefore implies survival in the full oriented graph. The p=0 and p=1 endpoints have deterministic infinite paths.

Check

The comparison is inclusion of path events, not equality of the two models. It proves a region above the directed-percolation threshold and its reflected counterpart only.

Remaining obstruction

It does not settle the interval between one-half and that threshold. The critical p=1/2 result does not extend by continuity alone.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0011 Brownian SIR survival in dimensions at least two

Source: Selected problems in probability theory | Question 6.1(ii), PDF pages 9-10

https://arxiv.org/pdf/2205.07318

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does the fully diffusing Poisson Brownian SIR model survive for sufficiently small positive removal rate when d>=2?

P0011-A001 | Counterexample to the usual pathwise lifetime monotonicity coupling

Argument or result

Consider a deterministic contact schedule with A initially infected: A-C at time .1, A-B at 1, C-B at 9, B-D at 10, and no other contacts. Baseline infectious lifetimes are L_A=.5, L_C=12, L_B=2. Then B is infected at 9 and infects D at 10. Multiply every lifetime by 3: A now infects B at 1; B is removed at 7, cannot be reinfected at 9, and D is never infected.

Check

Check both timelines: C remains infectious at 9 in each case; A has already died at 1 only in the baseline. The example refutes infection-set inclusion under scaled lifetimes on a fixed contact schedule.

Remaining obstruction

This is not a counterexample to survival monotonicity or to the Brownian epidemic conjecture. It blocks a direct attractiveness proof; a Brownian contact-support realization was not proved.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0012 Gaussian tensor type-2 bound

Source: Randomstrasse101: Open Problems of 2025 | Conjecture 16, Entry 8

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For symmetric order-r tensors T_i on R^d, p>=2 and independent standard Gaussians g_i, is E||sum_i g_i T_i||_(l_p) bounded by polylog(d,n) times O_(r,p)(d^(1/2-1/p) sqrt(sum_i ||T_i||_(l_p)^2))?

Scope

The unresolved range is p<2r apart from known matrix and other special cases. The source says p>=2r is proved. All parameters remain one problem.

P0012-A001 | Diagonal-tensor Gaussian estimate

Argument or result

For a diagonal homogeneous form T(x)=sum_j a_j x_j^r and 2<=p<=r, its supremum over the l_p unit ball is max_j |a_j|. Indeed sum |x_j|^r<=1 and a coordinate vector attains the maximum. For independent standard g_i, each diagonal coefficient of sum_i g_i T_i has variance at most sum_i ||T_i||^2, so the Gaussian maximum bound gives E||sum_i g_i T_i|| <= sqrt(2 log(2d)) (sum_i ||T_i||^2)^(1/2).

Check

The upper bound follows by the exponential-moment union bound and minimizing its parameter; coordinate vectors prove the norm identity even for even r by taking absolute value outside the form.

Remaining obstruction

This verifies a restricted diagonal family. Off-diagonal terms and the full p range require the tensor geometry absent here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0013 Expected Lovasz theta of a dense independent-edge graph

Source: Randomstrasse101: Open Problems of 2025 | Conjecture 17, Entry 9

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For G distributed as G(n,1/2), is E theta(G)=(1+o(1)) sqrt(n)?

Scope

Theta is the SDP with PSD matrix, trace one, and zero entries on graph edges.

P0013-A001 | Complement symmetry and uniform-integrability obstruction

Argument or result

The theta product inequality theta(G) theta(complement G)>=n and complement symmetry of G(n,1/2) give E theta>=sqrt(n) by arithmetic-geometric mean. A proof of concentration on a logarithmic scale would still need tail control: set Z_n to plus or minus (log n)/2, each with probability 1/(2 sqrt(n)), and zero otherwise. Then Z_n tends to zero in probability and variance, but E exp(Z_n) tends to 3/2.

Check

The example has a symmetric law and exp(Z_n) in [n^(-1/2),n^(1/2)]. Its expectation is 1-1/sqrt(n)+(1+n^(-1))/(2), which tends to 3/2. It is not a random-graph construction.

Remaining obstruction

Need a sharp upper bound and uniform integrability at the sqrt(n) scale; complement symmetry only yields the lower bound.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0014 Expected Lovasz theta of a random circulant graph

Source: Randomstrasse101: Open Problems of 2025 | Conjecture 18, Entry 9

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the graph on Z/nZ formed by independently including each unordered nonzero generator pair {k,-k} with probability 1/2, is E theta(G)=(1+o(1)) sqrt(n)?

Scope

The inverse-pair convention includes the order-two generator once when n is even; this differs substantively from independent edges.

P0014-A001 | Vertex-transitive complement identity

Argument or result

A circulant graph is vertex transitive, so theta(G) theta(complement G)=n. Its random model is complement invariant. For Z=log(theta(G)/sqrt(n)), Z has the law of -Z and E theta/sqrt(n)=E cosh Z. The conjecture is therefore equivalent to E(cosh Z-1) tending to zero.

Check

The exact identity is stronger than the general product inequality. The symmetric rare-tail example in P0013 has Z tending to zero in probability yet E cosh Z tending to 3/2, so concentration alone is insufficient.

Remaining obstruction

Must estimate the exponential tails of the logarithmic theta fluctuation for the dependent circulant adjacency structure.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0015 Random complex phase-retrieval injectivity at 4M-5 measurements

Source: Randomstrasse101: Open Problems of 2025 | Conjecture 19(a,b), Entry 10

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Let p_M be the probability that a complex Gaussian (4M-5)-by-M matrix gives an injective phaseless measurement map modulo global phase. Is p_M<1 for each meaningful M, and does p_M tend to zero?

Scope

M>=2 is needed for the matrix size to be meaningful. The two clauses are retained as one source conjecture, not counted twice.

P0015-A001 | Complete M=2 obstruction via Hermitian lifting

Argument or result

For M=2 the matrix has three complex Gaussian rows. Almost surely these span C^2. The real-linear measurement map from the four-dimensional Hermitian 2-by-2 space to R^3 has a nonzero kernel element H. H cannot be positive or negative semidefinite: a_i^* H a_i=0 for a spanning set would force H=0. Thus H has one positive and one negative eigenvalue and can be written xx^*-yy^* with noncollinear x,y. Their three intensity measurements agree. Hence p_2=0.

Check

Dimension gives a nonzero H without assuming generic rank three; the spanning condition rules out semidefinite kernel elements. Spectral decomposition independently verifies equal squared magnitudes.

Remaining obstruction

For M>2 a nonzero kernel matrix need not have rank at most two with one eigenvalue of each sign. The requisite low-rank intersection probability remains unresolved.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0016 Worst square-submatrix conditioning in a Gaussian frame

Source: Randomstrasse101: Open Problems of 2025 | Open Problem 21 and Definition 10.1, Entry 10

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a (2M-1)-by-M matrix of independent standard real Gaussians, determine the asymptotic size of omega(A), the minimum smallest singular value over row subsets whose complement has rank below M. In particular determine whether its decay is exponential and identify the exponential rate.

Scope

Almost surely every M-row subset is nonsingular, so the minimum reduces to M-row subsets. Interpret asymptotic size in probability; the source does not specify a convergence mode or a normalized logarithmic limit.

P0016-A001 | Union small-ball lower bound and one-normal upper bound

Argument or result

For a standard M-by-M real Gaussian matrix A, sigma_min(A)<=t implies that some column has distance at most sqrt(M)t from the span of the other columns: project a minimizing unit vector equation using a coordinate of magnitude at least 1/sqrt(M). Each such distance has the law |N(0,1)|, so P(sigma_min<=t)<=sqrt(2/pi) M^(3/2)t. There are at most 4^M M-row submatrices of a (2M-1)-by-M frame. Therefore its minimum omega exceeds exp(-(log 4+epsilon)M) with probability tending to one. Conversely fix M-1 rows and a unit normal u. The M remaining row projections are independent N(0,1), and omega is at most their minimum absolute value, giving omega<=C log(M)/M with probability tending to one for sufficiently large C.

Check

Full spark holds almost surely. Adding rows increases the Gram matrix, so the minimum relevant frame bound is attained among M-row subsets. Both bounds use union/inclusion inequalities and never assume independence among overlapping minors.

Remaining obstruction

The exponential lower bound and polynomial upper bound are far apart. Sharp extremes of the strongly dependent minors remain uncontrolled.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0017 Dimension-free log-concave Poincare or Cheeger constant

Source: Randomstrasse101: Open Problems of 2025 | Conjecture 30, Entry 13

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is there a universal positive lower bound for the Euclidean Cheeger constant of every isotropic log-concave probability measure, in every dimension?

Scope

Use outer boundary liminf in the standard definition; the source's displayed halfspace formula has an S/H notational mismatch. Thin-shell and slicing results are separate and reported resolved by this source.

P0017-A001 | Gaussian special case by orthogonal expansion

Argument or result

For standard Gaussian measure, expand a polynomial f in orthonormal multivariate Hermite polynomials with coefficients a_alpha. Then Var(f)=sum_(|alpha|>=1) a_alpha^2 <= sum |alpha| a_alpha^2 = E||gradient f||^2. Closure in Gaussian Sobolev norm extends the inequality to its natural domain.

Check

The constant is attained by any nonconstant linear function, checking both normalization and dimension independence.

Remaining obstruction

A general isotropic log-concave measure lacks this diagonal Hermite generator. This special case does not prove its uniform Poincare or Cheeger bound.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0018 Sharp polynomial-logarithmic expected norm of graph matrices

Source: Randomstrasse101: Open Problems of 2025 | Conjecture 31 and Definition 14.2, Entry 14

https://arxiv.org/pdf/2603.29571

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For each fixed graph shape alpha with specified left and right index sets, form M_alpha by summing injective labelings with products of independent Rademacher edge variables. Must E||M_alpha|| be Theta(n^f(alpha) (log n)^g(alpha)) for some shape-dependent exponents?

Scope

The target is the general existence of sharp exponents, not a separate problem for every shape. Definitions of overlapping or internal vertex sets must follow the full cited graph-matrix convention.

P0018-A001 | Moment expansion with a crude operator-norm bound

Argument or result

For every finite random graph matrix, E||M|| <= (E||M||_F^2)^(1/2). Expanding each entry as a sum of Rademacher edge monomials, cross terms vanish unless the two monomials use every edge with even combined multiplicity. For a fixed shape the remaining label identifications can be counted to obtain a polynomial upper bound in n.

Check

Orthogonality concerns equality of edge monomials, not equality of vertex labelings. Distinct labelings can produce the same edge set and must not be discarded.

Remaining obstruction

The Frobenius bound can lose powers of n and cannot decide the sharp operator norm or the existence of one power-log asymptotic form.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0019 Brownian exit approach directions in higher dimensions

Source: Krzysztof Burdzy: My favorite open problems | Problem 1

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For Brownian motion exiting a domain in dimension d>2, is the set of limiting approach directions at the exit point almost surely a sphere or a hemisphere?

Scope

The question is meaningful on finite exit; either condition on finite exit or restrict to bounded domains. Do not call infinite exit a counterexample to the intended geometric question.

P0019-A001 | Smooth-boundary directional containment

Argument or result

At an exit point z where the domain boundary is C^1 with outward unit normal n, local differentiability gives n dot (z-x)>=-o(|z-x|) for x in the domain approaching z. Thus every limit of (z-B_s)/|z-B_s| as s increases to the exit time lies in the closed hemisphere {u:n dot u>=0}.

Check

The sign is checked on a half-space, where the vector from an interior point to the exit point has positive outward-normal component.

Remaining obstruction

Containment does not prove that all hemisphere directions occur, and there is no supporting tangent half-space at the arbitrary boundaries in the full question.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0020 Finite-crossing arcs through a planar Brownian trace

Source: Krzysztof Burdzy: My favorite open problems | Problem 2(i)

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Almost surely, can every two points outside the trace of planar Brownian motion on [0,1] be connected by a Jordan arc meeting the trace at only finitely many points?

Scope

The quantifier is simultaneous over all pairs of outside points, not a single deterministic pair.

P0020-A001 | Reduction to component adjacency and access

Argument or result

If both endpoints lie in the same component of the complement of the Brownian trace, an open connected planar set is polygonally connected; removing loops gives a simple arc avoiding the trace entirely. If an arc crosses the trace finitely many times, its complementary intervals give a finite sequence of complement components with touching closures at successive crossing points.

Check

A closure contact alone does not ensure an accessible common boundary point. This is why the necessary finite-chain condition was not promoted to an equivalence.

Remaining obstruction

Need finite accessible chains for every pair of components, simultaneously for the Brownian trace.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0021 Interior remainder of planar Brownian trace

Source: Krzysztof Burdzy: My favorite open problems | Problem 2(ii)

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Delete from a planar Brownian trace on [0,1] all boundaries of connected components of its complement. Is the remaining set totally disconnected almost surely?

Scope

This is a different topological property from Problem 2(i), although a negative answer is stated to imply a negative answer there.

P0021-A001 | Failure of transferring total disconnectedness through the path map

Argument or result

A possible approach is to identify a totally disconnected set of Brownian times surviving boundary removal. That would not suffice: the Cantor function restricted to the middle-third Cantor set maps this totally disconnected compact set continuously onto [0,1]. Continuous images can be connected.

Check

The restriction is onto because every value of the Cantor function is attained on the Cantor set, including plateau values at endpoints.

Remaining obstruction

The spatial remainder needs direct Brownian geometry. Showing it contains no simple arc would also be weaker than showing every connected component is a singleton.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0022 Percolation dimension of a planar Brownian trace

Source: Krzysztof Burdzy: My favorite open problems | Problem 3

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is the infimum of Hausdorff dimensions of nontrivial Jordan arcs contained in a planar Brownian trace on [0,1] equal to one almost surely?

Scope

A nontrivial arc contains at least two distinct points. The lower bound one alone is not a solution.

P0022-A001 | Universal dimension lower bound for connecting sets

Argument or result

Any connected set containing distinct x and y has Hausdorff dimension at least one. Orthogonal projection onto their joining line is Lipschitz and has connected image containing the interval between the projections, so its dimension is at least one.

Check

This argument does not require rectifiability or a parametrized simple curve, and hence applies to the connectivity notion in a percolation-dimension lower bound.

Remaining obstruction

Polygonal approximations of a Brownian path are not subsets of its trace. The missing step is constructing sufficiently thin connecting subsets inside the trace itself.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0023 Efficient reflected-Brownian coupling in acute triangles

Source: Krzysztof Burdzy: My favorite open problems | Problem 4

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

In any acute triangle, can reflected Brownian motions from distinct starting points be coupled so their meeting-time tail decays at every exponential rate strictly below the first positive Neumann spectral rate?

Scope

Check the generator convention in Burdzy-Kendall (2000): the page calls its rate a Laplacian eigenvalue. A factor-two convention discrepancy is not a solution of the coupling question.

P0023-A001 | Spectral lower bound on coupling tails

Argument or result

Let phi be a bounded nonconstant Neumann eigenfunction with semigroup P_t phi=e^(-lambda t) phi. For any coalescent coupling with meeting time T, |P_t phi(x)-P_t phi(y)|<=2||phi||_infinity P(T>t). Choosing phi(x)!=phi(y) yields P(T>t)>=c e^(-lambda t).

Check

The exponent is the eigenvalue of the actual reflected-Brownian generator: using Delta rather than Delta/2 changes it by a factor of two.

Remaining obstruction

This proves that the spectral rate is an upper limit on coupling efficiency, but gives no coupling attaining arbitrarily close to that rate in an acute triangle.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0024 Failure of synchronous contraction in a bounded planar domain

Source: Krzysztof Burdzy: My favorite open problems | Problem 5(i)

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does some bounded connected smooth planar domain admit synchronously coupled reflected Brownian motions whose distance has positive limsup with positive probability?

Scope

The page states that any such domain must have at least two holes by existing results.

P0024-A001 | Synchronous contraction on convex domains

Argument or result

For two reflected motions driven by the same Brownian path, the noise cancels in their difference. With inward normals, d|X-Y|^2=2<X-Y,n(X)>dL_X-2<X-Y,n(Y)>dL_Y. Convexity makes both contributions nonpositive, so the distance is nonincreasing.

Check

Each sign follows from the supporting half-space inequality at the boundary point. The calculation uses finite-variation reflection terms and has no missing quadratic-variation term.

Remaining obstruction

Nonincrease alone does not prove convergence to zero. Nonconvex domains can have both signs and require control of the long-run reflection geometry.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0025 Synchronous reflection outside a disk

Source: Krzysztof Burdzy: My favorite open problems | Problem 5(ii)

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For synchronously coupled reflected Brownian motions in the exterior of a nondegenerate closed disk, can their distance have positive limsup with positive probability?

Scope

This unbounded exterior geometry differs from the bounded-domain existence question.

P0025-A001 | Exterior-disk reflection has both distance signs

Argument or result

At x=(R,0) on the inner circular boundary of the exterior domain, its inward-to-domain normal is (1,0). For y=(r,0), r>R, <x-y,n>=R-r<0; for y=(-r,0) it is R+r>0. Hence a reflection at X can either shrink or enlarge the synchronous separation.

Check

Both test locations lie in the exterior and the normal points into that same domain, fixing the otherwise easy sign error.

Remaining obstruction

A global monotone-distance proof is unavailable. The question turns on boundary occupation and angular correlations over indefinitely long times.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0026 Non-explosion of Fleming-Viot Brownian particles in arbitrary domains

Source: Krzysztof Burdzy: My favorite open problems | Problem 6 with 2024 update

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For N>=3 Brownian particles in any bounded open connected domain, killed at the boundary and immediately cloned from a uniformly selected survivor, are the successive death times almost surely non-accumulating?

Scope

N=2 is explicitly reported solved by Kwasnicki (2024). The earlier general proof had an irreparable error; it is not admissible evidence for the general result.

P0026-A001 | Attempted boundary-distance Lyapunov proof

Argument or result

The candidate Lyapunov function -sum_i log dist(X_i,boundary D) becomes singular when particles approach the boundary, but an Ito argument needs derivative bounds. Even for a smooth domain, the distance is nonsmooth on its medial set; for arbitrary domains there may be no uniformly controlled smooth substitute. At a cloning event another particle may be still closer to the boundary, so the jump is not uniformly favorable.

Check

The known N=2 result is excluded from this target. A pair comparison does not transfer automatically to N>=3 because a particle can clone the third particle rather than its proposed comparator.

Remaining obstruction

No Lyapunov drift bound uniform over arbitrary boundary geometry and all cloning configurations was obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0027 Rigidity of shy reflected-Brownian couplings

Source: Krzysztof Burdzy: My favorite open problems | Problem 7

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If reflected Brownian motions in a bounded connected domain can remain a positive distance apart forever with positive probability, must there be such a coupling with one path a fixed deterministic function of the other at each time?

Scope

Regularity sufficient to define the intended reflected Brownian motion needs to be fixed from the cited background. A metric-graph counterexample on the page is not a Euclidean-domain counterexample.

P0027-A001 | Explicit deterministic shy coupling on an annulus

Argument or result

If an isometry F preserves a reflected-Brownian domain and has inf_x |x-F(x)|>0, take Y_t=F(X_t). Orthogonal transformation of Brownian noise and equivariance of inward reflection show Y has the correct reflected law, while separation stays positive. On R<|x|<R_out, F(x)=-x yields |X_t-Y_t|>=2R.

Check

The annulus excludes the origin; a disk would have zero minimum displacement and would not satisfy this check.

Remaining obstruction

This verifies a nonempty deterministic class. It does not turn an arbitrary shy coupling into a single state map with the required path consistency.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0028 Bounded independent Brownian pieces and two-sided Brownian motion

Source: Krzysztof Burdzy: My favorite open problems | Problem 8; compare Burdzy-Scheutzow (2014) Problem 7.7

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Concatenate independent Brownian path segments cut at stopping times indexed by all integers, with lengths uniformly bounded by a deterministic constant and divergent sums in both directions. Is the resulting path two-sided Brownian after some random time shift?

Scope

The paper's related backward-Brownian question and its almost-sure random bound are recorded as related variants, not an extra count. Independent lengths alone are not the same as stopping times depending on their Brownian paths.

P0028-A001 | Stationary Palm construction in the identically distributed finite-mean case

Argument or result

For iid stopped Brownian segments with 0<E T<infinity, the cycle sequence is ergodic under shifting its index. A stationary occupation law is Q(F)=E[integral_0^T F(theta_s X) ds]/E T. Shifting by t changes the numerator by E[integral_T^(T+t) F(theta_s X) ds-integral_0^t F(theta_s X) ds]=0 because the concatenation rooted at the next cycle has the same Palm law. Tilting the iid cycle sequence by T_0/E T and choosing a uniform time in that cycle realizes this law; the forward filling lemma supplies that tilt by a finite randomized index shift.

Check

For each deterministic s, {s<T} is measurable from the segment up to s. Future Brownian increments followed by independent stopped segments retain the Brownian law and are independent of the past. Integrating over s gives a Brownian future under Q; stationarity then supplies a Brownian past and its independence. This is a proof outline pending full artifact and earlier-literature comparison, not a newly claimed theorem.

Remaining obstruction

The actual question allows independent non-identically distributed segments. There is then no invariant iid index law or common Palm mean to which this argument applies.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0028-A002 | Primary-source check of the iid finite-mean reduction

Argument or result

Burdzy-Scheutzow (2014), Theorem 5.1 explicitly proves that strongly decomposable processes with finite mean segment length are two-sided Brownian motions. P0028-A001 rederives this known special case and is not a novel solution.

Remaining obstruction

The independent non-identically distributed bounded-length question remains unresolved here.

Verification: source_theorem_checked | Independent review: none | Full question solved: False

P0029 Separation of two peaks in random square-grid labelings

Source: Krzysztof Burdzy: My favorite open problems | Problem 9

https://sites.math.washington.edu/~burdzy/open_mathjax.php

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Uniformly label an n-by-n square grid and condition on exactly two local maxima. Does their graph distance divided by n converge in distribution to zero?

Scope

This is the literal displayed question, despite the heading about repulsion. Verify against Twin peaks (2020) before correcting the sign or reporting a counterexample.

P0029-A001 | Exact small-grid count and an orientation reformulation

Argument or result

On the 2-by-2 grid (a four-cycle), exactly two peaks must be opposite and carry labels 3 and 4. There are two choices of diagonal, two orders of the top labels, and two orders of the bottom labels: eight of the 24 labelings. Their graph distance is always two. More generally orient each edge toward its larger label. Two peaks means precisely two sinks of the induced acyclic orientation; the number of labelings yielding a fixed orientation is its number of linear extensions.

Check

Adjacent vertices cannot both be peaks. In the four-cycle, a proposed peak with label 2 would have to exceed both other nonpeak labels, which is impossible; this checks the count without enumeration software.

Remaining obstruction

The small-grid calculation gives no large-n scale. The orientation sum requires sharp weighted counts of acyclic orientations with two prescribed sinks. The source title says repel while its displayed question asks distance/n -> 0; that wording discrepancy is retained, not silently corrected.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0030 Compatibility of forward and backward Brownian decompositions

Source: Forward Brownian motion | Problems 7.1-7.3, PDF pages 25-26

https://math.washington.edu/sites/math/files/documents/research/1302.6958.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does a decomposable forward Brownian motion which is also backward Brownian necessarily become two-sided Brownian after a random shift? The source also asks weaker related formulations using independence of past and future at a random time.

Scope

The three linked formulations are grouped as one compatibility problem. A general process that is both forward and backward Brownian without the extra assumptions is already counterexampled in the paper.

P0030-A001 | Common-splice compatibility obstruction

Argument or result

Two independent standard Brownian half-paths attached at the same deterministic time give a two-sided Brownian motion. For a general process certified separately as forward and backward Brownian, the certifying random times can differ. Moving either certificate to another random time can bias its marginal law, as the selection-density calculation in P0032 demonstrates.

Check

Independence of two halves at a proposed joining time is insufficient if their Brownian marginal laws have been lost there.

Remaining obstruction

No construction aligns both certificates at one time while preserving both marginal laws and their independence.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0031 Sparse first-passage pieces in a forward Brownian concatenation

Source: Forward Brownian motion | Problem 7.4, PDF page 26

https://math.washington.edu/sites/math/files/documents/research/1302.6958.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

In a two-sided independent concatenation, use unit-length Brownian segments except at increasingly rare negative-time indices where segments stop on first hitting -1. Characterize frequencies giving backward or two-sided Brownian motion.

Scope

The exact deterministic or random schedule is part of the classification; do not invent a schedule and count it as another sourced question.

P0031-A001 | Sparse exceptional hitting-time clock estimate

Argument or result

For a standard Brownian first hitting time tau of -1, reflection gives P(tau>t)=2 Phi(1/sqrt(t))-1 ~ sqrt(2/pi)t^(-1/2), hence E tau=infinity. If exceptional segment indices are 2^k, then sum_k P(tau_k>epsilon 2^k)<infinity. Borel-Cantelli yields tau_k/2^k ->0 almost surely. Splitting the sum into finitely many early terms and a geometrically weighted tail gives sum_(k<=K) tau_k/2^K ->0.

Check

Independence is not needed for this Borel-Cantelli upper bound. Infinite means do not contradict the almost-sure sparse-clock estimate.

Remaining obstruction

A negligible exceptional time fraction is an asymptotic statement; it does not construct an exact backward Brownian law after a finite random shift. The full frequency classification remains open here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0032 Embedding an equivalent Brownian law by adding a finite prefix

Source: Forward Brownian motion | Problem 7.5, PDF page 26

https://math.washington.edu/sites/math/files/documents/research/1302.6958.pdf

Historical status: source_says_open | Current status: unverified | Result: candidate_proof

Question

Given a process whose law on nonnegative time is equivalent to Wiener measure, can a random finite prefix be added, on an enlarged probability space if necessary, to produce a process with Wiener law?

Scope

No independence or Brownian stopping-time restriction is explicit in the source. The candidate proof uses a finite nonnegative integer random time depending on entire paths and auxiliary randomness.

P0032-A001 | Forward density matching on the ergodic two-sided Wiener shift

Argument or result

Complete candidate construction of a finite nonnegative integer prefix time for any suffix law absolutely continuous with respect to Wiener measure

Check

Two-state cyclic example checks the pushforward direction and normalization; finite-window rejection sampling gives an independent special-case construction

Remaining obstruction

No mathematical gap currently identified under the stated interpretation; source interpretation, measurability details, and earlier filling/embedding literature still under review

Verification: candidate_under_author_and_literature_checks | Independent review: none | Full question solved: False

P0033 Ergodicity of stationary Galton-Watson flows

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 17.18, printed p.575, PDF p.592

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: candidate_counterexample

Question

If a consistent equivariant Borel flow rule has a stationary probability measure equivalent to Galton-Watson law, must the associated subtree Markov chain be ergodic?

Scope

The definition permits zero child weights. Strict positivity is a separate sufficient hypothesis in Proposition 17.17. Exercise 17.22 only treats a merely absolutely continuous stationary measure.

P0033-A001 | Endogenous AND/OR tree coloring and conditional subtree selection

Argument or result

Candidate counterexample: GW itself stationary, with two invariant color classes of mass one half

Check

Exact second-moment recursion proves color measurability; full conditional subtree law proves stationarity; source definition expressly permits zero child weights

Remaining obstruction

No mathematical gap currently identified; full author recheck and prior tree-automaton literature pending

Verification: candidate_under_author_and_literature_checks | Independent review: none | Full question solved: False

P0034 Recurrence of the planar horizontal-new vertical-old walk

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 2.27, printed p.56

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Start at the origin in Z^2; after first visiting a site step horizontally with either sign equally likely, and after a repeat visit step vertically with either sign equally likely. Does any site receive infinitely many visits with positive probability?

Scope

The source indexing omits the first step; use its intended new-site rule also at time zero. The analogous four- and three-dimensional results do not decide this planar model.

P0034-A001 | Martingale estimates for the axis-switching walk

Argument or result

The axis of the next step is predictable from the past, while its sign is fair. Both coordinates are martingales, their cross quadratic covariation is zero, and E|X_n|^2=n. Bounded increments and Azuma give summable bounds for P(|X_n|>epsilon n), so X_n/n tends to zero almost surely.

Check

The predictable quadratic variation of the horizontal coordinate counts first visits, and the vertical one counts repeat visits; their sum is exactly n.

Remaining obstruction

Zero speed and a diffusive second moment do not imply recurrence. The occupation-dependent balance of the two coordinate clocks remains uncontrolled.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0035 Positive speed of inward-biased walks below the Cayley growth rate

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 3.11, printed p.91

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a Cayley graph of exponential growth rate b and 1<lambda<b, does the distance from the identity divided by time for RW_lambda converge almost surely to a strictly positive speed?

Scope

RW_lambda uses the source's radial conductances, favoring motion toward the identity for lambda>1. The case lambda=1 may have zero speed and is excluded.

P0035-A001 | Exact radial speed on a regular tree

Argument or result

On the (b+1)-regular rooted tree, away from the root the radial walk steps out with probability b/(b+lambda) and in with probability lambda/(b+lambda). For lambda<b its speed is (b-lambda)/(b+lambda)>0: the root is visited finitely often and the bounded-difference martingale law of large numbers applies to the radial drift.

Check

The speed tends to zero at lambda=b and to one at lambda=0, checking the normalization.

Remaining obstruction

General Cayley graphs have dead ends and nonuniform distance-level neighborhoods. Transience alone does not control time spent in traps.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0036 Percolation threshold on arbitrary geodesic Cayley spanning trees

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 3.12, printed p.91

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If T is any geodesic spanning tree rooted at the identity of a Cayley graph of growth rate b, must br(T)=b, equivalently must its Bernoulli bond percolation threshold be 1/b?

Scope

The branching-number/percolation equivalence is a theorem of the source. The question is stated there geometrically; its probability formulation is recorded transparently. Lexicographically minimal spanning trees satisfy it.

P0036-A001 | Equal level growth does not determine branching number

Argument or result

A geodesic spanning tree has exactly the Cayley graph sphere sizes at each level. But growth can exceed branching number for a bounded-degree tree: take a single infinite spine and at its nth vertex attach a finite b-ary tree of height n. Its level sizes have exponential rate sqrt(b)>1, while its only infinite ray is the spine and arbitrarily deep singleton cutsets give branching number one.

Check

At level m the largest attached contribution starts around n=m/2 and has order b^(m/2), up to constant factors and a geometric sum. Every attachment is finite, so no additional infinite ray appears.

Remaining obstruction

This is not a geodesic Cayley spanning-tree counterexample. It proves that matching sphere counts alone cannot establish the conjectured branching number.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0037 BK disjoint occurrence for weighted uniform spanning trees

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 4.10, printed p.125

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For increasing events A,B on edge subsets, is the probability of their disjoint occurrence under any weighted uniform spanning-tree law at most P(A)P(B)?

Scope

Disjoint witnesses use disjoint edge-coordinate sets. Arbitrary events are excluded; negative association for fixed disjoint coordinates is weaker. Goldman 2023 proves a special class generated by simple points.

P0037-A001 | BK for fixed conjunctions and a nonincreasing-event obstruction

Argument or result

For disjoint fixed edge sets F and H, the spanning-tree determinantal formula and Hadamard-Fischer inequality give P(F union H subset T)<=P(F subset T)P(H subset T). Thus fixed increasing conjunctions obey the desired inequality. The increasing condition matters: on a triangle, A={e in T} and B={f not in T}, e!=f, have disjoint coordinate witnesses but P(A intersection B)=1/3>2/9=P(A)P(B).

Check

There are three equiprobable spanning trees of a triangle, which checks the latter probabilities directly. The determinantal argument is for a positive semidefinite transfer-current kernel.

Remaining obstruction

General increasing disjoint occurrence is a union over adaptive witnesses. A fixed-coordinate negative-association inequality cannot simply be applied after that union.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0038 Transience preserved in Cayley percolation clusters

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 6.44, printed p.216

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is every infinite Bernoulli percolation cluster on a transient Cayley graph almost surely transient?

Scope

The claim concerns all infinite clusters simultaneously. Nonamenable and Euclidean cases have substantial existing results.

P0038-A001 | Percolation on regular trees

Argument or result

For bond percolation on the (b+1)-regular tree, descendants of a nonroot vertex form a Galton-Watson process with Binomial(b,p) offspring. Conditional on survival when bp>1, its branching number is bp almost surely, so the electrical-network branching-number criterion gives transience.

Check

The root has b+1 possible children but a finite root modification does not change transience. At bp<=1 the nondegenerate branching process dies out.

Remaining obstruction

The reduction uses independent descendant trees and does not extend to clusters with overlapping paths in a general transient Cayley graph.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0039 Positive-speed equivalence for Cayley percolation clusters

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 6.45, printed p.216

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does simple random walk on a Cayley graph have positive speed if and only if simple random walk on its infinite Bernoulli percolation clusters has positive speed almost surely?

Scope

Use graph distance in the cluster for the cluster speed as in the source context; compare with ambient speed carefully. Nonamenable cases are reported known.

P0039-A001 | Zero speed on every bounded-degree subexponential-growth subgraph

Argument or result

Let H be a connected subgraph of a bounded-degree graph G with |B_G(x,n)|=exp(o(n)). The Carne-Varopoulos bound gives P_x(X_n=y)<=2 sqrt(deg_H(y)/deg_H(x)) exp(-d_H(x,y)^2/(2n)). Summing over reachable vertices yields P(d_H(x,X_n)>=epsilon n)<=C exp(o(n)-epsilon^2 n/2), which is summable. Thus the intrinsic speed on H is zero almost surely.

Check

Reachability in H implies membership in B_G(x,n), but distances in the exponential bound remain intrinsic to H. Bounded degrees control the prefactor. This is a consequence of the classical heat-kernel bound, not a novelty claim.

Remaining obstruction

It settles the zero-speed equivalence for subexponential-growth Cayley graphs and their clusters only. Exponential-growth cases need different information.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0040 Uniqueness threshold below one in one-ended transitive graphs

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 7.27, printed p.258

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does every locally finite quasi-transitive graph with exactly one end have Bernoulli percolation uniqueness threshold $p_u<1$?

Scope

A one-ended finite-presentation class is known. The assertion $p_c<1$ alone is weaker and now resolved in the transitive setting.

P0040-A001 | Amenable Cayley special case and a cutset bottleneck

Argument or result

For an amenable Cayley graph with one end, uniqueness gives $p_u=p_c$. The known strict $p_c<1$ theorem for transitive graphs not roughly isometric to Z therefore gives $p_u<1$ in this special case. One-endedness excludes the two-ended roughly-Z case.

Check

This use is restricted to Cayley graphs; no unverified extension of the strict-critical-threshold theorem to every quasi-transitive graph is presumed.

Remaining obstruction

The remaining nonamenable quasi-transitive setting needs bounds on separating cutsets or cycle structure. A bounded cycle-basis argument is unavailable without its hypotheses.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0041 Nonuniqueness phase for nonamenable quasi-transitive graphs

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjectures 7.31 and 7.41, printed pp.260 and 265

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Must p_c(G)<p_u(G) for every nonamenable quasi-transitive graph? The source's related formulation asks whether the property p_c<p_u is preserved by rough isometries among quasi-transitive graphs.

Scope

The closely linked rough-isometry conjecture is grouped here to avoid counting a reformulation separately. Nonunimodular and many other cases are established.

P0041-A001 | Strict gap on regular trees

Argument or result

On a (b+1)-regular tree with b>=2, branching-process survival gives p_c=1/b. For every p in (1/b,1), infinitely many separated descendant subtrees contain infinite open clusters while their attachment edge is closed, using independent positive-probability events. Thus there are infinitely many clusters and p_u=1.

Check

The attachment events can be placed on an infinite collection of disjoint descendant cones. For p<=1/b there is no infinite cluster.

Remaining obstruction

This proves the strict gap in a fundamental nonamenable family. Nonamenability by itself does not provide the same independent cone decomposition.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0042 Bootstrap thresholds as a characterization of amenable groups

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 7.42, printed p.266

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is a finitely generated group amenable exactly when every Cayley graph from every finite generating set has bootstrap complete-occupation critical probability p(G,k) in {0,1} for every threshold k?

Scope

Initially each vertex is occupied independently with probability p; vacant vertices with at least k occupied neighbors become occupied. Criticality is for positive probability of eventual occupation of all vertices.

P0042-A001 | Bootstrap thresholds on the line and the three-regular tree

Argument or result

On Z the one-neighbor bootstrap threshold is zero, whereas the two-neighbor threshold is one because any initially vacant adjacent pair stays vacant. On the three-regular tree with threshold two, the final vacant set is the two-core of the initial vacant percolation. If its density q<=1/2, all vacant clusters are finite almost surely and leaf pruning empties them. If q>1/2, with positive probability a vertex has two surviving vacant branches, giving a bi-infinite vacant path that remains vacant. Hence the complete-occupation threshold is p=1/2.

Check

An infinite bi-infinite path supplies a stable set since each vertex has two vacant neighbors. Finite trees have an exposed leaf and are exhausted by the pruning process. Critical Binomial(2,1/2) clusters die out.

Remaining obstruction

This illustrates both the amenable endpoint behavior and a nonamenable interior threshold. It does not classify every generating set or every group.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0043 Absence of infinite clusters at critical percolation

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 8.15, printed p.285

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If G is quasi-transitive and p_c(G)<1, is there almost surely no infinite Bernoulli cluster at p_c(G)?

Scope

Do not count each lattice dimension as a separate problem. Known nonamenable and high-dimensional cases do not settle the entire assertion.

P0043-A001 | Critical branching-process extinction

Argument or result

For the regular tree at p=1/b, an infinite descendant cluster would have survival probability x satisfying x=1-(1-px)^b. For b>=2 strict concavity gives 1-(1-x/b)^b<x for every x>0, so only x=0 is possible.

Check

The strict inequality can also be obtained from the binomial generating function lying strictly above its tangent at one. This excludes the deterministic b=1 exception, whose p_c is one.

Remaining obstruction

Critical clusters in a general quasi-transitive graph are not independent branching trees. Their loop dependence is not captured by this fixed-point proof.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0044 Invariant monotone coupling of free and wired uniform spanning forests

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.6, printed pp.348-349

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For an infinite network, is there a coupling (W,F) of its wired and free uniform spanning forests with W subset F almost surely whose joint law is invariant under all network automorphisms?

Scope

Strassen gives an unconstrained monotone coupling. Sofic Cayley groups admit invariant couplings by Lyons-Thom 2016; arbitrary invariant percolations can fail this property.

P0044-A001 | Invariant monotone coupling under an amenable action

Argument or result

Start with a monotone coupling of the two invariant forest laws under a countable amenable action. Average its translates over a Folner sequence. The product space of two edge configurations is compact, the prescribed marginals are unchanged, and the order relation is closed. A subsequential weak limit is therefore still a monotone coupling and is invariant because translation changes each average by at most its Folner boundary ratio.

Check

The argument averages probability measures, not individual configurations, so convexity preserves the marginals and the monotone support. Compactness is available for a countable locally finite edge set.

Remaining obstruction

For a nonamenable acting group there is no Folner averaging sequence. Existence of an arbitrary monotone coupling does not by itself produce an invariant one.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0045 Rational expected free-forest degree for finitely presented groups

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.11, printed p.351

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a Cayley graph of a finitely presented group, is the expected degree at the identity in its free uniform spanning forest rational?

Scope

This is the first l2-Betti-number question via E degree=2+2 beta_1. Counterexamples to broader Atiyah statements must be checked for both first degree and finite presentation before exclusion.

P0045-A001 | Rational finite-volume values do not force a rational limit

Argument or result

For the standard free-group Cayley tree the free spanning forest is the entire tree, so its expected degree is the integer 2r. Finite exhaustion spanning-tree expectations are rational under unit conductances, but rational approximants can converge to an irrational number, as floor(n sqrt(2))/n demonstrates.

Check

A finite tree has only one spanning tree, so the free-group calculation is exact and requires no limiting interchange.

Remaining obstruction

The general finitely presented-group question needs the algebraic structure behind the first l2 Betti number; rationality of finite combinatorial counts is insufficient.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0046 Connecting a free uniform spanning forest by arbitrarily sparse invariant additions

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.12, printed p.351

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For any Cayley graph, its free uniform spanning forest F, and epsilon>0, is there an invariant connected random spanning subgraph omega containing F with P(e in omega minus F)<epsilon for every edge e?

Scope

Added edges may depend on F. Independent Bernoulli sprinkling is explicitly known to fail for some nonamenable graphs.

P0046-A001 | Connection to the cost equality obstruction

Argument or result

If the free forest is already connected, taking its own configuration gives the requested extension with no added edges. Otherwise, an invariant connected extension with each edge added with probability at most epsilon would have expected root degree at most E deg_F(o)+deg_G(o) epsilon. The standard invariant-graphing bound would then imply cost(Gamma)<=E deg_F(o)/2=1+beta_1(Gamma) after epsilon tends to zero.

Check

The root-degree estimate is the finite sum of per-edge addition probabilities. Combined with the known opposite cost inequality, the proposed construction would prove equality, not merely a weak approximation statement.

Remaining obstruction

No dependent sprinkling construction was found. This reduction explains why independent sprinkling or a generic connectivity argument does not settle the question.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0047 Deterministic component count of the free uniform spanning forest

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.28, printed p.359

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

On every infinite connected locally finite network, is the number of free uniform spanning-forest components almost surely a constant?

Scope

Automorphisms with an infinite orbit give a positive answer. The remaining arbitrary-network setting lacks that ergodic shortcut.

P0047-A001 | Why tail triviality alone does not fix the component count

Argument or result

An ergodic action under which the forest law is invariant makes its component count deterministic. On an arbitrary network there need not be such an action. Trying to substitute tail triviality fails at the logical step: adding one edge between two components can change the component count, so this statistic has not been shown to be measurable with respect to the edge tail sigma-field.

Check

The finite edge change can preserve acyclicity when it joins two different tree components, so acyclicity does not remove this obstruction.

Remaining obstruction

A forest-law-specific argument making the count constant is still needed in the absence of a useful invariant ergodic action.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0048 One or infinitely many free-forest components without unimodularity

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.29, printed p.360

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

On every transitive network, is the free uniform spanning forest almost surely connected or composed of infinitely many components?

Scope

The unimodular case was solved by Hutchcroft-Nachmias 2017 and Timar 2018. The unresolved scope retained here is nonunimodular.

P0048-A001 | Invariant ergodicity permits a finite component count larger than one

Argument or result

On the Cayley graph of Z with generators +/-1 and +/-2, retain exactly the edges of length two. The resulting deterministic spanning forest has two bi-infinite components, is translation invariant, and its point-mass law is ergodic. Thus those two properties alone cannot exclude a finite count of two.

Check

The two components are the even and odd integers; every retained vertex has degree two and neither component contains a cycle.

Remaining obstruction

This forest is not asserted to have a uniform-spanning-forest law. The target requires extra free-forest structure in the nonunimodular case.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0049 Infinitely many ends in all free-forest components without unimodularity

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.52, printed p.377

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

On a transitive network with distinct free and wired uniform spanning-forest laws, must every free-forest component have infinitely many ends almost surely?

Scope

The source's subsequent indistinguishability theorem settles the unimodular case. Haar-light components alone do not automatically refute an ends statement.

P0049-A001 | Missing mass-transport step in the nonunimodular case

Argument or result

In the unimodular setting, an expected-degree argument can establish the presence of infinitely ended components, and indistinguishability can promote a component property to all components. On a nonunimodular graph the unweighted root mass balance used in that argument is invalid; modular weights enter and light components need not receive the same balance.

Check

An existence statement about some components would remain weaker than the source assertion about every component even if the first step survived.

Remaining obstruction

No replacement weighted transport and no counterexample with the required FUSF/WUSF inequality were obtained. This is a failed proof route, not a new theorem.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0050 Singularity of distinct free and wired forest measures

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.58, printed p.385

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If the wired and free uniform spanning-forest measures differ on an infinite network, must they be mutually singular?

Scope

Known for trees and networks with an infinite automorphism orbit. Distinct probability measures need not be singular in general.

P0050-A001 | Distinct invariant ergodic laws are mutually singular

Argument or result

For a fixed countable group action, two distinct invariant ergodic probability measures are mutually singular. In the Lebesgue decomposition of one relative to the other, the absolutely continuous part is invariant. Its density is invariant and hence constant under the second ergodic law; ergodicity of the first forces its absolutely continuous mass to be either zero or one. Nonzero overlap therefore forces equality.

Check

The singular and absolutely continuous parts are preserved by the action by uniqueness of Lebesgue decomposition; this is the load-bearing invariance step.

Remaining obstruction

This handles forest laws admitting such an ergodic common action. Arbitrary networks may lack it; two distinct probability laws need not be singular in general.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0051 Zero-one law for all forest components having one end

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 10.59, printed p.386

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For both free and wired uniform spanning forests of an arbitrary infinite network, is the event that every component has one end of probability zero or one?

Scope

Both boundary conditions are grouped. Changing finitely many forest edges can merge components, so tail measurability needs a proof.

P0051-A001 | The all-one-ended event is not automatically a tail event

Argument or result

Take two disjoint one-ended infinite trees and add one edge joining them. Before the addition every component has one end; afterwards the connected tree has two ends, since deleting the joining edge separates two infinite pieces.

Check

The added edge creates no cycle and the change is finite. Thus the property can change within the class of spanning forests under a finite edge modification.

Remaining obstruction

Uniform-forest tail triviality cannot be invoked until a separate tail representation of the event is proved. The example is a logical obstruction, not a uniform-forest counterexample.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0052 Limiting bushes along a wired uniform-forest ray

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 10.60, printed p.386

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Along the unique ray from the identity in its wired uniform spanning-forest component, do the finite bushes at successive ray vertices, translated back to the identity, converge in distribution?

Scope

Restrict to Cayley graphs where the stated unique ray exists. The source records solutions for growth at least five and standard Z^2,Z^3,Z^4; the remaining low-growth Cayley cases must be checked individually before any promotion.

P0052-A001 | Stationarity along a chosen spine is an additional requirement

Argument or result

The WUSF viewed from an ordinary invariant root and the WUSF viewed from the nth vertex on its distinguished spine are different rooted laws. Moving along the spine is a configuration-dependent selection and need not preserve the original root law. Compactness or tightness, even if proved for the bush sequence, would only give subsequential limits.

Check

The elementary sequence of probability laws delta_0,delta_1,delta_0,delta_1 is tight but has no full limit, illustrating precisely the missing implication rather than modeling WUSF bushes.

Remaining obstruction

Need regeneration or asymptotic stationarity along the loop-erased spine. Known high-growth and standard-lattice cases in the source must be removed before claiming a new remaining case.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0053 One-ended wired minimal forests without unimodularity

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 11.13, printed p.401

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does each wired minimal spanning-forest component of a nonunimodular transitive graph have one end almost surely?

Scope

The accompanying request for useful nonhomogeneous sufficient conditions is retained as a related extension, not another count.

P0053-A001 | A failed largest-edge argument on an infinite path

Argument or result

A finite-cycle minimum-spanning-tree proof deletes the largest edge of a cycle. A proposed extension to a bi-infinite path fails because its supremum need not be attained: labels 1-1/(|k|+2) along k in Z approach one and have no maximum.

Check

The deterministic sequence only disproves the logical inference that every infinite path has a largest label. It is not an iid weighted-graph realization or a counterexample to the wired forest question.

Remaining obstruction

One-endedness needs an invasion-percolation or mass-transport argument that treats unattained suprema and the nonunimodular setting.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0054 Infinitely many ends in free minimal forests distinct from wired

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 11.16, printed p.402

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If a quasi-transitive graph has distinct wired and free minimal spanning-forest laws, does every free minimal-forest component have infinitely many ends almost surely?

Scope

The source and later Timar results establish unimodular cases; retain the genuinely remaining scope. All edges have independent continuous labels.

P0054-A001 | Regular-tree special case and resampling obstruction

Argument or result

On a regular tree every edge is a bridge and there are no finite cycles, so the free minimal spanning forest is the whole tree. It has infinitely many ends, providing a direct special case where free and wired forests differ. On a graph with cycles, inserting an edge can force deletion of another edge by the cycle-maximum rule.

Check

The free forest statement follows directly from its finite-cycle definition; it is not the wired definition, which also sees connections at infinity.

Remaining obstruction

A minimal forest is not a Bernoulli insertion-tolerant process. Arguments proving end properties for percolation do not transfer without controlling the forced deletions.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0055 Minimal-forest connectivity and uniqueness at p_u

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 11.18, printed p.403

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

On a nonamenable quasi-transitive graph with p_u<1, is the free minimal spanning forest connected almost surely exactly when Bernoulli percolation has a unique infinite cluster at p_u?

Scope

The source's stronger bijection between forest components and p_u clusters is grouped as a strengthening. One direction is already proved there.

P0055-A001 | The easy direction of the uniqueness connection

Argument or result

Under the p_u>p_c hypothesis in the source proposition, each FMSF component meets exactly one infinite p_u cluster. Consequently a connected FMSF can meet only one such cluster, so connectivity implies uniqueness at p_u in that setting.

Check

The condition p_u>p_c is retained; assuming it for every nonamenable graph would silently solve another admitted conjecture. The proposition does not say that one p_u cluster meets only one forest component.

Remaining obstruction

The reverse direction needs to exclude several free-forest components meeting the same infinite p_u cluster. The stronger component-bijection formulation remains unproved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0056 One or infinitely many minimal-forest components

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 11.19, printed p.403

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Must both free and wired minimal spanning forests on a quasi-transitive graph have either one or infinitely many components almost surely?

Scope

Free and wired are grouped. Distinct-law unimodular cases are reported resolved by Timar 2018.

P0056-A001 | Ergodic factor laws give constancy but not the value of the count

Argument or result

Minimal spanning forests are equivariant functions of iid continuous edge labels, so an ergodic label action makes their component count constant. That alone permits a finite constant larger than one: the invariant deterministic two-component forest on Z with generators +/-1,+/-2 from P0048 is a counterexample to the purely ergodic inference.

Check

The example is explicitly not a minimal-spanning-forest law. It separates the zero-one statement for invariant events from a classification of possible constant values.

Remaining obstruction

The usual insertion-tolerance route must account for cycle-breaking deletions. No argument excluding a finite count between two and infinity was obtained in the full quasi-transitive scope.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0057 Uniqueness-critical percolation versus invariant spanning trees

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 11.31, printed p.408

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a Cayley graph with $p_u<1$, is there an implication in either direction between multiple infinite clusters at p_u and the nonexistence of an invariant random spanning tree?

Scope

Each condition implies the free minimal spanning forest is disconnected, but a common consequence does not establish either implication.

P0057-A001 | Conditional implication through the component-bijection conjecture

Argument or result

If there is no invariant random spanning tree, the invariant FMSF cannot be connected, because a connected spanning forest would itself be such a tree. If the stronger conjecture in P0055 gave a bijection between free-forest components and infinite p_u clusters, that disconnection would imply multiple clusters at p_u.

Check

The first implication uses only the definition of an invariant spanning tree. The second is explicitly conditional on a separate unproved component correspondence.

Remaining obstruction

Two properties sharing FMSF disconnection as a consequence do not thereby imply one another. Neither unconditional implication in the source question is established here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0058 Indistinguishability of minimal spanning-forest components

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 11.33, printed p.409

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Are the components of both free and wired minimal spanning forests on a unimodular transitive graph indistinguishable by every invariant Borel component property?

Scope

Timar 2018 settles the free case when free and wired differ. The remaining equal-law or wired scope is the target.

P0058-A001 | Invariant ergodicity does not imply component indistinguishability

Argument or result

Construct a periodic forest on Z^2 using blocks of four horizontal rows. The first row is a bi-infinite horizontal path. The next three rows form a comb: the middle row is a horizontal spine, with each vertex in the two outer rows attached vertically to that spine. Repeat and choose a uniformly random vertical phase. The law is translation invariant and ergodic on its finite phase orbit, yet its components are distinguishable as lines and combs by the existence of degree-four vertices.

Check

Every vertex belongs to exactly one component and there are no cycles. If full square-lattice automorphism invariance is desired, also choose a uniform horizontal/vertical orientation; the full automorphism group is transitive on the resulting finite orbit.

Remaining obstruction

This is not a minimal-spanning-forest construction. It disproves only the shortcut from invariance and ergodicity to indistinguishability.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0059 Limiting bushes along a wired minimal-forest ray

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 11.34, printed p.409

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a Cayley graph not roughly isometric to Z, does the sequence of translated bushes along a ray in the identity's wired minimal spanning-forest component converge in distribution?

Scope

Uniqueness of that ray is itself tied to critical percolation. Keep the conditional structural assumptions explicit rather than selecting an unmentioned canonical ray.

P0059-A001 | Unique-ray and limiting-law issues must be separated

Argument or result

A bush decomposition first needs an almost surely unique infinite ray in the relevant component. Assuming that ray before proving the applicable one-endedness result would import another open assertion. Even when uniqueness is available, rooting successively along the ray is a configuration-dependent operation and does not automatically preserve the original rooted law.

Check

A convergent subsequence of bush laws would not establish a full limit. The period-two sequence example in P0052 shows the logical distinction.

Remaining obstruction

No regeneration theorem or coupling proving convergence along the wired minimal-forest ray was obtained. The preliminary one-endedness assumptions must be checked for each claimed case.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0060 Dimensions with a connected Euclidean minimal spanning forest

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 11.35, printed p.409

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For which dimensions d is the minimal spanning forest on Z^d connected almost surely?

Scope

Competing predictions put a transition near dimension six or eight. These dimension variants form one classification problem.

P0060-A001 | Exact one-dimensional connectivity

Argument or result

On Z every edge is a bridge, so the free minimal spanning forest retains all edges. For an edge with label u<1 to be excluded by a wired connection at infinity, both infinite half-lines would have to supply paths of edges with labels below u. Each such event has probability zero for iid uniform labels, since the probability for its first N edges is u^N. Thus the wired forest also retains every edge almost surely.

Check

There are only countably many edges, allowing the probability-one statement to hold simultaneously for all of them. The argument uses the unique paths on Z.

Remaining obstruction

The higher-dimensional connectivity classification depends on intersections of invasion structures and is not determined by the one-dimensional calculation.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0061 Full scaling limit of the square-lattice minimal spanning tree

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 11.36, printed p.410

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does the minimal spanning tree on epsilon Z^2 have a full distributional scaling limit as epsilon tends to zero, in the embedded-tree topology used for its known subsequential limits?

Scope

The source leaves the exact topology to the cited Aizenman-Burchard-Newman-Wilson framework. A triangular-lattice theorem does not settle square-lattice universality.

P0061-A001 | Exact universality under a monotone change of edge labels

Argument or result

On every finite graph with distinct edge weights, a strictly increasing transform preserves their order and hence preserves Kruskal selection and the minimum spanning tree. Independently, the cycle characterization retains exactly the same nonmaximal edges under that transform. For iid continuous distributions this identifies the law through ranks, or through the probability integral transform almost surely.

Check

For a continuous CDF that is not strictly increasing everywhere, flat intervals have zero probability of a sampled weight, so the rank argument still applies almost surely.

Remaining obstruction

This is label-distribution universality on a fixed graph. It does not compare triangular and square lattice geometries or prove tightness and uniqueness of a continuum scaling limit.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0062 Connectivity of minimal forests on graphs roughly isometric to trees

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 11.37, printed p.410

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If a connected locally finite graph is roughly isometric to a tree, must its free minimal spanning forest be connected almost surely?

Scope

This concerns minimal forests, not uniform forests; Pete-Timar's uniform-forest counterexample cannot be substituted.

P0062-A001 | Connected FMSF on trees of finite cycle blocks

Argument or result

Suppose an infinite cactus graph is built from bridges and edge-disjoint finite cycles. Its free minimal spanning forest retains every bridge and removes exactly the unique largest-weight edge of each cycle. Each resulting cycle block remains connected. A finite block path between any two vertices remains connected after these deletions, so the forest is globally connected and acyclic.

Check

A simple cycle in a cactus is one of its cycle blocks; there are no additional cycle constraints coupling two blocks. Continuous iid labels make every finite maximum unique almost surely.

Remaining obstruction

Uniformly bounded block diameters give a class roughly isometric to trees. General roughly-tree graphs can have overlapping cycles and need not admit this block decomposition.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0063 Singularity of binary and ternary harmonic codings of the Cantor set

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 16.7, printed p.542

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the middle-thirds Cantor set E, let nu_b be harmonic measure on the source's base-b coding tree T[b](E), pushed to E. Is nu_2 mutually singular with the Cantor-Lebesgue measure nu_3?

Scope

Use the coding-tree construction and boundary map in Section 16.2; distinct harmonic measures or dimensions alone do not prove singularity.

P0063-A001 | Likelihood-ratio strategy and its exact missing input

Argument or result

For product Bernoulli measures with different parameters, the strong law gives disjoint full-measure sets of limiting digit frequencies and hence mutual singularity. A possible analogous strategy for the two Cantor measures is a likelihood-ratio martingale along a common binary coding, with cumulative conditional relative entropy forcing its limit to vanish.

Check

Merely proving that two measures are unequal is insufficient: two different positive bounded densities relative to the same measure are still equivalent. The binary digits of Cantor-Lebesgue measure are not assumed iid.

Remaining obstruction

No divergent conditional-entropy estimate or martingale tail control was established for the actual harmonic measure versus base-three Cantor measure.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0064 Dimension drop for every nonuniform Galton-Watson flow

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Conjecture 17.23, printed p.580

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a nondegenerate supercritical Galton-Watson law of mean m under the chapter's moment assumptions, must every consistent equivariant flow rule different almost surely from UNIF have boundary-measure Hausdorff dimension below log m almost surely?

Scope

The source proves a stationary-measure sufficient condition. Difference on a GW-null set is not an admissible counterexample.

P0064-A001 | Exact dimension drop for the nonergodic same-color flow candidate

Argument or result

For the AND/OR color construction of P0033, let K be the number of children of the parent color and choose uniformly among them. Conditional on root color zero, P(K=1,2,3)=(3/5,3/10,1/10); conditional on color one it is (3/10,3/5,1/10). The full conditional-child selection lemma implies these K variables are iid along the selected ray within each color class. Hence the local dimension is h_0=(3/10)log 2+(1/10)log 3 or h_1=(3/5)log 2+(1/10)log 3 by the strong law. Both are strictly less than log(12/5), the Galton-Watson dimension.

Check

For color zero the unconditioned masses are .30 for two mixed children (K=1), .15 for two zero children (K=2), and .05 for three zero children (K=3); divide by color probability .5. For color one the corresponding masses are .15,.30,.05. Cylinder mass is exactly the product of reciprocal K values; Fubini transfers the ray statement to almost every tree and almost every flow ray.

Remaining obstruction

This author-checked special construction supports dimension drop and is not a counterexample to the full question. It inherits the P0033 coloring proof and remains without independent review.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0065 Monotonicity of harmonic dimension under inward bias

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 17.29, printed p.582

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a nondegenerate supercritical Galton-Watson tree without leaves and 0<=lambda<m, does harmonic-measure dimension of RW_lambda increase with lambda, and is the increase strict?

Scope

On a deterministic regular tree the dimension is constant, so retain the chapter's nondegeneracy assumptions for strictness. Both monotonic and strict clauses are grouped.

P0065-A001 | Regular-tree equality and a local-entropy countercheck

Argument or result

On a deterministic b-ary tree, symmetry makes harmonic measure uniform on children for every transient bias lambda<b, so its dimension is log b, independent of lambda. Thus strict monotonicity must exclude deterministic offspring. A local root-entropy proof also fails: attach regular trees of branching 2 and 4 as two root subtrees. Their escape probabilities are 1-lambda/2 and 1-lambda/4 for lambda<2; last-exit child probabilities are proportional to these values and become more unequal as lambda increases toward 2, so their entropy decreases.

Check

The two-child example is not an iid Galton-Watson tree. Its role is to disprove a pointwise local entropy monotonicity claim, not the global dimension conjecture.

Remaining obstruction

Global dimension averages local entropy under a bias-dependent law of the environment seen along the ray. Controlling that changing law remains necessary.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0066 Monotonicity of biased-walk speed on Galton-Watson trees

Source: Probability on Trees and Networks, paperback electronic version 19 August 2026 | Question 17.33, printed p.592

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For Galton-Watson trees with p_0=0, is the speed of RW_lambda nonincreasing as the bias parameter lambda increases?

Scope

The transient range is 0<lambda<m; beyond criticality the speed is zero. Positive-regeneration and large-bias partial results are not the complete monotonicity claim.

P0066-A001 | The changing stationary environment blocks a pointwise drift proof

Argument or result

For fixed offspring D, the radial drift (D-lambda)/(D+lambda) decreases with lambda, and on a regular tree it gives the exact speed. In a nondegenerate Galton-Watson tree the stationary distribution of the environment seen by the walker also changes with lambda, so differentiating the speed has an additional measure-change term.

Check

An elementary comparison shows the logical gap: drift at lambda=1,D=2 is 1/3, while at lambda=2,D=100 it is 98/102. Pointwise monotonicity in lambda cannot compare averages taken under different offspring laws.

Remaining obstruction

The example is not a Galton-Watson speed counterexample. No sign bound for the measure-change term, or alternative monotone coupling of speeds, was proved.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0067 Global Kuramoto synchronization of random cubic graphs

Source: Randomstrasse101: Open Problems of 2024 | Conjecture3, PDF p.3

https://arxiv.org/pdf/2504.20539v1

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is a uniformly random simple 3-regular graph globally synchronizing with probability tending to one? Global synchronization means that every local minimum of sum_(ij in E)(1-cos(theta_i-theta_j)) is a consensus configuration.

Scope

Use even n. Connectivity is necessary but not sufficient; degree-three deterministic examples do not refute an asymptotic random-graph assertion.

P0067-A001 | Global Kuramoto synchronization of random cubic graphs special-case or structural calculation

Argument or result

On the cubic prism C_m times K_2, m>=5, put theta_(j,a)=2 pi j/m for both layers. The sine forces along the two cycle neighbors cancel and rung forces vanish. The Hessian is a weighted Laplacian with cycle weights cos(2 pi/m)>0 and rung weights1, so it is positive definite modulo the global rotation. This is a nonconsensus strict local minimum modulo rotation.

Check

Both the vanishing gradient and Hessian signs were checked; the graph is connected and cubic. This is an exact deterministic obstruction to a connectivity-only proof.

Remaining obstruction

Must show that uniformly random cubic graphs avoid all such stable phase arrangements with probability tending to one; the prism family itself has vanishing probability.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0068 Subspectral detection from a growing number of synchronization frequencies

Source: Randomstrasse101: Open Problems of 2024 | OpenProblem7, PDF p.7

https://arxiv.org/pdf/2504.20539v1

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For uniform U(1) signal x in C^n with |x_i|=1 and independent Wigner noises, observe Y_k=(lambda/n)x^(k)(x^(k))*+W_k/sqrt(n), k=1,...,L_n. Find a scaling of distinct frequencies L_n for which a polynomial-time algorithm reliably detects signal at some fixed lambda<1.

Scope

The source also names finite groups. For a fixed finite group the number of distinct irreducible frequencies is bounded; a growing group or repeated-observation convention must be specified. The unambiguous U(1) case is the admitted target.

P0068-A001 | Subspectral detection from a growing number of synchronization frequencies special-case or structural calculation

Argument or result

If L independent observations repeated the same rank-one signal, averaging would divide noise standard deviation by sqrt(L) and reduce the effective spectral threshold. In the actual model the vectors x^(k) differ with k, so averaging Y_k yields a sum of different rank-one projectors. It is not a repeated observation of xx*.

Check

The noise-average variance calculation is exact. The signal identity fails already for phases whose squares are not a common multiple of their first powers.

Remaining obstruction

Need an algorithm exploiting algebraic relations between distinct powers. Repetition-based variance reduction solves a different experiment.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0069 Subexponential detection at fixed synchronization frequencies

Source: Randomstrasse101: Open Problems of 2024 | OpenProblem8, PDF p.7

https://arxiv.org/pdf/2504.20539v1

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

In the same multifrequency U(1) or specified finite-group Gaussian synchronization model, fix L independently of n. For signal strength between its statistical threshold and1, is reliable detection possible in time exp(n^delta) for some delta<1?

Scope

The statistical threshold and group must refer to the fixed model. This is a time-accuracy question distinct from growing the observation model in OpenProblem7.

P0069-A001 | Subexponential detection at fixed synchronization frequencies special-case or structural calculation

Argument or result

Try enumerating an oracle seed of m=o(n) signal coordinates and use their edges to estimate a new coordinate in one channel. After multiplying by known seed phases, the sum has signal amplitude lambda m/n and noise standard deviation sqrt(m/n), so signal-to-noise ratio is lambda sqrt(m/n), which tends to zero.

Check

The calculation uses the displayed matrix normalization; using unnormalized Wigner noise gives the same ratio. An oracle seed is stronger than an enumerated guessed seed.

Remaining obstruction

This only defeats a one-step seed-and-average method. Iterative amplification or a different subexponential algorithm is not ruled out.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0070 Kikuchi spectral thresholds as the polynomial degree increases

Source: Randomstrasse101: Open Problems of 2024 | Conjecture9, PDF p.8

https://arxiv.org/pdf/2504.20539v1

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For even tensor order r and fixed l>=r/2, index M by l-subsets of [n] and set M_lJ=lambda+Z_(l symmetric_difference J) when that difference has size r, and zero otherwise, with independent standard Gaussian Z_S. Does the normalized spectral-outlier threshold n^(r/4) lambda_critical(r,l) tend to zero as l tends to infinity after n tends to infinity?

Scope

The source suppresses the n-dependence of its threshold notation. Its literal inclusion of r=2 needs a separate check against matrix-PCA limits; no silent restriction to r>=4 is made and no r=2 counterexample is yet claimed.

P0070-A001 | Kikuchi spectral thresholds as the polynomial degree increases special-case or structural calculation

Argument or result

Each row has k=binomial(l,r/2)binomial(n-l,r/2) nonzero entries. For the normalized all-ones vector v, v*M(lambda)*v=lambda k+(k/binomial(n,r))sum_(|S|=r)Z_S. Thus its noise variance is k^2/binomial(n,r). Write M(0)=sum_S Z_S A_S; A_S^2 is the diagonal indicator of |l intersection S|=r/2, hence sum_S A_S^2=k I.

Check

Double-counting pairs (I,J) with symmetric difference S gives the Rayleigh coefficient. Squaring A_S toggles the same subset S twice and returns I exactly when that toggle preserves cardinality.

Remaining obstruction

A Gaussian matrix-series bound introduces sqrt(log binomial(n,l)) into the norm estimate. Removing that loss and controlling the actual outlier transition remain necessary; the r=2 scope caveat needs resolution.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0071 Polynomial Glauber mixing throughout the SK high-temperature regime

Source: Randomstrasse101: Open Problems of 2024 | OpenProblem15, PDF p.12

https://arxiv.org/pdf/2504.20539v1

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the Sherrington-Kirkpatrick Ising model with arbitrary external field, does single-site heat-bath Glauber dynamics mix in polynomial total-variation time for every fixed beta<1, with high probability over the Gaussian disorder?

Scope

Keep SK Hamiltonian normalization consistent. A July2026 primary paper proves results for sufficiently large external fields, and an August2026 preprint claims optimal mixing for beta<1/2 uniformly over fields; neither states the full beta<1 arbitrary-field conclusion.

Dated literature checks

[{"finding": "Large-field result, not arbitrary-field full regime", "url": "https://arxiv.org/abs/2607.06813"}, {"finding": "Preprint claims beta<1/2 uniform fields; full beta<1 remainder not resolved by its statement", "url": "https://arxiv.org/abs/2608.22159"}]

P0071-A001 | Polynomial Glauber mixing throughout the SK high-temperature regime special-case or structural calculation

Argument or result

At beta=0, spins are independent under the target product law. Couple two random-scan heat-bath chains using the same update coordinate and uniform draw. Once a coordinate has been updated it agrees forever. The probability some coordinate remains unupdated by t is at most n(1-1/n)^t<=n exp(-t/n), giving mixing time at most n log(n/epsilon) for every external field.

Check

The update probability of each spin may depend on its field, but at beta=0 it does not depend on neighboring spins, so agreement is preserved. This is a complete endpoint proof, with no novelty claim.

Remaining obstruction

For beta>0 a later update can destroy agreement through neighboring disagreements. The beta=0 coupon-collector argument does not control the whole replica-symmetric interval.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0072 Geodesic selection by reinforced ants

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | Section2, main conjecture, PDF pp.1-2

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

On a finite graph with nest N and food F, ants take edge-weighted random walks until hitting F. Reinforce either a uniformly chosen shortest N-F path in the visited trace, or the loop erasure of the reversed walk. Are all edges whose weights grow linearly contained in shortest N-F paths of the original graph?

Scope

The two reinforcement algorithms form one conjecture family here. The source proves series-parallel and losange cases, not all graphs.

P0072-A001 | Geodesic selection by reinforced ants structural argument

Argument or result

On a finite tree with one food vertex, every N-F walk has the same simple path after loop erasure, and the shortest path in its trace is that path too. Each edge of that path therefore receives exactly one increment per ant; every other edge receives zero. The normalized weights converge to its indicator without a probabilistic limit theorem.

Check

Uniqueness of the simple path verifies both algorithms separately. This endpoint is included in known special cases, not a new solution.

Remaining obstruction

Cycles permit different simple paths and histories affect their weights. The deterministic tree reduction gives no control of their competition.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0073 Deterministic limits for trace-reinforced ants

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblems1-2, PDF p.2; underlying Kious-Mailler-Schapira Conjecture1.1

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

When each ant reinforces every edge it visited at least once, do all edge weights divided by the number of ants converge almost surely to deterministic constants on general finite graphs, and does an appropriate edgewise statement hold on Z^2?

Scope

The finite and Z^2 versions are grouped. The underlying paper excludes multiple edges incident to food; the brief workshop statement omits this load-bearing restriction. Its wording main conjecture is interpreted using the immediately preceding trace-process paragraph.

P0073-A001 | Deterministic limits for trace-reinforced ants structural argument

Argument or result

If two parallel edges connect N directly to F, an ant crosses exactly one and reinforces it. Starting at weights a,b gives the classical Polya urn: the first normalized weight has Beta(a,b) limit with variance ab/((a+b)^2(a+b+1))>0. Thus the unrestricted multigraph formulation is false, and the underlying no-parallel-food assumption is necessary.

Check

The transition probability and one-increment update coincide exactly with the urn. This is a scope counterexample, not a refutation under the original restriction.

Remaining obstruction

For simple finite graphs the nonlinear trace-hitting probabilities still require a uniqueness/attraction argument. On Z^2, changing conductances additionally complicate the guarantee that every ant reaches food.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0074 Maximal degree and its location in the infinite-path fitness regime

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem6, PDF p.5

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For superlinear preferential attachment with iid positive fitness and attachment f(k,w)=w k^p or k^p+w, determine the growth of the maximum degree and the label attaining it in the regime where the limiting tree is locally finite and has one infinite ray.

Scope

Use a positive initial-degree convention; the displayed one-vertex degree-zero multiplicative formula otherwise has a zero denominator. The source model must be initialized consistently.

P0074-A001 | Maximal degree and its location in the infinite-path fitness regime structural argument

Argument or result

Let Delta_n be the maximum degree and I_n a maximizing label. If Delta_n tends to infinity and each fixed vertex has finite limiting degree, then I_n tends to infinity: for every fixed K, the maximum of the K finite limiting degrees is finite, and eventually Delta_n exceeds it.

Check

This is a pathwise implication and does not assume independence. It deliberately leaves divergence of Delta_n as an extra hypothesis.

Remaining obstruction

Neither the rate of Delta_n nor even its divergence follows from local finiteness alone. A deterministic infinite path has maximum degree2.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0075 Empirical degree scaling for superlinear attachment with fitness

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem7, PDF p.5

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the scaling and limiting empirical degree distribution of the superlinear preferential-attachment tree with fitness described in Section3, including the infinite-path regime.

Scope

A degree-count measure and a fitness-degree joint measure are different objects; the admitted target is the empirical degree distribution explicitly named in the source.

P0075-A001 | Empirical degree scaling for superlinear attachment with fitness structural argument

Argument or result

For every n-vertex tree, the empirical degree law pi_n satisfies sum_k k pi_n(k)=2-2/n, hence pi_n([K,infinity))<=2/K. Thus the unscaled empirical laws are tight on the positive integers. Any weak subsequential limit is a probability law with mean at most2; its mean can be smaller because degrees are not uniformly integrable.

Check

The handshaking identity proves the first statement. A star has empirical law tending to delta_1 while its empirical mean tends to2, confirming the moment caveat.

Remaining obstruction

Tightness supplies no unique limit or model-specific tail scaling. Escape of the first moment must be tracked separately.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0076 Multiple-attachment phase transition with superlinear fitness

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem8, PDF p.5

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

In the Section3 fitness model, replace one attachment per new vertex by m>=2 choices and determine how the transition between degree condensation and an infinite-path structure changes.

Scope

Sampling with or without replacement and sequential updating must be fixed; the source proposes a multigraph extension. One family is counted for all m, rather than parameter copies.

P0076-A001 | Multiple-attachment phase transition with superlinear fitness structural argument

Argument or result

After n arrivals with exactly m outgoing attachments each, total degree is 2m(n-1) up to the finite seed. Consequently no collection of vertices can each have degree fraction greater than a if it contains more than 2m/a vertices. This bounds the number of macroscopic hubs but permits several of them, unlike a single-star conclusion.

Check

The bound follows by summing nonnegative degrees; it remains valid with parallel edges and sequential updates.

Remaining obstruction

A tree embedding into independent genealogical birth processes is lost when a vertex has several parents. The degree budget does not identify the phase boundary.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0077 Fluctuations of fitness-reinforced sequence sums

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem9, PDF p.7

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Let a new positive value be sampled from mu with probability beta, and otherwise copy a previous value with probability proportional to value times multiplicity. With ess sup mu=1 and mu({1})=0, find the fluctuation limit of sum_(i<=n)(X_i-lambda_star), where lambda_star is the Section4 law-of-large-numbers constant.

Scope

The source equation beta integral x/(lambda_star-(1-beta)x) dmu=1 defines lambda_star when possible; otherwise lambda_star=1-beta. A universal square-root normalization is not assumed.

P0077-A001 | Fluctuations of fitness-reinforced sequence sums structural argument

Argument or result

Writing S_n=sum X_i and Q_n=sum X_i^2 gives E[X_(n+1)|F_n]=beta m_1+(1-beta)Q_n/S_n. Subtracting this predictable mean yields bounded martingale differences, but the centered sum also contains sum[beta m_1+(1-beta)Q_i/S_i-lambda_star].

Check

The conditional mean is obtained by summing x times its copying probability. Boundedness controls the martingale part without controlling the accumulated drift.

Remaining obstruction

A martingale CLT alone cannot settle the question: fluctuations of Q_n/S_n and the slowly relaxing drift may contribute at the same or larger scale.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0078 Boundary profile of the reinforced-sequence condensate

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem10 and displayed conjecture, PDF pp.7-8

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

When beta integral (1-x)^(-1) dmu<1 and mu(1-x,1) is regularly varying at zero with index alpha>1, determine a_n tending to infinity and c>0 for which n^(-1) sum f(a_n(1-X_i)) converges in probability to c integral_0^infinity f(x)e^(-x)x^alpha dx for compactly supported continuous f.

Scope

Convergence against compactly supported test functions does not by itself identify total condensate mass; mass at zero or infinity must be separately controlled.

P0078-A001 | Boundary profile of the reinforced-sequence condensate structural argument

Argument or result

Let rho=1-beta integral (1-x)^(-1)dmu be the mass missing from the bulk limit. If the claimed rescaled measure is tight on (0,infinity) and carries all that missing mass, integrating its density forces c=rho/Gamma(alpha+1). Without those two additional controls this identification is invalid.

Check

The integral of e^(-x)x^alpha equals Gamma(alpha+1); the deduction is explicitly conditional on no escaped mass.

Remaining obstruction

The source provides only vague convergence away from the boundary. Finding a_n and proving boundary tightness are the load-bearing missing steps.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0079 Growth of sums with unbounded reinforcement fitness

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem11, PDF p.8

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the same innovation/copying scheme with unbounded positive innovation law mu, determine the superlinear growth rate of S_n=sum_(i<=n) X_i as a function of the tail of mu.

Scope

No single rate can be asserted for every unbounded mu without describing its tail; the question is a distribution-dependent classification.

P0079-A001 | Growth of sums with unbounded reinforcement fitness structural argument

Argument or result

Let K_n count innovation draws and let Y_j be their iid values. Every innovation appears at least once, so S_n>=sum_(j<=K_n)Y_j and S_n>=max_(j<=K_n)Y_j, while K_n/n tends to beta almost surely. If E_mu Y=infinity, the truncated strong law and then monotone convergence give S_n/n tending to infinity from this lower bound alone.

Check

For a truncation L, divide the innovation sum by n and use the iid strong law; let L increase afterward.

Remaining obstruction

Finite-mean unbounded mu requires amplification by reinforcement; innovation lower bounds alone remain linear and cannot give the requested rate.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0080 Weak reinforcement on infinite WARM networks

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | Conjecture1, PDF p.10

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a bounded-degree infinite graph, bounded positive vertex firing rates, and WARM exponent 0<alpha<1, establish convergence of every normalized edge count to the intended unique positive equilibrium.

Scope

The source literally says unique equilibrium while allowing nonnegative coordinates. Boundary equilibria make that literal uniqueness false; the meaningful positive-attractor formulation is recorded explicitly, not silently substituted.

P0080-A001 | Weak reinforcement on infinite WARM networks structural argument

Argument or result

On Z with all firing rates1, set edge values alternately2 and0. At each vertex the only positive incident edge is selected with probability1; its two endpoints supply total rate2, while zero edges get rate0. This is an equilibrium for every alpha>0. Shifting gives a second one, and the uniform edge value1 is a third. Thus uniqueness among all nonnegative equilibria is false.

Check

All denominators equal2^alpha, so the example has no undefined 0/0 term. It does not contradict convergence from strictly positive initial counts.

Remaining obstruction

The positive-equilibrium uniqueness and stochastic attraction problem on general infinite graphs remains; no finite total Lyapunov function is available.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0081 Critical WARM on the integer line

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | Conjecture2, PDF p.10

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For WARM on Z with alpha=1 and all vertex firing rates1, prove that each edge count divided by time converges almost surely to1.

Scope

Keep the stochastic count process distinct from its deterministic equilibrium equation.

P0081-A001 | Critical WARM on the integer line structural argument

Argument or result

For any a,b>0 with a+b=2, the alternating edge configuration a,b,a,b,... solves the equilibrium equation: an a-edge receives a/(a+b) from each endpoint, totaling a, and similarly for b. Thus the ODE has a continuum of positive equilibria on the same graph.

Check

Substitution in the local rate formula verifies every edge. This extends the source even-cycle observation but concerns deterministic equilibria only.

Remaining obstruction

Convergence to1 cannot follow merely from uniqueness of ODE equilibria, which is false. One must exploit stochastic dynamics or spatial structure to select the uniform equilibrium.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0082 Sharp reinforcement threshold for WARM forests

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem12, PDF pp.10-11

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Improve the known sufficient threshold alpha>2 ensuring that every connected component of a stable finite-graph WARM equilibrium is a tree, allowing arbitrary positive firing rates.

Scope

The source identifies the uniform triangle threshold4/3 as a lower obstruction. A rederivation of it is not a new threshold theorem.

P0082-A001 | Sharp reinforcement threshold for WARM forests structural argument

Argument or result

At a positive equilibrium x, put s_v=sum_(e~v)x_e^alpha. The Hessian of the correctly rate-weighted Lyapunov function satisfies H[delta,delta]=(alpha-1)sum_e delta_e^2/x_e-alpha sum_v p_v/s_v^2 (sum_(e~v)x_e^(alpha-1)delta_e)^2. Along an even cycle choose alternating delta_e=+/-x_e^(1-alpha), zero elsewhere. Every vertex sum vanishes, giving a positive direction for alpha>1 and hence linear instability.

Check

The equilibrium identity x_e=sum_(v~e)p_v x_e^alpha/s_v reduces the diagonal term. Multiplication by the positive diagonal x preserves the Hessian inertia of the linearized gradient flow.

Remaining obstruction

This eliminates even cycles, not odd cycles or all odd unicyclic supports. It does not improve the universal forest threshold by itself.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0083 Sharp reinforcement threshold for WARM whiskers

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem13, PDF pp.10-11

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For finite WARM graphs with uniform vertex firing rates, improve the sufficient threshold alpha>25 for every surviving stable component to have diameter at most3.

Scope

This is a geometry question beyond the forest conclusion; all alpha values are grouped as one threshold problem.

P0083-A001 | Sharp reinforcement threshold for WARM whiskers structural argument

Argument or result

Consider a support path of four edges with reflection-symmetric values a,b,b,a and unit vertex rates. The endpoint equation is a=1+a^alpha/(a^alpha+b^alpha), so 1<a<2. The neighboring inner-edge equation is b=b^alpha/(a^alpha+b^alpha)+1/2. Summing gives a+b=5/2. Thus any such equilibrium reduces to a scalar equation with b=5/2-a.

Check

The total rate of the five vertices is5, equal to2a+2b. The central vertex contributes1/2 to each equal inner edge.

Remaining obstruction

Existence and stability of this scalar branch must still be analyzed, and excluding this symmetric path alone would not exclude all larger asymmetric trees.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0084 Finite surviving components in infinite superlinear WARM

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | Conjecture3, PDF p.11

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

With bounded-degree graph, firing rates bounded above and bounded away from zero, and alpha>1, prove edgewise convergence of normalized WARM counts and finiteness of every component of surviving edges.

Scope

The lower bound on firing rates is essential in the source: examples with decaying rates on trees have infinitely many infinite components.

P0084-A001 | Finite surviving components in infinite superlinear WARM structural argument

Argument or result

If every edge had survival probability at most q, this would not imply domination by iid Bernoulli percolation. For example, activate all edges simultaneously with probability q and none otherwise. Every marginal is q, yet there is an infinite component with probability q for every q>0 on an infinite connected graph.

Check

The example computes the marginals and connectivity exactly. It is a counterexample to a marginal-only proof method, not to WARM.

Remaining obstruction

A valid comparison needs conditional probability bounds or explicit dependence control, together with convergence of the edge weights; small one-edge probabilities are insufficient.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0085 WARM percolation with vanishing rates on lattices

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem14, PDF p.11

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Do bounded positive firing rates on Z^d, d>=2, which are not bounded away from zero, and some alpha>1 yield a positive probability of an infinite component of surviving WARM edges? If so, determine uniqueness.

Scope

The existence and uniqueness parts are grouped. Tree examples do not directly embed as bounded-length lattice edges.

P0085-A001 | WARM percolation with vanishing rates on lattices structural argument

Argument or result

There is no bounded-length embedding of a b-ary tree, b>=2, into Z^d with distinct vertices: its depth-r ball contains order b^r vertices, whereas images lie in a lattice ball of radius Lr with only O(r^d) vertices. Therefore transferring the known regular-tree construction requires unbounded subdivisions or a different geometry.

Check

This is the exponential-versus-polynomial volume comparison, independent of reinforcement details.

Remaining obstruction

Unbounded subdivisions change the local competition and rates along paths. The volume obstruction explains why a direct bounded-distortion transfer fails but does not exclude lattice WARM percolation.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0086 Recurrence of sublinear vertex-reinforced walk on Z

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem15, PDF p.12

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For nearest-neighbor vertex-reinforced random walk on Z, with transition weights w(1+local_time) and positive w satisfying w(x)=o(x^alpha) for some alpha<1, is the walk recurrent?

Scope

The source states positivity and the growth upper bound, without an explicit monotonicity hypothesis. Any use of monotonicity must be flagged.

P0086-A001 | Recurrence of sublinear vertex-reinforced walk on Z structural argument

Argument or result

For constant w the walk is simple symmetric random walk. Starting at1 and stopping at0 or m, the harmonic function h(k)=k/m gives probability1/m of hitting m before0. Letting m tend to infinity shows that the probability of avoiding0 forever is0; by the strong Markov property returns recur indefinitely.

Check

This is the standard gambler-ruin proof of the constant-weight case, not a new result. Constant weights satisfy the stated sublinear condition for any alpha in(0,1).

Remaining obstruction

For nonconstant reinforcement, position alone is not Markov and the gambler-ruin harmonic function no longer solves the conditional transition equation.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0087 Forced network urn convergence and diffusion scaling

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem16, PDF pp.14-16

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For Z_(n+1)=(1-r-f)Z_n+r X_(n+1)+f p with E[X_(n+1)|F_n]=Q Z_n, Q=A^T irreducible row-stochastic and X binary coordinatewise, determine nonconvergence and the continuous-time limit when r tends to0 and f is asymptotic to c r^2.

Scope

Assume r,f>0 and r+f<=1. The source permits 0<=p<=1, including extreme vectors; those have a different convergence answer. The displayed conditional mean alone does not specify the cross-coordinate noise covariance.

P0087-A001 | Forced network urn convergence and diffusion scaling structural argument

Argument or result

A complete fixed-step classification follows. On any event Z_n->z, the update forces X_n to converge to a binary vector x and hence eventually equal it. The bounded martingale-difference strong law gives x=Qz. The update limit gives z=(r x+f p)/(r+f). A binary coordinate of Qz can equal0 or1 only when all its positive-weight predecessors have the same x-value and matching p-value. Irreducibility then forces x and p to be the all-zero or all-one vector. Conversely if p=0, the positive stationary vector pi gives E[pi.Z_n]=(1-f)^n E[pi.Z_0], and Markov plus Borel-Cantelli yields Z_n->0; p=1 follows by complementing.

Check

The proof needs no conditional independence. The strong law follows from summable variances of martingale increments divided by n, then Kronecker. The maximum principle can also be checked by propagating the set of zero coordinates backwards through irreducibility.

Remaining obstruction

This resolves the fixed-step convergence subquestion under the stated assumptions, but not the whole diffusion question. Fast synchronization, an initial layer, and the covariance specification must be treated before a full solution label.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0088 Matching local limits of Bernoulli and Poisson-outdegree attachment

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem23, PDF p.21

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Relate the local weak limit of the Dereich-Morters Bernoulli preferential-attachment model, whose older vertices are chosen independently with probability f(in_degree)/(n-1), to an appropriately matched model with iid Poisson outgoing degree and attachment based on in-degree.

Scope

The matching of the affine rule and Poisson parameter is part of the question. Early probabilities exceeding1 need a consistent initialization or cap.

P0088-A001 | Matching local limits of Bernoulli and Poisson-outdegree attachment structural argument

Argument or result

Conditional on current degrees, the Bernoulli outgoing count has mean Lambda_n=sum_i p_ni. Its total variation distance to Poisson(Lambda_n) is at most sum_i p_ni^2 by the elementary independent Bernoulli-to-Poisson coupling. If max_i p_ni->0 and Lambda_n stays bounded, this error tends to0.

Check

Use sum p_i^2<=max p_i sum p_i. This verifies the count approximation only; the vertex targets retain degree-dependent marks.

Remaining obstruction

A matching outgoing count does not prove local graph convergence. Target choices, ancestor types, and dependencies across successive births must be coupled over an explored neighborhood.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0089 Local limits for nonaffine attachment with multiple edges

Source: Aletti et al.: Open problems in stochastic reinforcement models; March2023 workshop, published2025 | OpenProblem24, PDF p.21

https://www.matrix-inst.org.au/wp_Matrix2016/wp-content/uploads/2023/Aletti.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine local weak limits of preferential-attachment graphs with fixed outgoing degree m>=2 and nonaffine attachment function f, extending the known tree and affine cases.

Scope

State a nonexplosive/noncondensed class of f before expecting a locally finite limit; the source asks for a theory beyond the affine case, not one arbitrary universal formula.

P0089-A001 | Local limits for nonaffine attachment with multiple edges structural argument

Argument or result

With m=1 and equal attachment weights, the graph is a uniform recursive tree and its empirical degree law is tight by the tree degree identity. With m>=2 the degree budget still gives uniform-root degree tightness, since its mean is at most2m plus a vanishing seed term.

Check

Markov gives P(deg(U_n)>K)<=2m/K+o(1) without assumptions on f.

Remaining obstruction

Tightness of the root degree alone does not ensure tightness of radius-two neighborhoods: a uniformly selected leaf can attach to a hub of diverging degree. Condensation and size bias must be controlled.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0090 Multistate pivot-tree criterion for interpretation

Source: Random tree recursions: which fixed points correspond to tangible sets of trees? | Section6, PDF p.39, first paragraph

https://arxiv.org/pdf/1808.03019v2

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the two-state interpretation criterion of Theorem1.7 to finite color sets of size at least3, covering critical and supercritical pivot trees even without positive regularity.

Scope

A deterministic automaton acts on child-color counts. Interpretability is measurability of root color from the uncolored Galton-Watson tree; positive-mass colors and the standing offspring assumptions are retained.

P0090-A001 | Multistate pivot-tree criterion for interpretation structural argument

Argument or result

For a finite-type pivot branching process with mean matrix M and spectral radius less than1, E[population at level n] tends to0, hence P(level n nonempty)<=E[size] tends to0. This proves extinction and explains the subcritical direction already covered by Theorem1.8.

Check

Jordan normal form shows M^n->0 when its spectral radius is below1, including reducible matrices.

Remaining obstruction

At spectral radius1 the first moment need not decay, and reducible type classes can behave differently. The established subcritical argument does not settle critical/supercritical interpretation.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0091 Infinite-state tree-automaton interpretation criteria

Source: Random tree recursions: which fixed points correspond to tangible sets of trees? | Section6, PDF pp.39-40, second paragraph

https://arxiv.org/pdf/1808.03019v2

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the random-tree interpretation and pivot-process theory to infinite state spaces, so it can handle recursive distributional models beyond finite colors.

Scope

The source asks for an extension framework. Finite-type spectral statements cannot be copied unchanged to a countable type space.

P0091-A001 | Infinite-state tree-automaton interpretation criteria structural argument

Argument or result

Consider a countable-type branching process in which each type k individual has deterministically one child of type k+1. It survives forever, although every finite truncation of its mean matrix is nilpotent and every fixed type eventually disappears.

Check

The population is exactly one at every level. The top-left K by K shift matrix has Kth power zero.

Remaining obstruction

Finite truncations falsely suggest extinction because population escapes through types. Any infinite-state criterion needs tightness or a suitable operator topology, not only finite truncation eigenvalues.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0092 Randomized tree automata and endogeny

Source: Random tree recursions: which fixed points correspond to tangible sets of trees? | Section6, PDF pp.39-40, second paragraph

https://arxiv.org/pdf/1808.03019v2

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the interpretation/pivot theory when each vertex has iid auxiliary randomness and its state is a function of its child states and that local randomness.

Scope

Endogeny must be relative to the tree plus all vertex innovations; it is a different sigma-field from the bare uncolored tree.

P0092-A001 | Randomized tree automata and endogeny structural argument

Argument or result

Let each vertex state be its own independent Bernoulli(p) innovation, ignoring its children. The pivot tree is empty and the state is measurable from the marked tree, but for $0<p<1$ it is independent of the unmarked tree and cannot be a measurable function of that tree.

Check

Conditional variance given the bare tree is $p(1-p)>0$; conditional variance given innovations is0.

Remaining obstruction

A literal extension retaining only the unmarked-tree sigma-field is false. A useful theorem must specify the innovation sigma-field and then handle nontrivial child dependence.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0093 Infinitely many fixed points of a finite-color automaton

Source: Random tree recursions: which fixed points correspond to tangible sets of trees? | Section6, PDF p.40, third paragraph

https://arxiv.org/pdf/1808.03019v2

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a finite color set of size at least3 and a deterministic child-count automaton under the paper standing offspring assumptions, can the distributional map have infinitely many fixed points?

Scope

The one-child identity example is excluded by the genuine-branching assumption. Empty-color embeddings are not used to manufacture examples.

P0093-A001 | Infinitely many fixed points of a finite-color automaton structural argument

Argument or result

If the offspring distribution has finite support bounded by D, each coordinate of the distributional map is a polynomial of degree at most D in the color probabilities. Fixed points are therefore a compact real algebraic subset of the probability simplex. In contrast with one variable, polynomial equations may have curves of zeros, so analyticity alone cannot prove finiteness.

Check

For example, the algebraic system x-y=0 in a square has a whole line of zeros; this illustrates the logical gap without asserting that this system comes from an automaton.

Remaining obstruction

The nonnegative multinomial coefficients and deterministic count assignments constrain which algebraic sets can occur. Neither an admissible positive-dimensional example nor a general exclusion is obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0094 Prescribing rogue and interpretable automaton fixed points

Source: Random tree recursions: which fixed points correspond to tangible sets of trees? | Section6, PDF p.40, third paragraph

https://arxiv.org/pdf/1808.03019v2

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Given a finite prescribed configuration of two-state fixed points with rogue/interpretable labels, determine whether a deterministic tree automaton and offspring distribution can realize it, including the monotone-automaton restriction.

Scope

Endpoint and multiplicity constraints are part of realizability. The request is not permission to assign arbitrary labels that contradict the pivot criterion.

P0094-A001 | Prescribing rogue and interpretable automaton fixed points structural argument

Argument or result

For a differentiable increasing distributional map Psi, let g(p)=Psi(p)-p. At an isolated simple fixed point, crossing from g>0 to g<0 gives Psi prime<1, while the opposite crossing gives Psi prime>1. Consecutive simple interior crossings must alternate signs. Thus a prescribed list of simple attractive/repulsive crossings cannot have two consecutive crossings of the same orientation without an intervening zero.

Check

This is the intermediate value theorem and the derivative sign at a simple root; it makes no claim for tangencies or nondifferentiable maps.

Remaining obstruction

The derivative restriction is necessary, not sufficient for an automaton realization or rogue classification in degenerate cases. Constructing compatible multinomial maps remains open here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0095 Automaton-level conditions for a unique fixed point

Source: Random tree recursions: which fixed points correspond to tangible sets of trees? | Section6, PDF p.40, fourth paragraph

https://arxiv.org/pdf/1808.03019v2

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Find structural conditions on a tree automaton itself that force uniqueness of its distributional fixed point, complementing the source condition on a first-order interpretation.

Scope

The condition must involve the offspring law as appropriate; the same rule can undergo bifurcations when that law changes.

P0095-A001 | Automaton-level conditions for a unique fixed point structural argument

Argument or result

Suppose that for each offspring count D the rule depends on a fixed set of at most k(D) child positions, and E k(D)=c<1. Couple child colors optimally with disagreement probability dTV(mu,nu); the union bound over that fixed relevant set gives dTV(Psi(mu),Psi(nu))<=c dTV(mu,nu). Iteration is a contraction and yields a unique fixed point on the finite simplex.

Check

The relevant set must be fixed independently of the child configuration; a pointwise bound on pivotal children alone would not justify this global coupling. For symmetric count automata the crude k(D)=D criterion reduces to E D<1.

Remaining obstruction

For supercritical offspring the crude contraction constant exceeds1. A useful structural criterion must exploit much finer properties than the number of children.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0097 An explicit nontrivial FPP time constant and limit shape

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions1 and37, PDF pp.11 and141

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Identify an explicit positive axial time constant for a nondegenerate iid nearest-neighbor passage-time law on Z^d, or explicitly determine a nontrivial compact limit shape.

Scope

The survey itself calls Question37 a rephrasing of Question1; only one problem is counted. A zero time constant from a supercritical zero-edge atom or the deterministic l1 ball is excluded.

P0097-A001 | An explicit nontrivial FPP time constant and limit shape structural argument

Argument or result

If every edge time is multiplied by c>0, every finite path cost and every passage time is multiplied by c, so the time constant is multiplied by c and the limit shape is dilated by1/c. Thus scaling a known degenerate example cannot produce the missing nontrivial shape.

Check

The equality holds configuration by configuration before taking any limit. Positive-time and nondegeneracy restrictions exclude the zero-time percolation examples.

Remaining obstruction

Homogeneity supplies a transformation law, not a computable nondegenerate base model. Integrable directed last-passage formulas apply to a different optimization problem.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0098 Strict FPP comparison in higher dimensions at a large minimum atom

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question3, PDF p.13; Definition2.8 and Theorem2.12, pp.12-13

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend strict comparison of axial time constants for strictly ordered passage-time laws in increasing-concave order to d>2 when the minimum support value r>0 has mass at least the oriented-percolation threshold.

Scope

More variable means integral phi dF <= integral phi dG for every increasing concave phi. The claim is axial; equality inside an oriented percolation cone cannot be ignored in statements about arbitrary directions.

P0098-A001 | Strict FPP comparison in higher dimensions at a large minimum atom structural argument

Argument or result

For the special coupling tau'_e=tau_e+delta with delta>0, every path from0 to ne1 has at least n edges. Its primed cost is at least its original cost plus delta n. Taking infima and limits gives mu'(e1)>=mu(e1)+delta.

Check

The bound does not depend on a minimizing path or an upper bound on its length. It proves an actual strict gap for this special stochastic ordering.

Remaining obstruction

General increasing-concave order, including mean-preserving spreads, has no pointwise additive coupling. The argument does not handle the remaining comparison class.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0099 Strict convexity of iid FPP limit shapes

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question4, PDF p.21; Section2.8

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For continuous iid passage-time distributions satisfying the shape-theorem assumptions, prove that the deterministic limit shape is strictly convex.

Scope

This qualitative absence of boundary segments is distinct from a quantitative uniform-curvature estimate. Continuity means no atoms; finite moment and positive-speed assumptions must accompany a compact shape.

P0099-A001 | Strict convexity of iid FPP limit shapes structural argument

Argument or result

Finite geodesic uniqueness does not imply strict convexity of an asymptotic norm. For example, the strictly convex norms ||x||_p converge uniformly on compact sets to ||x||_infinity as p tends to infinity, and the latter unit ball has flat faces. Even stronger finite approximations can lose strictness in the limit.

Check

The inequality ||x||_infinity <= ||x||_p <= d^(1/p)||x||_infinity verifies the convergence directly.

Remaining obstruction

This is a counterexample to an inference about limits, not an iid FPP counterexample. A uniform quantitative gap in the triangle inequality is missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0100 Excluding axial facets and cubical FPP shapes

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions5,8,9, PDF pp.21 and24

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Rule out a cube as an iid FPP limit shape and, more strongly, rule out a boundary segment perpendicular to e1 at the axial boundary point; exhibit a nondegenerate law with no such axial segment.

Scope

These nested axial-flatness formulations are grouped. Convexity, coordinate-reflection symmetry, and a nonempty interior are known but do not settle them.

P0100-A001 | Excluding axial facets and cubical FPP shapes structural argument

Argument or result

The l_infinity unit ball satisfies every elementary symmetry and convexity condition of an FPP limit shape and has axial facets. The l1 unit ball also satisfies those conditions but has an axial corner. Thus the shape theorem's convexity and lattice symmetries permit both geometries.

Check

Both balls are compact with nonempty interior and invariant under coordinate permutations and sign changes. Their intersections with the supporting plane x1=1 distinguish the facet from the corner.

Remaining obstruction

One needs a property specific to iid optimization that excludes the cubical candidate; deterministic convex geometry alone supplies none.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0101 Classification of flat FPP boundary pieces

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions6-7, PDF p.24; definition of M_p, p.22

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For planar iid FPP, show that the only flat boundary portions are the oriented-percolation-cone portions associated, after rescaling, with a positive minimum edge weight carrying a sufficiently large atom.

Scope

M_p in the source normalizes the minimum support value to1 and its atom to p>=oriented p_c. The two source questions concerning flats outside the cone and converse classification are grouped.

P0101-A001 | Classification of flat FPP boundary pieces structural argument

Argument or result

If tau_e>=1, then every path from0 to x costs at least ||x||_1, so the limit shape lies inside the l1 ball. An oriented path consisting entirely of unit-time edges attains this lower bound exactly. This explains the mechanism generating the known cone flats.

Check

The lower bound is pathwise. Along an oriented path, its length equals the l1 displacement, so equality is exact.

Remaining obstruction

The mechanism proves why these flats occur, not why no alternative mechanism creates a flat segment away from that cone.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0102 Strong Busemann limits for subadditive arrays and FPP

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions10 and32, PDF pp.26 and113

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Find useful conditions ensuring almost-sure limits of X_(m,N)-X_(n,N), including FPP differences of passage times to receding hyperplanes.

Scope

The survey explicitly translates the FPP limit question into the general-array question. These are one family; a weak subsequential or averaged limit is not the requested pathwise limit.

P0102-A001 | Strong Busemann limits for subadditive arrays and FPP structural argument

Argument or result

Let S be simple symmetric random walk and X_(m,n)=|S_n-S_m|. This is an integrable stationary ergodic subadditive array. For N with S_N>=2, X_(0,N)-X_(1,N)=S_1; for S_N<=-2 it equals -S_1. Both events occur infinitely often almost surely, so this Busemann difference does not converge.

Check

Subadditivity is the triangle inequality; stationarity and ergodicity come from the iid increments. Recurrence in both directions proves the two subsequential values, which differ because S_1 is+1 or-1.

Remaining obstruction

Kingman's usual hypotheses are insufficient. FPP triangle bounds yield compact subsequences, but an additional uniqueness or directional structure is needed for a full limit.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0103 Uniform curvature of continuous-law FPP shapes

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question11, PDF p.31

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove that every continuous iid passage-time law with a compact limit shape has a uniform constant C>0 such that 1-mu(z)>=C min(mu(z-z1),mu(z-z2))^2 for boundary points z1,z2 and z=(1-lambda)z1+lambda z2.

Scope

This is the source's geometric definition, avoiding ambiguity in printed curvature-exponent conventions. It is quantitatively stronger than strict convexity.

P0103-A001 | Uniform curvature of continuous-law FPP shapes structural argument

Argument or result

Strict convexity alone cannot imply this inequality. On the l_p unit ball for p>2, take z1=((1-t^p)^(1/p),t), z2=((1-t^p)^(1/p),-t), and their midpoint z. Then 1-||z||_p is asymptotic to t^p/p, whereas ||z-z_i||_p^2=t^2. Their ratio tends to0.

Check

The expansion (1-u)^(1/p)=1-u/p+o(u) gives the asserted ratio. The l_p ball is strictly convex for p>1.

Remaining obstruction

This is a norm counterexample to a proof strategy, not an FPP counterexample. Uniform lower curvature must be proved from the random model.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0104 Subdiffusive and KPZ passage-time fluctuation exponents

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions13 and15, PDF pp.34 and40; Section3

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Under suitable nondegenerate iid passage-time assumptions, prove a fluctuation exponent strictly below1/2, identify the planar exponent as1/3, and establish a positive exponent in a suitable dimension.

Scope

These closely related exponent-range targets are grouped. Different definitions using variances or stronger concentration must be specified before claiming equivalence.

P0104-A001 | Subdiffusive and KPZ passage-time fluctuation exponents structural argument

Argument or result

A variance upper bound of order n/log n does not establish an exponent below1/2. Indeed log(n/log n)/(2log n) tends to1/2. Sums of floor(n/log n) independent mean-zero variance-one variables realize that variance order in an elementary model.

Check

The logarithm calculation is exact in the limit; the example has additive variances. It is not asserted to be FPP.

Remaining obstruction

Known sublinear-variance estimates leave a polynomial-exponent gap. An additional power saving, and a matching lower bound for identification, are required.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0105 High-dimensional limit of the FPP fluctuation exponent

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question14, PDF p.34

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine whether the dimension-dependent passage-time fluctuation exponent chi(d) tends to zero as d tends to infinity.

Scope

Existence and definition of chi(d) are themselves part of the scope; decay of the time constant is not decay of its fluctuation exponent.

P0105-A001 | High-dimensional limit of the FPP fluctuation exponent structural argument

Argument or result

Embedding Z^d into Z^(d+1) with shared weights gives T_(d+1)<=T_d and hence mu_(d+1)<=mu_d. This ordering says nothing comparable about variance: a Bernoulli random variable is bounded above by the constant1 but has larger variance.

Check

The passage-time inequality follows because the higher-dimensional infimum has more admissible paths. The variance counterexample has values1/4 and0 at Bernoulli parameter1/2.

Remaining obstruction

A distributional comparison controlling centered fluctuations across dimensions is missing. Monotonicity of means cannot supply it.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0106 Polynomial overlap of independent planar FPP geodesics

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question16, PDF p.42

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Show that two independent geodesics from0 to ne1, sampled in independent iid planar environments with the same law, have expected edge overlap at least n^a for some a>0.

Scope

Independent environments and a fixed measurable geodesic-selection rule are intended; two samples in one unique-geodesic environment would be identical.

P0106-A001 | Polynomial overlap of independent planar FPP geodesics structural argument

Argument or result

Let E_r be the 8r+4 edges leaving the square[-r,r]^2, for r=0,...,n-1, and p_e=P(e belongs to the selected geodesic). Every path to ne1 crosses E_r, so sum_(e in E_r)p_e>=1. Independence and Cauchy-Schwarz give E|Gamma intersect Gamma'|=sum_e p_e^2 >= sum_(r=0)^(n-1)1/(8r+4) >=(1/8)log(2n+1).

Check

The sets E_r are disjoint; corners contribute two outward edges and total8r+4 is exact. Tonelli justifies the infinite sum. Integrating1/(8x+4) gives the final lower bound.

Remaining obstruction

This establishes only a logarithmic overlap lower bound. Improving shell-level spreading to a power requires model-specific concentration of geodesic locations.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0107 Critical Bernoulli FPP growth in three dimensions

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question19, PDF p.71

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For iid edge times0 with probability p_c(Z^3) and1 otherwise, determine whether E T(0,boundary B(n)) is comparable to log n.

Scope

The target is undirected edge FPP at the three-dimensional bond threshold; planar critical exponents cannot be transferred.

P0107-A001 | Critical Bernoulli FPP growth in three dimensions structural argument

Argument or result

Collapse every connected zero-edge cluster to a vertex and retain unit-cost edges between the resulting vertices. Passage time is precisely graph distance in this random quotient graph, counting the fewest positive edges needed to reach the target. In the original graph it is bounded above by the number of positive edges on a fixed n-step coordinate path.

Check

Every original path projects to a quotient path with the same number of unit-cost crossings, and every finite quotient path lifts through finite zero-edge connections. This proves equality even when cluster sizes are unbounded.

Remaining obstruction

The logarithmic estimate requires quantitative geometry of the cluster-adjacency quotient. The planar argument based on nested dual circuits has no direct three-dimensional counterpart.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0108 Sharp width threshold for the thin-cylinder FPP CLT

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question20, PDF p.72

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the largest power alpha for which standardized passage times constrained to Z times[-n^alpha,n^alpha]^(d-1) converge to a standard normal law.

Scope

The source defines gamma_F(d) by this threshold under a finite2+delta edge-time moment. A sufficient exponent from the source theorem is not an optimality result.

P0108-A001 | Sharp width threshold for the thin-cylinder FPP CLT structural argument

Argument or result

At width0 the graph is a line, and the passage time to ne1 is exactly the sum of n iid edge weights. For a nondegenerate finite-variance law the classical iid CLT gives normalization sqrt(n Var(tau)) and mean n E(tau).

Check

Any line path containing a backtracking excursion can have that excursion removed without increasing its nonnegative cost, leaving the unique coordinate segment.

Remaining obstruction

For growing width, competing paths introduce correlations and varying block sizes. The width0 reduction does not control how rapidly an approximately independent block decomposition deteriorates.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0109 Existence of finite FPP geodesics without distributional assumptions

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question21, PDF p.73

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For every iid finite nonnegative passage-time distribution in d>=3, prove that the infimum over finite paths between any two vertices is attained.

Scope

Infinite passage-time edges are not included in this formulation. Non-iid weighted graphs can fail attainment and therefore cannot settle the iid question by analogy.

P0109-A001 | Existence of finite FPP geodesics without distributional assumptions structural argument

Argument or result

If all edge times are at least a>0, a fixed connecting path of cost M bounds any competing path length by M/a; only finitely many simple paths of that length start at a fixed vertex, so a minimum exists. Without a positive lower bound, a deterministic ladder can fail: give each vertical edge from height j-1 to j on both rails weight2^(-j), and the rung at height k weight3*2^(-k). A route across at height k costs2+2^(-k), with infimum2 unattained.

Check

For a finite ladder route attaining maximum height k, vertical cost is at least2(1-2^(-k)) and at least one rung has cost at least3*2^(-k); this gives the same strict lower bound. Embed the ladder into Z^d with all other edges cost100 to preserve failure.

Remaining obstruction

The ladder environment is deterministic and spatially inhomogeneous, so it is only a counterexample to a general compactness inference. The iid critical case requires additional probabilistic structure.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0110 An iid FPP law with proved KPZ exponent relation

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question22, PDF p.89; equation4.19

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Exhibit a nondegenerate iid undirected FPP law with well-defined fluctuation and wandering exponents satisfying chi=2xi-1.

Scope

Conditional proofs assuming suitably strong exponents exist do not exhibit a law for which those assumptions hold. Exactly solvable directed last-passage models are outside the target.

P0110-A001 | An iid FPP law with proved KPZ exponent relation structural argument

Argument or result

If a geodesic deviates transversely by n^xi and the norm has quadratic curvature, the deterministic excess norm over a displacement of length n has order n*(n^(xi-1))^2=n^(2xi-1). This yields the proposed balance only after assuming random fluctuations are of the same order.

Check

The power follows from norm homogeneity and the square of the relative transverse displacement. A curvature exponent kappa would instead give kappa*xi-(kappa-1).

Remaining obstruction

Both the required exponent existence and error control when replacing random passage times by the norm remain unproved for a concrete iid law here. The balance calculation is conditional, not a proof.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0111 Infinitely many planar FPP geodesic rays for continuous laws

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions23-24, PDF p.91

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove that continuous iid planar passage-time distributions have infinitely many pairwise eventually disjoint geodesic rays; the source also asks for an example outside the percolation-cone class.

Scope

The universal continuous-law question and its weaker example target are grouped. Distinct rays mean sharing at most finitely many vertices, not merely having different first edges.

P0111-A001 | Infinitely many planar FPP geodesic rays for continuous laws structural argument

Argument or result

Finite geodesic existence and finite lattice degree yield at least one ray: among geodesics to points tending to infinity, choose an infinite subsequence with a common first edge, then a common second edge, and continue. Every finite segment of the diagonal limit is minimizing.

Check

At each stage finitely many choices ensure an infinite subsequence. Path lengths tend to infinity, and any shortcut of a limiting finite segment would contradict minimization in all sufficiently long selected paths.

Remaining obstruction

This compactness proof gives one ray, not many eventually disjoint rays. Endpoint sequences in different directions could still share the same limiting tail without additional shape information.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0112 Full-sequence convergence of FPP geodesics to a ray

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question25, PDF p.92

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For iid passage-time laws with unique finite geodesics, determine whether the geodesics from0 to ne1 converge locally to a single infinite ray as n tends to infinity.

Scope

Convergence means stabilization of every finite initial segment. Existence of some subsequential ray is strictly weaker.

P0112-A001 | Full-sequence convergence of FPP geodesics to a ray structural argument

Argument or result

The space of infinite rooted paths in a locally finite graph has a compact product topology after adjoining finite-path endpoints. Thus long finite geodesics admit subsequential limits. But the abstract sequence of paths alternating between two disjoint infinite branches has two subsequential limits and no full limit.

Check

On a rooted binary tree, alternating the all-left and all-right depth-n paths supplies an explicit sequence illustrating the failure of compactness to imply uniqueness.

Remaining obstruction

The example is not an iid lattice FPP counterexample. A uniqueness-of-limit or directional-coalescence argument is missing for the specified endpoint sequence.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0113 Almost-sure escape from a fixed midpoint in FPP

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question26, PDF p.92

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

In uniquely geodesic planar iid FPP, prove that no fixed vertex belongs to geodesics from-ne1 to ne1 for infinitely many n.

Scope

This is the stronger almost-sure formulation printed in the survey. The known BKS result that the probability of midpoint membership tends to zero is not by itself this statement.

P0113-A001 | Almost-sure escape from a fixed midpoint in FPP structural argument

Argument or result

If a fixed vertex v belongs to infinitely many such geodesics, both endpoint distances from v tend to infinity. A diagonal subsequence of the two oriented halves produces a doubly infinite geodesic through v. Therefore absence of all bigeodesics would imply the target.

Check

Finite degree supplies the diagonal subsequence and every limiting finite segment is a minimizing segment of a selected geodesic. Conversely, probabilities tending to0 cannot justify a Borel-Cantelli conclusion: independent events with probability1/n occur infinitely often.

Remaining obstruction

Neither a summable midpoint bound nor unconditional absence of bigeodesics is obtained. Later midpoint papers must be compared against this exact almost-sure quantifier.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0114 Uniqueness of planar directional geodesics in every fixed direction

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions27-28, PDF p.95

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Show that every deterministic direction belongs to the Licea-Newman full-measure set D: from a fixed lattice vertex there are almost surely not two distinct geodesic rays with that direction.

Scope

The axial case and all-direction case are grouped. This is at-most-one, not an assertion that a ray in that direction exists. The null event may depend on the fixed direction.

P0114-A001 | Uniqueness of planar directional geodesics in every fixed direction structural argument

Argument or result

A full-measure deterministic set may omit the axis, for example[0,2pi) without0. Also, a random exceptional set can be nonempty almost surely while excluding each fixed angle almost surely: a singleton containing an independent uniform angle has this property.

Check

The first example checks the deterministic exceptional-set issue. In the second, P(U=theta)=0 for every fixed theta but the singleton is never empty.

Remaining obstruction

Fubini-based almost-every-direction arguments do not settle even a prescribed exceptional angle. A geometric argument sensitive to that direction is needed.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0115 Coalescence of directional FPP rays in higher dimensions

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question29, PDF p.97; Theorem4.22, p.96

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the planar coalescence conclusion to higher dimensions: for a fixed direction with almost-sure uniqueness of a directional ray from each vertex, must directional rays from different vertices coalesce?

Scope

The source asks whether its two-dimensional theorem extends. The relevant directional-uniqueness hypothesis must be stated rather than silently assumed in higher dimensions.

P0115-A001 | Coalescence of directional FPP rays in higher dimensions structural argument

Argument or result

Translation invariance and one outgoing edge at every vertex do not force coalescence. Orient every edge line in the positive e1 direction in Z^d. There is exactly one forward ray from each vertex, but rays on different parallel lines never meet.

Check

The construction is deterministic and translation invariant. All rays have the same asymptotic direction, so directional agreement alone adds no coalescence.

Remaining obstruction

This is a forest counterexample to a proof strategy, not the iid FPP model. Planar shielding and ordering used in the source proof do not extend automatically.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0116 Growth of the infinite-descendant part of the FPP geodesic tree

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question30, PDF p.97

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Let f(r) count edges meeting the Euclidean circle of radius r that belong to infinitely many finite geodesics from0. Determine whether f(r) tends to infinity, and at what rate.

Scope

Assume a unique rooted finite-geodesic tree. Source questions about how many rays exist and this radial growth function are related but the latter requests additional spatial rate information.

P0116-A001 | Growth of the infinite-descendant part of the FPP geodesic tree structural argument

Argument or result

In a locally finite rooted tree, an edge has infinitely many descendants exactly when it lies on an infinite ray, by the finite-branching compactness lemma. If the tree has infinitely many ends, then for every k one can choose k rays with disjoint tails outside some fixed ball; each tail crosses every sufficiently large circle. Consequently f(r)>=k eventually.

Check

The forward implication is Konig's lemma applied below the edge; the reverse is immediate. A finite selection of distinct ends separates beyond finitely many branch vertices.

Remaining obstruction

This proves divergence conditional on infinitely many ends, without a rate. It supplies no unconditional ray multiplicity and no bound on where k rays separate.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0117 Absence of bi-infinite geodesics in planar iid FPP

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question31, PDF p.98

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For nondegenerate continuous iid planar passage-time laws in the usual positive-speed regime, determine whether a bi-infinite geodesic exists.

Scope

The survey's short question sits in the uniquely geodesic random setting. Deterministic equal weights and the one-dimensional lattice have obvious geodesic lines and do not resolve the intended planar question.

P0117-A001 | Absence of bi-infinite geodesics in planar iid FPP structural argument

Argument or result

Taking geodesics between-ne1 and ne1 gives candidate long paths, but a compactness argument produces a bi-infinite limit through a bounded region only if their distance from that region is tight along a subsequence. Paths can escape every fixed ball while connecting endpoints on opposite sides.

Check

An explicit lattice path from(-n,0) up to(-n,n), across to(n,n), then down to(n,0) avoids every fixed ball for all large n; it illustrates the missing compactness anchor, without being asserted to be a geodesic.

Remaining obstruction

Finite-geodesic existence does not give the required anchoring. Proving or ruling out such anchoring under iid passage weights remains unresolved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0118 Noncircular low-dimensional Eden limit shape

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question33, PDF p.130; model definition pp.128-129

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove that the Eden model's limit shape is not a Euclidean ball in the dimensions left open by existing high-dimensional results, in particular dimension2.

Scope

The source defines an Eden version choosing a boundary vertex uniformly. Edge-exponential infection generally chooses vertices at rates proportional to infected neighbors; these versions must not be identified without checking the time change.

P0118-A001 | Noncircular low-dimensional Eden limit shape structural argument

Argument or result

With independent rate-one clocks on currently available boundary vertices, the next added vertex is uniform and the holding time is exponential with rate equal to their number. In edge-clock growth, a vertex with k infected neighbors has k active edge clocks and hence total rate k. A boundary vertex with two infected neighbors is selected twice as often as one with a single neighbor in that version.

Check

The minimum of m independent rate-one exponentials has rate m and a uniform minimizing index. Summing k independent clock hazards gives rate k.

Remaining obstruction

This identifies the correct finite-state generator and prevents a model substitution, but the angular growth rate does not close as a function of the current radius. No anisotropic limit-shape inequality is obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0119 Noncoexistence of unequal-speed Richardson infections

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Questions34-35, PDF pp.134-135

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Show that two-type Richardson growth on Z^d, d>=2, from distinct finite seeds cannot coexist indefinitely at unequal infection speeds; establish the proposed monotonicity in relative speed if that route is valid.

Scope

The source assumes an ordering of the two parameters before asking monotonicity. Speed versus mean-passage-time conventions must be fixed. The universal noncoexistence and monotonicity route are grouped.

P0119-A001 | Noncoexistence of unequal-speed Richardson infections structural argument

Argument or result

A common-clock monotone comparison of one infection's absolute size does not imply monotonicity of coexistence, because the latter requires both species to remain infinite. On Z with seeds at0 and1, every site at most0 can only be reached first through the left seed and every site at least1 through the right seed; with positive finite passage times both species survive for every speed ratio.

Check

The seeds block any nearest-neighbor path across the interface once colors are permanent. Both outward rays are infinite, independent of relative rates.

Remaining obstruction

The one-dimensional example shows the essential role of bypass geometry and is excluded from the intended d>=2 conclusion. A coupling controlling the joint survival event in higher dimensions remains missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0120 Strict interface speedup in half-plane inhomogeneous FPP

Source: Auffinger, Damron and Hanson: 50 years of first-passage percolation | Question38, PDF p.142

https://arxiv.org/pdf/1511.03262

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Characterize the pairs of iid half-plane edge laws F_minus,F_plus for which vertical growth along their interface is strictly faster than in either homogeneous model.

Scope

Greater speed means a smaller directional time constant. The exact convention for edges on the interface must be fixed; finite detours in the constant-law check below do not affect its conclusion.

P0120-A001 | Strict interface speedup in half-plane inhomogeneous FPP structural argument

Argument or result

For constant times a on one half-plane and b on the other, let c=min(a,b)>0. Every path with vertical displacement n costs at least cn. A path taking a bounded horizontal detour into the cheaper half-plane, moving n steps vertically, then returning costs cn+O(1). Thus the interface time constant is exactly c, so there is no strict speedup beyond both homogeneous models.

Check

The lower bound uses path length at least n; the explicit path gives a matching upper bound after division by n.

Remaining obstruction

Random switching between regions may exploit complementary favorable segments. The deterministic comparison cannot characterize when that advantage is strictly positive.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0121 Uniform Lipschitz height models and their interfaces

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problems2.2-2.3, PDF pp.15-16

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Establish Gaussian-free-field scaling and SLE4 interface limits for the source's uniform odd-integer triangular-lattice heights, and for its uniform real-valued Lipschitz heights with the appropriate boundary amplitude.

Scope

The two genuinely different height ensembles are conservatively grouped as one universality family. Integer heights have nearest-neighbor difference0 or2 and boundary values plus/minus1; real heights have difference at most1 and tuned boundary values plus/minus lambda.

P0121-A001 | Uniform Lipschitz height models and their interfaces structural reduction or obstruction

Argument or result

Conditioning an interior height on all other heights gives a uniform law on its feasible interval, intersected with the odd integers in the discrete model. The interval endpoints are the maximum of neighbor heights minus the slope bound and the minimum plus the bound. Its conditional mean is the midpoint of these extrema, not generally the arithmetic mean of neighboring heights.

Check

For real slope bound1 and six neighboring heights0,0,0,0,0,1, the feasible interval is[0,1], so the conditional mean is1/2 while the neighbor average is1/6. These neighbor values satisfy their mutual triangular-edge constraints.

Remaining obstruction

The discrete Gaussian harmonic-observable proof therefore does not transfer by replacing a Gaussian field with a uniform Lipschitz field. One must first control the effective large-scale field and then the discontinuous operation of extracting its zero interface.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0122 SLE4 from spectral truncations of the Gaussian free field

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.4, PDF p.16

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Project a planar Dirichlet Gaussian free field onto Laplace eigenmodes with eigenvalue magnitude at most R, add the harmonic extension of suitably chosen two-arc boundary values, and prove that the connecting zero interface converges to SLE4 as R tends to infinity.

Scope

The cutoff is spectral and nonlocal. Distributional convergence of the field itself is already built into the construction and is not the requested interface convergence.

P0122-A001 | SLE4 from spectral truncations of the Gaussian free field structural reduction or obstruction

Argument or result

Even uniform convergence of smooth functions does not make zero sets continuous. On a fixed square, h_n(x,y)=1/n has no zero set, whereas its uniform limit0 vanishes everywhere. With changing signs, n^(-1)sin(nx) also converges uniformly to0 but has many macroscopic vertical zero components.

Check

The sup norms are at most1/n. For the sinusoidal example the zero lines are x=k*pi/n wherever they lie in the square; their number diverges.

Remaining obstruction

These examples do not satisfy the prescribed random-field law or boundary data; they invalidate only the proposed continuous-mapping argument. Uniform control of near-zero topology and a characterization of the selected interface are missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0123 High-temperature triangular Ising interfaces converge to SLE6

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.5, PDF p.17, high-temperature portion

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For every fixed inverse temperature strictly between0 and the triangular Ising critical inverse temperature, prove convergence of the two-boundary-arc spin interface to chordal SLE6.

Scope

The critical-temperature SLE3 component has been resolved in later literature and is not admitted here. The independent-spin beta=0 endpoint is also known. Fixed positive high-temperature correlations are the target.

P0123-A001 | High-temperature triangular Ising interfaces converge to SLE6 structural reduction or obstruction

Argument or result

The high-temperature Gibbs law is not a small total-variation perturbation of independent spins uniformly over a growing domain. For n independent two-spin blocks with coupling beta>0, the average within-block product concentrates at tanh(beta), whereas at beta=0 it concentrates at0. Choosing a threshold halfway between these means distinguishes the two product laws with probability tending to1.

Check

Each block product is plus/minus1 with mean tanh(beta); its variance is at most1. Chebyshev applied to n blocks proves asymptotic separation without a central limit approximation.

Remaining obstruction

Thus finite-range or exponentially decaying local dependence does not justify global total-variation replacement. A renormalization or observable argument controlling crossing geometry is still needed for the actual triangular model.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0124 Critical planar random-cluster interfaces for general cluster weight

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problems2.6 and2.11, PDF pp.17-18

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the square-lattice FK model at p=sqrt(q)/(1+sqrt(q)), q in(0,4), establish the predicted chordal SLE limit with kappa=4*pi/arccos(-sqrt(q)/2).

Scope

The q=2 FK-Ising case is already resolved and excluded from the unresolved scope. Square bond percolation is q=1 and is grouped here instead of being counted again as Problem2.11.

P0124-A001 | Critical planar random-cluster interfaces for general cluster weight structural reduction or obstruction

Argument or result

Given all edges except e, the FK conditional probability that e is open is p if its endpoints are already connected, and p/[p+q(1-p)] otherwise. At the proposed self-dual p these become sqrt(q)/(1+sqrt(q)) and1/(1+sqrt(q)). They straddle1/2 for q>1.

Check

Opening e changes its Bernoulli odds by p/(1-p); if it merges components it also changes the cluster-count weight by1/q. Normalizing the two possible weights gives the displayed conditional probabilities.

Remaining obstruction

Self-duality supplies symmetry but not an independent percolation exploration law. The dependence on long-range connectivity prevents a direct reuse of the q=1 or q=2 observable argument.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0125 Conformal limits of the critical and dense O(n) loop model

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.7, PDF p.17

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the hexagonal O(n) loop model, establish the predicted SLE limits of a boundary-to-boundary strand at the critical edge weight and in the dense phase.

Scope

The source's whole parameter family is one problem. Previously admitted self-avoiding walk at n=0 is cross-referenced, and known Ising cases at n=1 are not new open items. The remaining interacting-loop ensembles form this entry.

P0125-A001 | Conformal limits of the critical and dense O(n) loop model structural reduction or obstruction

Argument or result

Deleting all closed loops from a configuration does not give a self-avoiding walk with weights depending only on its length. The marginal weight of a strand gamma is proportional to x^|gamma| times the partition function of loops compatible with gamma. Two equal-length strands may leave different complementary regions and therefore different loop partition functions.

Check

Summing the joint weight x^(|gamma|+total loop length)*n^(number of loops) over compatible loop families factors out x^|gamma| and leaves exactly the stated complementary partition function.

Remaining obstruction

The environmental partition-function factor is the missing ingredient in a reduction to self-avoiding walk. Its conformal asymptotics have not been derived here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0126 Planar uniform lattice-tree diameter and scaling limit

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.8, PDF p.18

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the expected Euclidean diameter growth and the rescaled random-set limit of the uniform n-vertex lattice tree containing the origin, in particular on the planar square lattice.

Scope

A uniform embedded n-vertex tree is not a uniform spanning tree of a fixed region or a critical Galton-Watson tree. High-dimensional spread-out results do not settle the planar model.

P0126-A001 | Planar uniform lattice-tree diameter and scaling limit structural reduction or obstruction

Argument or result

Every such planar tree has Euclidean diameter at most n-1. If its diameter is D, all its vertices lie in an axis-aligned square of side at most2D around any chosen vertex, containing at most(2D+1)^2 lattice points. Hence D is at least(sqrt(n)-1)/2, deterministically.

Check

The upper bound follows from connectedness and at most n-1 edges on any simple tree path. The lower bound counts the containing square and applies to every connected n-vertex subset, independently of the measure.

Remaining obstruction

The interval between square-root and linear growth is too wide to identify the exponent. Expected-diameter normalization also gives no tightness of the full tree geometry by itself.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0127 Efficient approximately uniform sampling of lattice trees

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.9, PDF p.18

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Give a provably efficient algorithm to sample approximately uniformly from the n-vertex lattice trees containing the origin, or prove an appropriate computational obstruction.

Scope

An ergodic chain with uniform stationary distribution is not automatically a polynomial-time sampler. The meaning of approximation must specify, for example, total variation error.

P0127-A001 | Efficient approximately uniform sampling of lattice trees structural reduction or obstruction

Argument or result

All eligible trees lie in the radius n-1 lattice ball. Enumerating all subsets of its edges, retaining exactly the connected acyclic n-vertex subsets containing0, and choosing a uniformly random retained index gives an exact finite sampler. On Z^2 the ambient edge count is O(n^2), so the naive running time is at most exponential in n^2.

Check

Every tree vertex is reached by at most n-1 unit steps from0, proving the finite containment. Acyclicity plus connectedness and n vertices characterizes the required objects, with each edge set counted once.

Remaining obstruction

This supplies existence and exactness but fails efficiency. The later Fredes-Marckert survey constructs uniform-stationary chains; their existence alone does not provide a polynomial mixing bound.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0128 Cardy conformal invariance for Voronoi percolation

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.12, PDF p.19

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove the analogue of Smirnov's crossing-probability theorem for critical planar Voronoi percolation, with the corresponding conformally invariant interface limit.

Scope

Annealed Poisson geometry and independent fair cell colors are intended. Invariance under changing a background metric, known earlier, is a different statement from Cardy's formula.

P0128-A001 | Cardy conformal invariance for Voronoi percolation structural reduction or obstruction

Argument or result

A planar conformal map f sends a homogeneous Poisson process of intensity lambda to an inhomogeneous process whose intensity at w is lambda/|f'(f^(-1)(w))|^2. Thus exact rotation and translation symmetry of the original process do not produce exact invariance under arbitrary conformal maps.

Check

The expected image count in a Borel set A equals lambda times the area of f^(-1)(A); the change-of-variables formula gives the stated Jacobian. Disjoint-set independence is preserved, so this identifies the transformed process.

Remaining obstruction

One would need to prove that a smoothly varying local intensity does not affect the limiting critical crossing law. The Jacobian computation exposes that missing universality step without proving it.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0129 Cardy conformal invariance for Boolean continuum percolation

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem2.13, PDF p.19

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Establish Smirnov-type conformally invariant critical crossing limits for the planar Poisson Boolean model of equal-radius disks.

Scope

Varying the disk radius at intensity1 and varying intensity at radius1 are related by a Euclidean scaling and are one problem, not two corpus entries.

P0129-A001 | Cardy conformal invariance for Boolean continuum percolation structural reduction or obstruction

Argument or result

Scaling space by1/r maps intensity1 centers with disk radius r to radius-one disks whose centers have intensity r^2. Therefore all Euclidean-scale behavior depends on the dimensionless product lambda*r^2 for general intensity lambda. Under a nonaffine conformal map, however, each disk is deformed and its local size depends on position.

Check

An area A in scaled coordinates corresponds to original area r^2 A, yielding mean center count r^2 A. This proves the exact parameter equivalence.

Remaining obstruction

The scalar scaling relation does not control spatially inhomogeneous radii and noncircular images under conformal maps. Proving their irrelevance at criticality remains the obstruction.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0130 Sharp planar one-arm probability beyond its exponent

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem3.1, PDF p.20, one-arm remainder

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Improve the critical triangular-percolation one-arm probability R^(-5/48+o(1)) to sharp multiplicative asymptotics, at least bounds by constant multiples of R^(-5/48).

Scope

Du-Gao-Li-Zhuang resolve sharp half-plane and planar polychromatic multi-arm asymptotics; those resolved portions of the 2006 question are excluded. Their abstract does not include the planar one-arm event.

P0130-A001 | Sharp planar one-arm probability beyond its exponent structural reduction or obstruction

Argument or result

Quasi-multiplicative information and a logarithmic exponent need not imply a constant asymptotic amplitude. For alpha>0 and0<a<alpha, define f(R)=R^(-alpha)*exp(a*sin(log R)), R>=1. Then f is decreasing, has logarithmic exponent-alpha, and obeys exp(-3a)<=f(RS)/(f(R)f(S))<=exp(3a), but R^alpha f(R) has no limit.

Check

The derivative of log f with respect to log R is-alpha+a*cos(log R)<0. The quasi-multiplicative bound follows from three sine terms each bounded by1; subsequences with sine plus/minus1 give different amplitudes.

Remaining obstruction

This is a scalar obstruction to extracting an asymptotic constant from coarse multiplicativity. It is not a percolation counterexample, and even constant-factor bounds for the specified probability are not proved by it.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0131 Exact critical percolation chemical-distance exponent

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem3.3, PDF p.20

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the exponent alpha for the expected length of the shortest left-right critical percolation crossing of a square of side R, conditional on a crossing existing.

Scope

The Euclidean connection event, the number of edges in a shortest open path, and its conditional expectation are distinct quantities. The source requests the exponent itself, not merely improved upper or lower bounds.

P0131-A001 | Exact critical percolation chemical-distance exponent structural reduction or obstruction

Argument or result

A shortest crossing may be chosen self-avoiding by deleting loops. Its length is at least a constant times R and at most the number of sites in the square, which is O(R^2). Thus any exponent defined by the stated expectation must lie in[1,2].

Check

The lower bound is net horizontal displacement divided by the maximum horizontal displacement per edge. The upper bound counts sites on a self-avoiding path, so it holds on every configuration with a crossing.

Remaining obstruction

These geometric bounds do not quantify the critical detours around closed clusters. Interface dimensions are not automatically dimensions of shortest open crossings.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0132 Measurable loop erasure of planar Brownian motion

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem6.1, PDF p.24; Berestycki-Sauzedde2026, Section7 first question

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

In the joint scaling limit of a planar random walk and its loop erasure, determine whether the SLE2 path is a measurable function of the Brownian path.

Scope

The 2026 loop-addition theorem establishes the coupling and its chronological Brownian law; the authors explicitly leave inverse measurability open. Existence of a coupling does not imply a deterministic factor.

P0132-A001 | Measurable loop erasure of planar Brownian motion structural reduction or obstruction

Argument or result

For random elements(G,W) in standard Borel spaces, G is measurable from W exactly when the regular conditional distribution Law(G|W) is a point mass almost surely. An arbitrary measurable choice of a point in each fiber is insufficient: if G is standard normal and W=|G|, choosing the positive preimage changes the joint law, whose conditional signs remain fair.

Check

The forward implication follows by conditioning G=g(W). For the reverse, a probability kernel supported on Dirac measures identifies a measurable atom through a countable separating family. In the normal example symmetry gives conditional sign probability1/2 for W>0.

Remaining obstruction

No degeneration of the conditional SLE law given the complete Brownian chronology has been shown. The new forward map does not provide this missing conditional-law argument.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0133 Torus spanning-tree dual homotopy distribution

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem6.2, PDF p.24

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the limiting distribution of the homotopy classes of the three edge-simple noncontractible closed paths in the dual complement of a uniform spanning tree on a fine toroidal square grid.

Scope

The question concerns the joint winding information, not only the number of dual cycles. The geometry of the torus is an input. Later line-bundle determinant methods require a separate exact-scope comparison.

P0133-A001 | Torus spanning-tree dual homotopy distribution structural reduction or obstruction

Argument or result

On a toroidal N-vertex square grid, E=2N and there are N faces. Removing the N-1 primal tree edges leaves N+1 dual edges. The dual complement is connected, so its first Betti number is(N+1)-N+1=2. This explains two independent topological cycles but does not itself specify the triple of simple-loop homotopy classes.

Check

Euler's formula on the torus gives V-E+F=0. Dual connectivity follows because a separating primal-dual cut would correspond to a cycle contained in the primal tree, which is impossible. The cycle-rank formula is E-V+1 for a connected finite graph.

Remaining obstruction

Rank2 determines neither winding slopes nor their probabilities. A scalar Laplacian determinant counts all spanning trees without resolving this joint topological information.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0134 A nontrivial exact Hausdorff gauge for SLE

Source: Schramm: Conformally invariant scaling limits, overview and collection of problems, 2006v2 | Problem7.1, PDF p.25

https://arxiv.org/pdf/math/0602151

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Find a nontrivial Hausdorff gauge measure for SLE_kappa, kappa<8, which is sigma-finite and preferably finite on compact subarcs, as intended by the source's natural-parametrization question.

Scope

Literally, the zero measure is sigma-finite; the source context requires a useful nonzero measure. Rezaei proves that the critical power gauge r^(1+kappa/8) gives zero Hausdorff measure. Minkowski content is not automatically a Hausdorff measure.

P0134-A001 | A nontrivial exact Hausdorff gauge for SLE structural reduction or obstruction

Argument or result

Hausdorff dimension alone cannot select the exact gauge. For any set of dimension d, a power exponent s>d gives zero Hausdorff measure, while s<d gives infinite measure; at s=d either zero, finite positive, or infinite behavior can occur. Multiplying r^d by a slowly varying factor can change the critical measure while leaving the dimension unchanged.

Check

The two off-critical conclusions are the defining threshold property of Hausdorff dimension. At the critical exponent, a smooth interval has finite positive H^1, while a countable union of disjoint intervals with divergent total length has infinite H^1; critical zero examples exist among dimension-one Cantor sets.

Remaining obstruction

One needs fine local occupation estimates, not just dimension or existence of Minkowski content. The predicted logarithmic correction has not been proved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0135 Almost-sure fibers of chronological loop addition

Source: Berestycki and Sauzedde: Brownian paths as loop-decorated SLEs,2026 | Section7, Almost sure(non-)injectivity

https://arxiv.org/pdf/2602.04673

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine whether the chronological attachment map from a typical SLE2 path and the intersected Brownian loops to the resulting Brownian path has a unique preimage almost surely, or instead infinitely many.

Scope

Global noninjectivity is already noted by the authors. The question concerns typical fibers in the specified support. It is related to, but stronger than, recovering only the SLE path measurably.

P0135-A001 | Almost-sure fibers of chronological loop addition structural reduction or obstruction

Argument or result

Global noninjectivity cannot settle almost-sure invertibility. The map f:R->R with f(0)=f(1)=0 and f(x)=x otherwise is noninjective, yet it is injective off the two-point set and therefore injective on a full-measure set for a continuous input law. Conversely, x->|x| has two typical preimages for a symmetric continuous law.

Check

A finite set has probability0 under an atomless law. For the absolute-value map, every y>0 has two preimages of equal conditional probability under a standard normal input.

Remaining obstruction

One must exhibit an ambiguity mechanism occurring on a positive-probability or full-probability family of loop configurations. A deterministic exceptional pair or a measurable selection theorem does not do this.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0136 Chronological Brownian loop addition in three dimensions

Source: Berestycki and Sauzedde: Brownian paths as loop-decorated SLEs,2026 | Section7, Dimension3

https://arxiv.org/pdf/2602.04673

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the chronological loop-decoration representation to the three-dimensional scaling limit of loop-erased random walk, yielding the full parametrized Brownian path rather than only equality of random ranges.

Scope

The source explains that its planar deterministic continuity hypothesis fails in dimension3. A probabilistic continuity statement at typical configurations could still suffice.

P0136-A001 | Chronological Brownian loop addition in three dimensions structural reduction or obstruction

Argument or result

A fixed planar circle intersecting a line in R^3 becomes disjoint from that line after an arbitrarily small translation in the perpendicular direction. For example, gamma is the x-axis segment[-2,2] and ell is the unit circle in the xy-plane; ell+(0,0,epsilon) avoids gamma for every epsilon nonzero while converging uniformly to ell.

Check

Every point of the translated loop has z-coordinate epsilon and every point of gamma has z-coordinate0. At epsilon0 the circle meets gamma at(plus/minus1,0,0).

Remaining obstruction

Attaching or omitting a loop of fixed diameter changes the resulting path macroscopically. Typical random intersections need a stability argument beyond deterministic uniform closeness; the example does not refute such an almost-sure statement.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0137 A loop-erased representation of a unit Brownian loop

Source: Berestycki and Sauzedde: Brownian paths as loop-decorated SLEs,2026 | Section7, SLE2 loop analogue

https://arxiv.org/pdf/2602.04673

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Describe a unit Brownian loop as an SLE2-type loop decorated by an independent Brownian loop soup, with a precise law and chronology.

Scope

A closed loop has no distinguished endpoint ordering. Rooting conventions, duration conditioning, and the law of the proposed SLE loop must be fixed together.

P0137-A001 | A loop-erased representation of a unit Brownian loop structural reduction or obstruction

Argument or result

Even in a finite closed walk, rooting before chronological loop erasure can discard all information: the standard loop erasure of a finite walk ending at its starting vertex is the single starting vertex. When the terminal return is processed, it erases the whole currently retained path.

Check

The standard stack algorithm retains distinct vertices and, upon revisiting a vertex already on the stack, removes everything after its previous occurrence. The final return to the first vertex therefore leaves only that vertex.

Remaining obstruction

A nontrivial loop analogue cannot simply apply the open-path loop-erasure algorithm to an arbitrarily rooted closed Brownian path. A cyclic or marked-edge construction and its resulting sampling bias remain to be established.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0138 Law of loop-decorated SLE beyond kappa2

Source: Berestycki and Sauzedde: Brownian paths as loop-decorated SLEs,2026 | Section7, Other values of kappa

https://arxiv.org/pdf/2602.04673

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Identify the chronological path law obtained by decorating chordal SLE_kappa with Brownian loops at restriction intensity lambda=(8-3*kappa)*(6-kappa)/(2*kappa), including its proposed interpretation as partially erased Brownian motion.

Scope

Only parameters giving a nonnegative soup intensity and an applicable attachment construction are meaningful. The interval2<=kappa<=8/3 interpolates the Brownian kappa2 endpoint and undecorated SLE8/3 endpoint.

P0138-A001 | Law of loop-decorated SLE beyond kappa2 structural reduction or obstruction

Argument or result

At kappa2 the displayed intensity is1; at kappa8/3 it is0. Agreement of the filled hull with a restriction law cannot identify the ordered path: a circle traversed clockwise or counterclockwise has the same trace and hull, and arbitrary monotone time changes also preserve the hull.

Check

Substitution gives lambda2=(8-6)*(6-2)/4=1 and lambda8/3=0. The two circle trajectories differ as parametrized maps while having identical ranges.

Remaining obstruction

Restriction formulas control avoidance of sets, not chronological finite-dimensional distributions or quadratic variation. A path-level stochastic characterization is missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0139 Rough-path continuity of chronological loop attachment

Source: Berestycki and Sauzedde: Brownian paths as loop-decorated SLEs,2026 | Section7, Continuity in rough path topologies

https://arxiv.org/pdf/2602.04673

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove appropriate Holder, p-variation, or rough-path continuity estimates for the chronological loop-attachment map, possibly restricted to the cusp conditions proposed by the authors.

Scope

The precise topology must include all necessary time and iterated-integral information. Continuity in the source's existing path topology does not imply continuity of rough-path lifts.

P0139-A001 | Rough-path continuity of chronological loop attachment structural reduction or obstruction

Argument or result

Concatenate m positively oriented circles of radius m^(-1/2), each based at0, over a unit time interval. Their sup norm tends to0 and, for p>2, their first-level p-variation is at most C_p*m^(1/p-1/2), which also tends to0. But their total signed area is m*pi*m^(-1)=pi.

Check

Each circle contributes O(r^p) to the p-variation power, and splitting a partition at returns to0 changes the bound by only a p-dependent factor. Areas add over closed concatenated loops and equal pi*r^2 per circle.

Remaining obstruction

This proves that first-level convergence alone cannot control the second iterated integral. It does not rule out continuity with properly enhanced input loops or the source's additional cusp restrictions.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0141 First-order distance constants in finite-variance preferential attachment

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem2.3, PDF p.74

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For affine preferential attachment with m>=2 and delta>0, prove convergence of typical graph distance/log n and diameter/log n to explicit positive constants a<b.

Scope

Two-sided logarithmic bounds are weaker than convergence to constants. The positive-delta regime is the finite-variance degree regime; logarithmic-logarithmic formulas from negative delta do not apply.

P0141-A001 | First-order distance constants in finite-variance preferential attachment structural reduction

Argument or result

For every connected graph, the distance D(U,V) between two uniform vertices is bounded by its diameter, so if the two normalized quantities converge in probability to constants a,b then necessarily a<=b. Strict inequality does not follow: a complete graph has typical distance and diameter both1 with probability tending to1.

Check

If a>b, choose epsilon<(a-b)/2; simultaneous convergence would force D(U,V)>diameter with probability tending to1, a contradiction. The complete-graph calculation checks the strictness issue separately.

Remaining obstruction

This gives only the necessary weak ordering. The preferential-attachment age correlations and extreme peripheral vertices must be quantified to identify constants and show a strict gap.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0142 Fluctuations of preferential-attachment distances and diameter

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem2.4, PDF p.76

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the centered fluctuation scales and limiting laws for typical distances and diameter in affine preferential-attachment graphs with m>=2.

Scope

Parameter regimes are one source problem, not separate corpus variants. Tightness and full-sequence convergence must be distinguished because graph distances are integer-valued.

P0142-A001 | Fluctuations of preferential-attachment distances and diameter structural reduction

Argument or result

If integer-valued X_n is centered by real c_n, every centered value lies in Z-c_n. Along a subsequence with fractional parts c_n tending to r, any weak limit is supported on Z-r. Two subsequences whose fractional parts have different limits can therefore have incompatible translated-lattice limits.

Check

For a closed set staying a positive distance from Z-r, its probability is eventually0; the portmanteau theorem puts any weak limit on Z-r. The deterministic example X_n=floor(c_n) gives centered value minus the fractional part.

Remaining obstruction

A claimed continuous full-sequence limit without additional scaling would require checking this lattice obstruction. No centered distance law for the actual attachment model is obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0143 Second-order extremal hopcount in exponential complete-graph FPP

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.1, PDF pp.99-100

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine fluctuations of the longest minimum-weight path, both from a fixed root and over all vertex pairs, beyond the first-order e log n and alpha_star log n scales.

Scope

The rooted shortest-path tree is a random recursive tree, for which substantial height refinements already exist. The all-pairs fluctuation target is retained; known rooted results are not new solutions.

P0143-A001 | Second-order extremal hopcount in exponential complete-graph FPP structural reduction

Argument or result

For a random recursive tree, F_n(z)=E sum_v z^(depth(v)) equals product_(j=1)^(n-1)(1+z/j). For z>1, P(height>=k)<=z^(-k)F_n(z)<=exp(z H_(n-1)-k log z). Taking k=a log n,z=a gives an upper exponent a-a log a. A union bound over n roots adds1, explaining alpha_star log(alpha_star)-alpha_star=1.

Check

When vertex n is added, its parent is uniform among n-1 vertices, so the expected generating function is multiplied by1+z/(n-1). The bound log(1+z/j)<=z/j yields the display.

Remaining obstruction

This recovers a first-order upper bound only. It loses dependence between extreme paths and the logarithmic corrections needed for a fluctuation law.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0144 Conservative weighted-distance fluctuations in heavy-tailed configuration models

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.2, PDF p.142

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Describe the fluctuations of the minimal total weight between two typical vertices of an infinite-variance configuration model when the associated weighted branching process is conservative.

Scope

The source explicitly says the question concerns random fluctuations beyond the leading quantile-sum scale. Later leading-order weighted-distance results must not be mistaken for a complete centered limiting law.

P0144-A001 | Conservative weighted-distance fluctuations in heavy-tailed configuration models structural reduction

Argument or result

For nonnegative weights, a path sum is at least its maximum edge, so the FPP distance dominates the minimax path value. The latter is exactly the smallest level t at which the two vertices are connected using edges of weight at most t. This follows by comparing the same path family under the two objectives.

Check

A connection at level t supplies a path with every edge at most t, and any path with maximum edge at most t supplies such a connection; taking infima proves the minimax/percolation identity.

Remaining obstruction

The lower bound tracks the largest obstruction but discards accumulated small costs. In the conservative regime infinitely many layers can contribute, so the minimax comparison does not determine centered sum fluctuations.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0145 Hopcount universality in infinite-variance weighted configuration models

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.3, PDF p.142

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the scaling and possible central limit laws for the number of edges on a minimum-weight path between typical vertices of an infinite-variance configuration model.

Scope

The target is hopcount, not total weight or unweighted distance. Separate edge-weight laws may lead to different behaviors but are one classification problem.

P0145-A001 | Hopcount universality in infinite-variance weighted configuration models structural reduction

Argument or result

If all weights lie in[a,b] with0<a<=b, the hopcount H of a minimizing path and graph distance d satisfy d<=H<=(b/a)d: a shortest unweighted path costs at most bd, whereas the chosen path costs at least aH. Monotone transformations do not preserve hopcount: paths with weights(1,4) and(1.8,1.8,1.8) reverse their cost ordering when weights are squared.

Check

The original costs are5 and5.4; the squared costs are17 and9.72. Both paths can be realized as the two branches of a finite cycle, so the comparison is exact.

Remaining obstruction

A bounded-factor comparison gives no fluctuation law, and it disappears for weights approaching0. The exploration process must jointly control accumulated cost and generation depth.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0146 Weak-to-strong disorder universality in complete-graph FPP

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.4, PDF p.149 and discussion p.154

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Complete the classification of complete-graph first-passage percolation with varying edge-weight distributions, including the strong-disorder regime related to the minimum spanning tree.

Scope

The source highlights weights E_e raised to s_n and the regimes not covered by its theorem. A value comparison with a bottleneck problem does not prove convergence of path geometry.

P0146-A001 | Weak-to-strong disorder universality in complete-graph FPP structural reduction

Argument or result

For any positive edge labels a_e and any simple path of at most n-1 edges, max a_e^s <= sum a_e^s <=(n-1)max a_e^s. Taking infima over paths and s-th roots shows that the FPP value lies between the minimax value M and(n-1)^(1/s)M. If s/log n tends to infinity, this ratio bound tends to1.

Check

The lower bound holds for every path; applying the upper bound to a minimax path gives the reverse comparison. No exchange of a random limit and an infimum is used.

Remaining obstruction

Nearly equal objective values may be attained by very different paths. This deterministic estimate does not identify the hopcount crossover or a metric-space scaling limit.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0147 Fluctuations of weighted flooding and diameter on configuration models

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.5, PDF pp.150-151

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Identify centered fluctuation laws for exponential-weight flooding time and weighted diameter in the configuration-model regimes of Theorem3.37.

Scope

The source assumes a convergent degree law with a uniformly bounded2+epsilon moment and positive mass at the minimum degree. If the graph is disconnected, distances must be restricted to the relevant giant component; infinite distances are not silently included.

P0147-A001 | Fluctuations of weighted flooding and diameter on configuration models structural reduction

Argument or result

On any fixed graph, the minimum incident exponential edge weight at a degree-d vertex is Exp(d). For a collection of m pairwise nonadjacent degree-d vertices these minima use disjoint edge sets and are independent. Hence P(d max X_v-log m<=x)=(1-exp(-x)/m)^m tends to exp(-exp(-x)).

Check

The minimum of d independent rate-one exponentials has tail exp(-dt). Nonadjacent vertices cannot share an incident edge. Multiplying their distribution functions proves the extreme-value limit.

Remaining obstruction

This identifies one endpoint-delay component of the extremes. The bulk connecting distances and their dependence on the extreme endpoints are not controlled, so no full flooding or diameter limit is claimed.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0148 General edge-weight laws for random-graph flooding and diameter

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.6, PDF p.152

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend weighted flooding-time and diameter asymptotics from exponential weights to general iid nonnegative edge-weight laws on the configuration model.

Scope

Both the lower tail used by fast paths and the upper tail of the minimum weight leaving a peripheral vertex can matter. This is one law-classification problem.

P0148-A001 | General edge-weight laws for random-graph flooding and diameter structural reduction

Argument or result

At a degree-d vertex the minimum incident weight X has tail P(X>t)=(1-F(t))^d. Over m nonadjacent such vertices, P(max X_v<=t)=[1-(1-F(t))^d]^m. For weights E^s this gives max X_v asymptotic to(log m/d)^s in probability.

Check

For E^s, P(X>t)=exp(-d*t^(1/s)); setting t=((1 plus/minus epsilon)log m/d)^s and using the displayed maximum distribution gives matching upper and lower probability bounds.

Remaining obstruction

Endpoint delays alone can change scale, but the fastest route through the bulk is a different optimization. Matching these two mechanisms for general F is still missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0149 Extreme hopcounts for general random-graph degree and weight laws

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem3.7, PDF p.152

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the first-order extreme hopcount from one root and over all pairs for general configuration-model degree laws, and extend beyond exponential edge weights.

Scope

The known regular-graph constants use a branching-generation large-deviation calculation. Replacing the degree by its mean is not justified for extremes.

P0149-A001 | Extreme hopcounts for general random-graph degree and weight laws structural reduction

Argument or result

On a graph with maximum degree Delta, there are at most n*Delta*(Delta-1)^(k-1) oriented simple k-edge paths. With iid weights Y, the chance any such path has total weight at most t is at most this count times exp(a*t)*(E exp(-aY))^k for every a>0.

Check

After the first edge there are at most Delta-1 nonbacktracking choices. Independence along a simple path and exponential Markov inequality bound its lower-tail cost. The union bound needs no independence between different paths.

Remaining obstruction

The bound is too crude when a few hubs make Delta large and does not automatically constrain minimizing paths with unknown total cost. Degree-biased branching and extreme endpoint selection require sharper control.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0150 Minimal assumptions for heavy-tailed critical metric convergence

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.2, PDF p.218; Theorem4.39

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Remove the extra degree restrictions used to upgrade heavy-tailed critical configuration-model components from Gromov-weak convergence to Gromov-Hausdorff-Prokhorov convergence.

Scope

The printed problem heading says finite third moment, but its referenced theorem and surrounding text concern infinite third moment, tau in(3,4). Bhamidi et al2020 improve the global lower-mass bound under slightly stronger assumptions; the minimal-assumption remainder requires exact comparison.

P0150-A001 | Minimal assumptions for heavy-tailed critical metric convergence structural reduction

Argument or result

Let X_n=[0,1] with the ordinary metric and probability measure(1-1/n)delta_0+(1/n)Lebesgue. Every fixed finite iid sample from this measure lies at0 with probability tending to1, so its distance matrices converge to those of a point. Yet the spaces all have diameter1 and cannot converge in Gromov-Hausdorff distance to a point.

Check

The probability that k sampled points all select the delta_0 component is(1-1/n)^k. The Gromov-Hausdorff distance from a compact space to a point is half its diameter. Each X_n still has full-support measure.

Remaining obstruction

This is a precise failure of weak measured convergence to control thin appendages. A uniform lower bound on masses of fixed-radius balls is the missing type of estimate.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0151 Critical window for fixed-outdegree preferential attachment

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.4, PDF p.220

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For affine preferential attachment with a fixed m>=2 edges per arriving vertex and delta>0, identify the percolation scaling window and largest critical component sizes.

Scope

The percolation threshold itself is covered by later work2312.14085. The2025 growing-graph window theorem concerns models with independently attached edges; it is not automatically the fixed-outdegree model asked here.

P0151-A001 | Critical window for fixed-outdegree preferential attachment structural reduction

Argument or result

Local weak convergence alone cannot determine macroscopic cluster sizes. A single cycle on n vertices and a disjoint union of cycles whose lengths are about sqrt(n) have the same local weak limit, the two-sided infinite path, but their largest components have orders n and sqrt(n).

Check

For every fixed radius r, once each cycle length exceeds2r+1 all rooted radius-r neighborhoods are identical. Their total component sizes remain explicitly different.

Remaining obstruction

This prevents deriving the critical window from the Polya-point local limit alone. A growing-scale exploration argument must retain age and fixed-outdegree dependencies.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0152 Kernel criteria for critical inhomogeneous random-graph universality

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.5-4.6, PDF pp.222-223

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Characterize critical component universality for general inhomogeneous random-graph kernels, including the proposed role of the positive principal eigenfunction belonging to L3.

Scope

The general question and the eigenfunction criterion are grouped. Rank-one and finite-type theorems do not cover every irreducible unbounded kernel.

P0152-A001 | Kernel criteria for critical inhomogeneous random-graph universality structural reduction

Argument or result

L3 integrability of the principal eigenfunction does not itself give a spectral gap. Partition[0,1] into blocks of masses p_i tending to0, choose epsilon_i>0 tending to0, and let m=sum p_i epsilon_i. On blocks i,j set kappa=(1-epsilon_i)1{i=j}/p_i+epsilon_i epsilon_j/m. This symmetric positive kernel has row integral1, so its principal eigenfunction is the constant1 in L3. For the centered indicator of block i, the Rayleigh quotient tends to1 as epsilon_i tends to0.

Check

The row integral is1-epsilon_i+epsilon_i=1. The Dirichlet energy of the block indicator is at most p_i epsilon_i, while its centered L2 norm squared is p_i(1-p_i). Thus the spectral gap is at most epsilon_i/(1-p_i), tending to0. Positivity of the kernel makes the constant eigenfunction unique up to scale.

Remaining obstruction

The example is an obstruction to using an unstated spectral-gap consequence of the conjectured criterion. It is not asserted to refute random-graph universality; the component scaling for this explicit kernel remains to be analyzed.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0153 Vacant-set percolation threshold on general configuration models

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.8, PDF p.225

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Identify the critical time for percolation of the vacant set left after a random walk runs for order n time on a configuration model, including degree laws with diverging second moment.

Scope

Time normalization, the giant component, and initial distribution must match the source. The suggestion that the threshold vanishes for infinite second moment is a question, not an established consequence of ordinary percolation.

P0153-A001 | Vacant-set percolation threshold on general configuration models structural reduction

Argument or result

For simple random walk on a finite connected graph, stationary mass at v is d_v/sum_w d_w. This controls expected occupation counts in stationarity, but it does not identify the probability of ever visiting v: repeated returns can account for many occupation times. In particular P(visit v by time t)<=E[number of visits by time t], with no matching lower bound from this mean alone.

Check

The stationary law follows from detailed balance pi_u/d_u=pi_v/d_v on each edge. The displayed visit bound is Markov's inequality for the nonnegative integer occupation count.

Remaining obstruction

Degree-biased occupation cannot be substituted for independent vertex deletion. One needs escape probabilities and joint avoidance estimates to determine the critical remaining branching structure.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0154 Metric scaling of critical random-walk vacant components

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.9, PDF p.225

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Obtain metric-space scaling limits for the largest critical vacant-set components of general configuration models, including heavy-tailed degree sequences.

Scope

Random-regular vacant-set metric limits are already established. The general degree law and possible suppression of high-degree hubs form the unresolved extension.

P0154-A001 | Metric scaling of critical random-walk vacant components structural reduction

Argument or result

A high degree need not imply large escape capacity. Attach K leaves to a vertex v that has just one additional edge leading into a fixed transient infinite tree. Every walk step from v into a leaf returns to v with probability1; only the remaining edge can contribute to escape. Thus d_v*P_v(no return after time0) is at most1, uniformly in K.

Check

Condition on the first step from v. Each leaf contributes0 to the sum of neighbor escape probabilities, and the one transient branch contributes at most1. This is the usual unit-conductance singleton capacity calculation.

Remaining obstruction

The example is not a configuration-model limit theorem. It shows why deleting hubs based only on degree can fail: escape geometry, not stationary occupation alone, controls avoidance.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0155 Erdos-Renyi component-size limits for high-dimensional tori

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.10, PDF p.226

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For nearest-neighbor percolation on growing d-dimensional tori with fixed d>6 in the critical window around p_c(Z^d), prove convergence of the ordered component sizes to the appropriate shifted and rescaled multiplicative-coalescent law.

Scope

Correct window centering and constants are part of the source question. Results for a growing-dimensional hypercube, or order-of-magnitude estimates, are not automatically this fixed-d torus theorem.

P0155-A001 | Erdos-Renyi component-size limits for high-dimensional tori structural reduction

Argument or result

Two-sided order bounds cannot identify a scaling law: random variables X_n=n^(2/3)A_n, where A_n alternates between two different bounded positive distributions, always have the predicted magnitude and are tight after scaling, but have different subsequential limits.

Check

Take A_n identically1 for even n and identically2 for odd n. The normalized variables have no weak limit despite uniform upper and lower deterministic bounds.

Remaining obstruction

For the percolation model one must identify the joint exploration process and the window shift, rather than only establish component-size tightness. No such identification is proved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0156 Configuration-model minimum spanning-tree metric limits

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.11-4.12, PDF p.231

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Identify and prove the metric-measure scaling limits of minimum spanning trees on configuration models with minimum limiting degree at least3, across finite-third-moment and heavy-tailed degree regimes.

Scope

The two source regimes are grouped. The random3-regular case is resolved by Addario-Berry and Sen1810.03802; it is excluded from the open remainder. The law of iid continuous edge weights is not an extra family of distinct problems.

P0156-A001 | Configuration-model minimum spanning-tree metric limits structural reduction

Argument or result

For a fixed finite graph with distinct edge weights, applying any strictly increasing function to every weight leaves the minimum spanning tree unchanged. Kruskal's algorithm inspects edges in the same order and makes the same cycle decisions. Consequently, with iid continuous weights, the distribution of the unweighted MST geometry depends only on a uniform random edge ordering.

Check

An iid continuous law gives no ties almost surely and every ordering has probability1/|E|!. The greedy cycle criterion depends only on the ordering, which independently verifies distributional invariance.

Remaining obstruction

The remaining difficulty is the graph ensemble and the global near-critical component-merger geometry. Rank invariance does not establish tightness, scaling exponents, or the limiting tree.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0157 Two competing largest-component scales in barely subcritical configuration models

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem4.13, PDF p.234

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Under convergence of third degree moments, prove that the largest barely subcritical configuration-model component has order max{d_max/epsilon_n, epsilon_n^(-2)log(n epsilon_n^3)} in the regime specified by the source.

Scope

The asymptotic regime must have epsilon_n tending to0 with n epsilon_n^3 large; otherwise the displayed logarithm is not the intended barely-subcritical scale. Infinite-third-moment results are a different regime.

P0157-A001 | Two competing largest-component scales in barely subcritical configuration models structural reduction

Argument or result

A branching-process approximation already explains the high-degree candidate scale. If the root has d children and every later vertex has mean offspring1-epsilon, then expected total progeny is1+d/epsilon: each child starts a subcritical tree of expected size1/epsilon.

Check

Generation expectations are d(1-epsilon)^(k-1) for k>=1; Tonelli and the geometric series yield1+d/epsilon.

Remaining obstruction

Expectation at a high-degree root is not a high-probability largest-component estimate. Uniform concentration, collision errors, and the competing extreme size of many ordinary components remain to be controlled.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0158 Ordered-phase susceptibility exponent on random trees and graphs

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem5.1, PDF pp.286-287

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove that the susceptibility critical exponent gamma_prime on the ordered side of the locally tree-like ferromagnetic Ising transition equals1 under the source's degree assumptions.

Scope

The high-temperature exponent gamma=1 and the random-regular ordered-phase case are already known. The source seeks the missing upper bound for general degree laws with a nontrivial finite critical inverse temperature.

P0158-A001 | Ordered-phase susceptibility exponent on random trees and graphs structural reduction

Argument or result

At zero cavity field, the expected generation-k contribution to susceptibility is proportional to nu^(k-1)*(tanh beta)^k, yielding a geometric series only when nu*tanh beta<1. Above criticality that same bound diverges, so it cannot prove the desired finite susceptibility asymptotic in the plus state.

Check

The ratio of consecutive generation terms is nu*tanh beta, which equals1 at criticality and exceeds1 on the ordered side. Nonzero random cavity fields supply extra attenuation factors in the exact source formula.

Remaining obstruction

A sharp lower bound on that attenuation, averaged along the size-biased spine and uniformly near criticality, is missing. Analytically continuing the high-temperature geometric sum would give an invalid negative denominator.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0159 Quenched non-Gaussian critical spin limits on heterogeneous random graphs

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem5.2, PDF p.305

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the critical non-classical total-spin limit theorem from annealed generalized random graphs to quenched configuration models or quenched generalized random graphs.

Scope

Quenched means fixing the sampled graph before taking its spin law. Matching critical exponents or annealed pressure does not establish a quenched distributional limit.

P0159-A001 | Quenched non-Gaussian critical spin limits on heterogeneous random graphs structural reduction

Argument or result

For environment G and partition functions Z_G(h), the averaged quenched moment generating function is E[Z_G(h)/Z_G(0)], whereas the annealed one is E Z_G(h)/E Z_G(0). These are generally different. If two equally likely environments have(Z0,Zh)=(1,1) and(2,2*cosh h), the expressions are(1+cosh h)/2 and(1+2*cosh h)/3.

Check

The two finite Boltzmann systems can be realized by one zero-spin state versus two states with spins plus/minus1. The formulas then follow directly from normalization. This is a normalization example, not a graph-model counterexample.

Remaining obstruction

The annealed integral identity weights graph environments by their partition functions. Removing this tilt while preserving critical fluctuations is the unresolved step.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0160 Fixed-temperature metastability for general configuration models

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem5.4, PDF p.313

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove metastable behavior of low-temperature ferromagnetic Ising dynamics on configuration models as n tends to infinity with inverse temperature fixed and large.

Scope

The source contrasts this with taking temperature to0 at fixed n. Later random-regular metastability results are known; the full general configuration-model extension is retained.

P0160-A001 | Fixed-temperature metastability for general configuration models structural reduction

Argument or result

A deterministic expansion condition gives an energy barrier. If every vertex set S of size at most n/2 has at least h|S| boundary edges, and the Hamiltonian is-J sum_edges sigma_u sigma_v-B sum_v sigma_v, then changing S from minus to plus costs2J|boundary S|-2B|S|, at least2(Jh-B)|S|. Every single-spin path from all minus to all plus crosses a set of size floor(n/2).

Check

Each boundary edge changes its interaction contribution by2J and each flipped spin changes its field contribution by-2B. The number of plus spins changes by1 per step, forcing the intermediate set size.

Remaining obstruction

A positive energy barrier is not by itself a fixed-temperature exit-time theorem because exponentially many configurations contribute entropy. One must compare beta times the barrier with entropy and control the metastable basin, with expansion estimates valid for the sampled graph.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0160-A002 | Energy plus entropy conductance bound

Argument or result

Let G have odd n and edge expansion |boundary S|>=h|S| for all |S|<=n/2. Use H(sigma)=-J sum_edges sigma_u sigma_v-B sum_v sigma_v with J>0,B>=0,Jh>B, and Gibbs law pi proportional to exp(-beta H). Consider any reversible single-spin continuous-time chain with each individual spin-flip rate at most1. Put A={number of plus spins< n/2}. Global spin reversal maps A bijectively onto its complement and does not decrease Gibbs weight, so pi(A)<=1/2. An A-to-complement jump must start with(n-1)/2 plus spins, whose energy exceeds the all-minus energy by at least(Jh-B)(n-1). There are at most2^n such states and total exit rate at most n per state. Since pi(A) is at least the all-minus weight, the conductance Phi=Q(A,A-complement)/pi(A) is at most n*2^n*exp[-beta(Jh-B)(n-1)]. Starting with pi conditioned on A, the time-s law is bounded above by pi/pi(A), by Markov-kernel positivity and stationarity. Consequently the expected number of A-to-complement jumps by time t is at most t*Phi. The probability of still being in A at time t is at least1-t*Phi, giving total variation distance at least1/2-t*Phi. Convexity of total variation implies the same lower bound for the worst starting state. Therefore t_mix(1/4)>=1/(4*Phi), at least exp[beta(Jh-B)(n-1)]/(4*n*2^n). This is exponential in n when beta(Jh-B)>log2. The calculation accounts explicitly for entropy. It is a deterministic expansion-based mixing lower bound; it does not prove a metastable exit-time distribution, or expansion for every configuration-model degree law.

Check

Boundary-layer counting and stationary domination give separate energy and time estimates.

Remaining obstruction

No general random-graph expansion theorem or metastable exit-time distribution is proved.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0161 Potts free energy for general configuration-model degrees

Source: Van der Hofstad: Stochastic processes on random graphs, Saint-Flour2017 | Open Problem5.5, PDF p.314

https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the ferromagnetic Potts Bethe free-energy and phase-diagram results from regular locally tree-like graphs to general configuration-model degree laws.

Scope

Can and van der Hofstad2503.17636 establish additional regular-graph and parameter-region results. A lower Bethe bound is not equality throughout the general heterogeneous phase diagram.

P0161-A001 | Potts free energy for general configuration-model degrees structural reduction

Argument or result

For a finite forest at zero field, Z=q^(number of components)*(exp(beta)+q-1)^(number of edges), by successively summing leaf colors. Applying that formula to a d-regular locally tree-like graph as if it were a forest predicts pressure log q+(d/2)log[(exp(beta)+q-1)/q]. For d>2 and large beta this is below beta*d/2, while the q monochromatic states alone imply pressure at least beta*d/2+log(q)/n.

Check

Leaf elimination gives one factor exp(beta)+q-1 per edge and q for each remaining component root. In a connected graph the monochromatic states have weight exp(beta*|E|), so their contribution is an exact lower bound.

Remaining obstruction

Local tree neighborhoods do not erase the linear number of global cycle constraints. Correct Bethe fixed-point selection, beyond the symmetric forest expression, remains essential.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0162 Fourier entropy versus total influence

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.5, Fourier Entropy-Influence Conjecture

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Bound the Shannon entropy of the squared Fourier coefficients of every Boolean function by a universal constant times its total influence.

Scope

The weaker min-entropy formulation is grouped here. Gonzalez,MacManus,Pereyra2606.00246 still calls the full statement a conjecture and proves additional classes.

P0162-A001 | Fourier entropy versus total influence exact calculation or structural obstruction

Argument or result

Discarding the Boolean constraint destroys the claim: the probability measure uniform on the n singleton subsets has entropy log2(n) but mean subset size1. It cannot be the squared spectrum of a Boolean function when n>=2.

Check

If all Fourier mass were on singletons, f=sum_i a_i*x_i with a_i=plus/minus1/sqrt(n). In f^2 the coefficient of x_i*x_j is2*a_i*a_j, nonzero for distinct i,j, whereas Boolean f requires f^2=1. This independently exhibits the violated Fourier convolution identity.

Remaining obstruction

Entropy versus mean cardinality alone is insufficient. Any proof must use the nonlinear identities imposed by f^2=1; no dimension-free inequality exploiting all of them is obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0163 Polynomial Fourier concentration for DNF formulas

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.4, Mansour Conjecture

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a size-s DNF, concentrate all but epsilon of its squared Fourier mass on at most s raised to O(log(1/epsilon)) coefficients.

Scope

The weaker polynomial-in-s formulation is grouped. Lecomte-Tan2109.04525 improve read-dependent bounds, not the full assertion.

P0163-A001 | Polynomial Fourier concentration for DNF formulas exact calculation or structural obstruction

Argument or result

Let C_j be the indicator of clause j, and encode the DNF by f=1-2*product_j(1-C_j). Each consistent conjunction has Fourier l1 norm1, and any product of conjunction indicators is either another conjunction or zero. Inclusion-exclusion gives L=sum_S |fhat(S)|<=2^(s+1)-1. Keeping coefficients of magnitude at least epsilon/L leaves squared tail at most epsilon and uses at most L^2/epsilon coefficients.

Check

For discarded coefficients, sum a_S^2 <=(epsilon/L)*sum |a_S|<=epsilon. For retained coefficients, their number times epsilon/L is at most L. Thus the explicit exponential bound is valid.

Remaining obstruction

This handles every DNF but is exponentially larger than the requested polynomial bound. The expansion ignores cancellations among overlapping clauses.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0164 Gaussian lower bound for noise stability of unbiased thresholds

Source: Kamath,Kumar,Manurangsi: When Majority Fails,2026 | Section1.4, Conjecture1; O-Donnell2012 p.5 weaker consequences

https://arxiv.org/pdf/2604.06590

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For an unbiased linear threshold f and rho in(0,1), prove Stab_rho(f)>=(2/pi)*arcsin(rho).

Scope

This is the authors refined conjecture, not the false finite-majority comparison. The degree-one Fourier lower bound2/pi and small-noise consequence are grouped as parts of this family.

P0164-A001 | Gaussian lower bound for noise stability of unbiased thresholds exact calculation or structural obstruction

Argument or result

For f=sign(S), S=sum_i w_i X_i with sum w_i^2=1 and no zero ties, Holder gives E|S|>=(E S^2)^(3/2)/(E S^4)^(1/2)>=1/sqrt(3), since E S^4=3-2*sum w_i^4<=3. Cauchy-Schwarz applied to E[fS]=sum_i w_i*fhat(i) yields sum_i fhat(i)^2>=1/3. Therefore Stab_rho(f)>=rho/3.

Check

Interpolate L2 between L1 and L4: 1=||S||2 <=||S||1^(1/3)||S||4^(2/3). Fourier noise stability is sum_S rho^|S|*fhat(S)^2, with nonnegative terms for rho>0. This verifies the bound without assuming regular weights.

Remaining obstruction

The elementary bound is weaker even than the established robust Khintchine bounds and far below the proposed arcsine curve. A central limit argument gives the arcsine law only when all normalized weights vanish; it does not compare arbitrary nonregular thresholds.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0165 Optimal odd functions for many-player correlation distillation

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.6, Optimality of Majorities for Non-Interactive Correlation Distillation

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

When r independent noisy copies of a common uniform bit string use the same odd Boolean function, is the agreement probability maximized by majority on some odd subset of coordinates?

Scope

The number of players and the noise parameter are one optimization problem. This differs from two-party erasure distillation and from minimizing threshold stability.

P0165-A001 | Optimal odd functions for many-player correlation distillation exact calculation or structural obstruction

Argument or result

Write rho=1-2*epsilon and m(x)=T_rho f(x). Conditional independence gives agreement probability E[((1+m)/2)^r+((1-m)/2)^r]. For r=2 this is(1+E m^2)/2, and for r=3 it is(1+3E m^2)/4. Since odd f is unbiased, Parseval gives E m^2=sum_nonemptyS rho^(2|S|)*fhat(S)^2<=rho^2. Dictators attain both bounds.

Check

Expand the two binomials; odd powers cancel. A dictator has m=rho*x_1 and equality in the Fourier bound. For r=4 the expression becomes(1+6E m^2+E m^4)/8, displaying a genuinely additional moment.

Remaining obstruction

The exact r=2,3 calculation does not optimize the higher moments jointly for r>=4. No argument shows that a subset-majority maximizes all those combinations.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0166 Influential coordinates of bounded low-degree polynomials

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.7, Aaronson-Ambainis Conjecture

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For f on the uniform discrete cube taking values in[-1,1] and having degree at most d, prove some L2 influence is at least(Var(f)/d)^C for a universal C.

Scope

Bhattacharya2402.13952 and its March2026 revision concern random restrictions. That does not settle every original polynomial.

P0166-A001 | Influential coordinates of bounded low-degree polynomials exact calculation or structural obstruction

Argument or result

For d=1, write f=a0+sum_i a_i*x_i. Evaluating at coordinate signs that align with a_i and at their opposites gives |a0|+sum_i|a_i|<=1. Hence Var(f)=sum_i a_i^2<=max_i|a_i|, so max_i Inf_i(f)=max_i a_i^2>=Var(f)^2.

Check

The two extreme cube values are a0 plus/minus sum|a_i|. L2 influence for an affine polynomial is exactly a_i^2. The proof is dimension-free and uses the full range constraint.

Remaining obstruction

In higher degree, cancellation among monomials prevents this direct l1-coefficient bound. The argument gives no polynomial-in-d control of the largest influence.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0167 Minimum correlation of symmetric Gaussian sets

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.8, Symmetric Gaussian Problem

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For correlated standard Gaussian vectors with prescribed rho and set measures mu,nu, minimize P(X in A,Y in B) under central symmetry A=-A.

Scope

The proposed centered-ball optimizer is not the general optimization question. Heilman1511.00382 gives negative results for ball/complement maximizers in dimension at least2; this does not identify every optimum.

P0167-A001 | Minimum correlation of symmetric Gaussian sets exact calculation or structural obstruction

Argument or result

For fixed A, put h=T_rho 1_A. The objective is integral_B h dgamma. Its minimum over sets of measure nu is attained by a lower-level set of h, with a suitable fraction of a level set if necessary. Since A is symmetric, h is even, so B can also be chosen symmetric.

Check

If B selects positive measure at a higher value of h while omitting equal measure at a lower value, exchanging the two decreases the integral. Applying the threshold at the nu-quantile proves the characterization. Gaussian measure is nonatomic, including after passing to antipodal pairs.

Remaining obstruction

This eliminates one set but leaves a nonlinear optimization over A. Symmetry alone does not make its smoothed indicator radial or force a ball.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0168 Noise-optimal equal-volume Gaussian partitions

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.9, Standard Simplex Conjecture

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove that equal-volume regular-simplex partitions maximize Gaussian noise stability for positive correlation and minimize it for negative correlation, for every admissible number of cells.

Scope

The signs of correlation and cell numbers form one source conjecture. Heilman2608.00333 claims a substantial three-cell correlation range; it does not cover all cell counts or every positive correlation.

P0168-A001 | Noise-optimal equal-volume Gaussian partitions exact calculation or structural obstruction

Argument or result

For four cells in dimension at least3, partition by the signs of the first two coordinates. Its stability is[1/2+arcsin(rho)/pi]^2, because the two Gaussian coordinate pairs are independent and each pair has sign-agreement probability1/2+arcsin(rho)/pi.

Check

Represent a correlated pair as(G1,rho*G1+sqrt(1-rho^2)*G2). Rotational symmetry in the plane gives sign-disagreement angle arccos(rho)/pi. The product of the two coordinate agreement events supplies the displayed four-cell benchmark.

Remaining obstruction

This computes a non-simplex competitor exactly but does not order it against every partition. At rho=0 every equal-volume partition has stability1/q, so independence is not a uniqueness argument.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0169 Linear Fourier mass versus Boolean degree

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.9, Linear Coefficients versus Total Degree

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Bound the sum of degree-one Fourier coefficients of a degree-d Boolean function by sqrt(d), or by the sharper majority-on-d-bits value proposed in the source.

Scope

The stronger and weaker proposed constants are grouped. Jha1611.01029 and Wang2020/002 discuss partial cases and reformulations; neither title is treated as a complete resolution.

P0169-A001 | Linear Fourier mass versus Boolean degree exact calculation or structural obstruction

Argument or result

Under independent input bits with common mean t, F(t)=E_t f is a polynomial of degree at most d bounded by1 in absolute value for t in[-1,1]. Its derivative at0 is sum_i fhat(i). However, boundedness and degree alone cannot force the desired sqrt(d) bound: for odd d the Chebyshev polynomial T_d is bounded by1 on this interval and |T_d prime(0)|=d.

Check

The biased expectation of the character on S is t^|S|. Also T_d(t)=cos(d arccos t), whose derivative at0 has magnitude d for odd d.

Remaining obstruction

A proof through one-variable biased expectations must exploit additional constraints inherited from a Boolean function. Chebyshev is an obstruction to the relaxed method, not a Boolean counterexample.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0170 Optimal limited independence for Gaussian polynomial thresholds

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.9, k-wise Independence for PTFs

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the smallest k(d,epsilon) for which every k-wise independent standard-Gaussian vector fools every degree-d polynomial threshold to error epsilon.

Scope

The joint distribution of every k coordinates must be product Gaussian. Matching finitely many mixed moments alone is a weaker property and is not substituted for this assumption.

P0170-A001 | Optimal limited independence for Gaussian polynomial thresholds exact calculation or structural obstruction

Argument or result

Take n=k+1 signs uniformly distributed subject to product_i sign_i=1, independent of n independent absolute standard Gaussians R_i. Then X_i=sign_i*R_i has k-wise independent standard-Gaussian coordinates. Yet product_i X_i>0 almost surely, while the same product for fully independent Gaussians is positive with probability1/2.

Check

Every assignment of any k signs has exactly one extension satisfying the parity constraint, so its probability is2^-k. Independent magnitudes then yield independent Gaussian marginals on that subset. The distinguishing polynomial has degree k+1.

Remaining obstruction

For epsilon<1/2 this proves the necessary bound k>=d. It is weaker than known quantitative lower bounds and supplies no sharp dependence on epsilon.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0171 Polynomial small bias sufficient to fool DNF events

Source: O Donnell: Open problems in analysis of Boolean functions | PDF pp.9-10, epsilon-biased Sets for DNFs

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For each fixed error delta>0, determine whether every distribution whose nonconstant parity biases are at most s^(-C_delta) fools every size-s DNF to error delta.

Scope

The2025 Generalised Linial-Nisan counterexample controls marginals of bounded-size coordinate sets; its theorem alone does not assert small bias for every parity. It is not treated as a resolution of this different source statement.

P0171-A001 | Polynomial small bias sufficient to fool DNF events exact calculation or structural obstruction

Argument or result

For a single conjunction C of k literals, its Fourier coefficients have magnitude2^-k on subsets of those k coordinates. An epsilon-biased distribution therefore changes E C by at most epsilon*(1-2^-k). Inclusion-exclusion for a union of s clauses gives the valid but crude error at most(2^s-1)*epsilon, since each nonempty intersection of consistent clauses is another conjunction.

Check

The constant coefficient contributes no expectation difference, while all other coefficients are paired with parity biases of magnitude at most epsilon. Expanding the union into its2^s-1 nonempty clause intersections proves the stated global bound without assuming independence of clauses.

Remaining obstruction

To obtain polynomial rather than exponential bias requirements, one needs a substantially more economical approximation or cancellation argument. Small errors for individual clause probabilities do not by themselves control their union.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0172 Sharp erasure-distillation bound from the Gaussian limit

Source: Kamath,Kumar,Manurangsi: When Majority Fails,2026 | Section1.4, Conjecture2

https://arxiv.org/pdf/2604.06590

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For unbiased Boolean f and coordinate-retention probability p in(0,1/2), prove E|E[f(X) given the retained coordinates]|<=(2/pi)*arcsin(sqrt(p)).

Scope

The false finite-majority comparator is excluded. This is the2026 authors refined conjecture. Requiring the optimizing second output to be unbiased is a distinct constraint; the source formulation here uses the absolute conditional expectation.

P0172-A001 | Sharp erasure-distillation bound from the Gaussian limit exact calculation or structural obstruction

Argument or result

Identify the conditional expectation with the multilinear extension f(Z), where each Z_i is0 with probability1-p and otherwise a fair sign. Orthogonality gives E f(Z)^2=sum_S p^|S|*fhat(S)^2<=p for unbiased f. Thus Cauchy-Schwarz yields E|f(Z)|<=sqrt(p). A parity on m coordinates instead has exact value p^m.

Check

If S and T differ, the expectation of Z_S Z_T vanishes because a differing coordinate has mean0. For S=T it equals p^|S|. Parseval and the absent constant coefficient give the bound. A parity survives only when all m relevant coordinates are retained.

Remaining obstruction

The L2 estimate loses the proposed arcsine constant. Conditional distributions of the smoothed Boolean function, beyond variance alone, must be controlled.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0173 Asymptotic influence bounds for polynomial thresholds

Source: O Donnell: Open problems in analysis of Boolean functions | PDF p.3, weaker Gotsman-Linial formulations

https://arxiv.org/pdf/1204.6447

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove total influence at most C*d*sqrt(n), or at least C_d*sqrt(n), for every degree-d polynomial threshold function on n uniform bits.

Scope

The exact symmetric-maximizer conjecture is false, with Chapman2108.02288 giving later counterexamples. The weaker asymptotic bounds explicitly posed in the source are grouped here.

P0173-A001 | Asymptotic influence bounds for polynomial thresholds exact calculation or structural obstruction

Argument or result

The desired C*d*sqrt(n) bound holds when the polynomial is a product of d affine linear forms with no cube zeros. Indeed its sign is a product of d threshold functions. If that product changes on a coordinate flip, at least one factor changes, so its influence is at most the sum of the factor influences. For a threshold g, unateness gives Inf_i(g)=|ghat(i)| and hence I(g)<=sqrt(n)*sqrt(sum_i ghat(i)^2)<=sqrt(n).

Check

For a monotone coordinate of a Boolean function, its discrete derivative is either0 or1 after the conventional half-difference, so its expectation equals its flip probability. Cauchy-Schwarz and Parseval then give the threshold bound. A union bound gives the product estimate.

Remaining obstruction

A general multivariate polynomial need not factor into real affine forms. The argument supplies a genuine subclass result, not a reduction for arbitrary polynomial thresholds.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0174 Sharp support-diameter threshold for free Brownian unimodality

Source: Hasebe,Ueda: Large time unimodality for classical and free Brownian motions | Problem 3.3, arXiv PDF p.6; journal p.359

https://arxiv.org/pdf/1710.08240

Historical status: source_says_open | Current status: unverified | Result: candidate_complete_proof

Question

Find the optimal universal C such that every compactly supported law mu becomes unimodal after free convolution with the variance-t semicircle law for all t >= C times the square of its support diameter.

Scope

The source prints 2<=C<=4, but its own symmetric-Bernoulli threshold t=4 and diameter 2 imply only C>=1. The candidate proof derives both bounds independently. Manuscript header 2021 differs from the arXiv revision date 2018.

P0174-A001 | Centered fourth-moment diameter inequality and reweighted Cauchy-profile critical points

Argument or result

Candidate sharp answer C=1. For any centered law Q of support diameter D, with a=E X^2 and b=E X^4, a<=D^2/4 and b+3a^2<=D^2*a. Reweighting mu by ((x-u)^2+R^2)^(-2) at a stationary Cauchy profile gives y=R^2<=3a when the second derivative is nonnegative, and 1/F=(b+2ay+y^2)/(a+y)<=D^2. Equality forces symmetric endpoint Bernoulli and a strict fourth-order maximum. Thus no local minimum has F<=1/D^2; each level 1/t for t>=D^2 has at most two roots. Biane parametrization and the source level-set lemma imply unimodality. Symmetric Bernoulli gives sharpness.

Check

Second analytic route differentiates v_t directly at critical points. For symmetric Bernoulli on {-c,c}, v_t second derivative at 0 equals (4*c^2-t)/(t*sqrt(t-c^2)), so t<4*c^2=D^2 gives a positive density local minimum. The centered moment inequality follows independently from E[(X-Xprime)^4]<=D^2*E[(X-Xprime)^2]. These are same-author checks, not independent review.

Remaining obstruction

No mathematical gap currently identified in the full written candidate. Later-literature novelty is unverified and there is no independent proof review. Check the reweighting normalization, Jensen direction, equality case, and exact source density assumptions before promotion.

Verification: author_checked_complete_candidate | Independent review: none | Full question solved: False

P0175 Symmetric unimodality under free addition

Source: Hasebe-Ueda: Large time unimodality, published version | Conjecture 3.5

https://alea.math.cnrs.fr/articles/v15/15-15.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is the free additive convolution of two symmetric unimodal probability laws unimodal?

Scope

The older AIM free-analysis question is the same family and is not counted again. A semicircle factor and some freely infinitely divisible subclasses are established cases.

P0175-A001 | Symmetric unimodality under free addition structural calculation

Argument or result

The classical proof by mixing symmetric unimodal components does not transfer by linearity: free convolution is not affine even on symmetric unimodal laws. Let mu=(delta_0+S(0,2))/2 and nu=S(0,1). The sixth moment of mu free-added to nu is67, whereas the mixture of the two component free convolutions has sixth moment70.

Check

For mu, m2=1,m4=4,m6=20, so its free cumulants are k2=1,k4=2,k6=3. Adding nu changes k2 to2 only, giving m6=3+6*2*2+5*2^3=67. The affine mixture would be(S(0,1)+S(0,3))/2, with m6=(5+135)/2=70. Both mixture components and mu are symmetric unimodal.

Remaining obstruction

This disproves a proposed proof mechanism, not the unimodality conjecture. The actual free convolution in this example is covered by the known semicircle-factor theorem.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0176 Existence of nontrivial freely strongly unimodal laws

Source: Hasebe-Ueda: Large time unimodality, published version | Problem 3.11

https://alea.math.cnrs.fr/articles/v15/15-15.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does a non-point-mass law exist whose free additive convolution with every unimodal law is unimodal?

Scope

The source already excludes all nondegenerate finite-variance laws. The remaining possible examples must have infinite variance.

P0176-A001 | Existence of nontrivial freely strongly unimodal laws structural calculation

Argument or result

An explicit infinite-variance candidate also fails: the standard Cauchy law. Let nu=(1/20)*delta_0+(19/20)*Uniform[0,1000], a unimodal law with mode0. Cauchy free convolution equals classical convolution, whose density derivative, multiplied by pi, is -x/[10*(1+x^2)^2]+19/20000*[1/(1+x^2)-1/(1+(x-1000)^2)]. Its signs at0,1,250,1000 are respectively positive,negative,positive,negative, so the density has at least two peaks.

Check

At1 the negative term is-1/40 and the positive term is less than19/40000. At250, writing A=62501 and B=562501, the derivative is -25/A^2+475/(A*B), positive because19*A>B. At1000 both displayed contributions are negative. The initial law has a single atom at its mode and a constant density to its right.

Remaining obstruction

This gives a concrete Cauchy counterexample, consistent with the known strong-unimodality criterion, but does not exclude every infinite-variance law.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0177 Location of Brownian modes at large time

Source: Hasebe-Ueda: Large time unimodality, published version | Problem 4.6

https://alea.math.cnrs.fr/articles/v15/15-15.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Estimate the mode of classical Brownian motion started from a law with a finite exponential quadratic moment, and determine analogous estimates for free Brownian motion.

Scope

The source gives a growing interval of radius sqrt(t)/2 once unimodality holds. Classical and free versions of this mode-location request are grouped.

P0177-A001 | Location of Brownian modes at large time structural calculation

Argument or result

For the classical mode, let m be the initial mean and mu_k=E[(X-m)^k]. The source exponential-moment assumption yields the asymptotic expansion mode(t)=m-mu_3/(2*t)+(mu_5-6*mu_2*mu_3)/(8*t^2)+O(t^-3). In particular the mode tends to the initial mean, and the leading displacement has sign opposite the third central moment.

Check

Set s=1/t and H(h,s)=E[(X-m-h)*exp(-s*(X-m-h)^2/2)]. The exponential quadratic moment makes H analytic near(0,0), with H_h=-1 there. The implicit function theorem gives h=a*s+b*s^2+O(s^3). Expanding H gives a=-mu_3/2 and b=mu_5/8-3*mu_2*mu_3/4. The associated density stationary point is a strict local maximum since H_h remains negative; the source large-time unimodality identifies it with the unique global mode.

Remaining obstruction

This is an author-checked asymptotic partial answer, with no novelty claim. It does not provide a uniform explicit finite-time error constant or solve the free-Brownian half.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0178 Critical moment condition for eventual stable-process unimodality

Source: Hasebe-Ueda: Large time unimodality, published version | Conjecture 5.5, journal p.373

https://alea.math.cnrs.fr/articles/v15/15-15.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a classical or free strictly alpha-stable process, alpha in(0,2), does a finite absolute(2+alpha)-moment of the initial law ensure eventual unimodality under the corresponding convolution?

Scope

Use the published wording: classical convolution with classical stable laws, free convolution with free stable laws. The archived preprint wording differs. All stability indices are one source conjecture; the Cauchy case is already established.

P0178-A001 | Critical moment condition for eventual stable-process unimodality structural calculation

Argument or result

A separated-atom counterexample of the source type cannot retain the critical moment while making its usual tail mechanism arbitrarily strong. If mu=sum_k w_k*delta_(a_k) and sum_k w_k*|a_k|^(2+alpha)<infinity, then w_k*|a_k|^(2+alpha) tends to0. Thus it cannot diverge along a subsequence, which is the condition used to make an isolated atom dominate the far-tail derivative of the main stable density.

Check

Every term of a convergent nonnegative series tends to0. In the Cauchy case alpha=1, the derivative of the bulk kernel at a_k has order |a_k|^-3, while the local atom contributes order w_k at any fixed smoothing time; their ratio is of order w_k*|a_k|^3. This matches the critical exponent in the exact known construction.

Remaining obstruction

Failure of that counterexample mechanism is not a uniform derivative estimate for arbitrary initial laws. Asymmetric stable laws and the free cases still require global control.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0179 Uniform permutation representations of the Gieseking group

Source: The strong convergence phenomenon,2025 survey | Section 6.1, p.64

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Do uniformly sampled permutation pairs satisfying U^2*V^2=V*U converge strongly, after removing the common constant vector, to the regular representation of the Gieseking group?

Scope

Surface groups are already covered by the source. The later2602.11905 free-product theorem concerns a different relation and does not settle this hyperbolic3-manifold group.

P0179-A001 | Uniform permutation representations of the Gieseking group structural calculation

Argument or result

Independent-permutation moment formulas cannot be inserted unchanged. Taking determinants in the relation forces sign(U)=sign(V). More fundamentally, the relator U^2*V^2*U^-1*V^-1 is identically I, so its normalized trace is1, whereas the corresponding nontrivial reduced free-group word has trace0 in the free regular representation.

Check

The determinant equation is1=sign(V)*sign(U). The relator identity follows by multiplying U^2*V^2=VU on the right by U^-1*V^-1. Removing the constant vector leaves the identity on the remaining space, so the trace discrepancy persists.

Remaining obstruction

The correct nonfree group trace respects the relator, but no asymptotic enumeration or outlier estimate for uniform constrained pairs is obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0180 Optimal spectra of random symmetric-group Cayley graphs

Source: The strong convergence phenomenon,2025 survey | Section 6.2, p.65

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For a fixed number of uniform random generators of S_N, do all nontrivial eigenvalues apart from the sign representation approach the free-group spectral bound? Develop the associated high-dimensional representation convergence.

Scope

The weaker expansion question and the stronger arbitrary-irrep formulation are grouped with the optimal-gap family. The trivial and sign representations must both be removed.

P0180-A001 | Optimal spectra of random symmetric-group Cayley graphs structural calculation

Argument or result

The sign obstruction has an exact positive probability: when all r generator signs agree, the corresponding adjacency eigenvalue has absolute value2r, above2*sqrt(2r-1) for r>=2. This event has probability2^(1-r). After removing it, pointwise convergence for each selected irrep still cannot justify the supremum over all irreps: in a triangular array with one uniformly chosen bad coordinate per row, every coordinate is bad with probability1/m_N tending to0, yet the row maximum is bad surely.

Check

The sign variables are independent fair signs for uniform permutations. The array example verifies the quantifier failure exactly and indicates the need for uniform probability bounds, rather than just increasing the range of an individual representation theorem.

Remaining obstruction

No uniform tail estimate over the high-dimensional irreducible representations is proved.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0181 Fixed-degree random Cayley graphs with optimal spectral gap

Source: The strong convergence phenomenon,2025 survey | Section 6.2, pp.65-66

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Find finite-group sequences whose fixed-number uniform random Cayley generators have asymptotically optimal nontrivial spectral gap, particularly finite special linear groups.

Scope

Different field and rank regimes are grouped; growing-number-of-generator theorems do not answer the fixed-degree question.

P0181-A001 | Fixed-degree random Cayley graphs with optimal spectral gap structural calculation

Argument or result

Abelian groups cannot supply even a uniform fixed-degree gap in a basic test case. For arbitrary r generators a_i of Z/pZ, divide the r-dimensional phase cube into M^r boxes with M^r<p. Two of the p points k*(a_1,...,a_r)/p fall in one box. Their nonzero difference k gives a nontrivial character for which all generator phases have distance at most2*pi/M from zero. Hence its adjacency eigenvalue is at least2r-4*pi^2*r/M^2.

Check

Use cos(x)>=1-x^2/2 in the eigenvalue2*sum_i cos(2*pi*k*a_i/p). Taking M tending to infinity at order p^(1/r) forces the gap to0, regardless of how the generators were chosen.

Remaining obstruction

The argument identifies a structural obstruction to abelian candidates but supplies no optimal bound for the nonabelian special-linear families.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0182 Strong convergence of high-dimensional representations of a fixed compact group

Source: The strong convergence phenomenon,2025 survey | Section 6.3, p.66

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For independent Haar elements of a fixed compact semisimple group, do their growing irreducible representations converge strongly to free Haar generators, starting with symmetric powers of SU(2)?

Scope

The source already establishes weak convergence in broad generality. The SU(2) test case and broader fixed-group question are one family.

P0182-A001 | Strong convergence of high-dimensional representations of a fixed compact group structural calculation

Argument or result

The one-generator SU(2) case has an elementary complete answer. A Haar element has eigenvalues exp(plus/minus i*theta), with theta/pi irrational almost surely. Its N-th symmetric power has eigenvalues exp(i*(N-2k)*theta),0<=k<=N. These become dense on the unit circle, so every Laurent-polynomial norm converges to the supremum on the circle, the regular Z norm.

Check

Irrational rotation makes the first N+1 multiples of2*theta asymptotically dense; multiplying the set by exp(i*N*theta) preserves its maximal circular gap. Uniform continuity then proves norm convergence for each Laurent polynomial. Geometric sums give convergence of every nonzero normalized trace moment to0.

Remaining obstruction

For two or more independently sampled group elements, their representation eigenbases interact; the scalar circle-density argument no longer controls joint polynomial norms.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0183 Deterministic strong convergence from LPS generators

Source: The strong convergence phenomenon,2025 survey | Section 6.4, pp.66-67

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Do the explicit LPS-type finite-group generators yield strong convergence to free generators, beyond their established adjacency spectral gap?

Scope

This is the explicit deterministic model in the source, not a request to rediscover deterministic Ramanujan graphs.

P0183-A001 | Deterministic strong convergence from LPS generators structural calculation

Argument or result

Weak freeness plus the usual adjacency norm is insufficient in general. Start with any strongly convergent r-unitary model and append a one-dimensional block in which every generator equals z=exp(i*theta), with cos(theta)=sqrt(2r-1)/r. This block does not change limiting traces and its adjacency eigenvalue is exactly2*sqrt(2r-1). Yet the phase-twisted adjacency polynomial has value2r on that block, above its free norm2*sqrt(2r-1).

Check

Normalized trace weight of the added block tends to0. Multiplying each free Haar generator by a fixed phase preserves the free law and norm. Thus both asserted facts hold while a different linear polynomial has a persistent outlier.

Remaining obstruction

This is an obstruction to inferring strong convergence from one spectral-gap theorem; the constructed scalar blocks are not the actual LPS representation, so the source conjecture is not refuted.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0184 Strong convergence of modular-group actions on finite projective lines

Source: The strong convergence phenomenon,2025 survey | Section 6.4, p.67

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does the PSL_2(Z) action on P^1(F_q), restricted to constants-orthogonal vectors, converge strongly to the regular group representation for every fixed tuple of integer matrices?

Scope

The diffusion-operator formulation and arbitrary-polynomial formulation are equivalent in the source and are counted once. Random homomorphisms from C2*C3 are a different model.

P0184-A001 | Strong convergence of modular-group actions on finite projective lines structural calculation

Argument or result

Weak convergence follows directly from fixed points. For a fixed nonidentity matrix X in PSL_2(Z), its fixed points on P^1(F_q) solve a projective quadratic equation. Except for finitely many primes where X becomes scalar, there are at most two fixed points. The normalized trace in the q-dimensional constants-orthogonal representation is therefore(fix(X)-1)/q, which tends to0.

Check

For X represented by entries a,b,c,d, finite fixed points satisfy c*z^2+(d-a)*z-b=0, with the point at infinity handled projectively. A nonzero quadratic on the projective line has at most two zeros. Applying this to each word yields the regular trace moments. Faithfulness of the regular trace then gives the lower polynomial-norm bound by even moments.

Remaining obstruction

Moment convergence gives no matching upper bound on operator norms; sparse exceptional eigenvalues remain the exact obstruction.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0185 Ramanujan inequalities for growing families of matrix polynomials

Source: The strong convergence phenomenon,2025 survey | Section 6.5, pp.67-68

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Can permutation models satisfy the free polynomial norm upper bound with no error for every polynomial of degree at most q(N) tending to infinity, or for each fixed polynomial along a chosen model?

Scope

The per-polynomial weaker question and simultaneous-degree version are grouped. The source excludes a bound simultaneously for every degree at fixed dimension.

P0185-A001 | Ramanujan inequalities for growing families of matrix polynomials structural calculation

Argument or result

The one-generator amenable case satisfies the exact inequality at every degree: take an N-cycle permutation and remove its constant eigenvector. Its remaining eigenvalues lie on the unit circle, so the norm of any Laurent polynomial is at most its supremum on that circle, exactly the norm in the regular representation of Z.

Check

A normal finite matrix is unitarily diagonalizable, and Laurent-polynomial norms are maxima over its eigenvalues. This proves the bound without an asymptotic error or probability qualifier.

Remaining obstruction

For r>=2 the free reduced group algebra is noncommutative and has no finite-dimensional faithful representation. The circle argument gives no construction for any diverging degree window in that setting.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0186 Largest coefficient dimension compatible with strong convergence

Source: The strong convergence phenomenon,2025 survey | Section 6.6, p.68

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the optimal growth of matrix-coefficient dimension D_N for strong convergence of polynomials in Gaussian or Haar random matrices.

Scope

Known subexponential sufficiency and failure at sufficiently large exp(C*N^2) leave a gap; Gaussian and Haar versions are grouped. New q-Gaussian model statements are not automatically the same fixed-model question.

P0186-A001 | Largest coefficient dimension compatible with strong convergence structural calculation

Argument or result

Expected outlier counts can fail as a method even when the extreme eigenvalue converges in probability. Let E_N have probability p_N tending to0, and take X_N=1_(E_N)*I_(D_N). Then its norm tends to0 in probability, but the expected number of eigenvalues above1/2 equals D_N*p_N. With D_N=exp(N) and p_N=exp(-N/2), that count diverges.

Check

The event norm(X_N)>1/2 is exactly E_N, whereas the count on that event is exactly D_N. This isolates multiplicity rather than a failure of norm convergence.

Remaining obstruction

The example does not refute strong convergence in the requested models. It explains why a trace-statistic proof needs a different mechanism once coefficient dimension creates large multiplicities.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0187 Optimal edge-fluctuation scales for arbitrary random-matrix polynomials

Source: The strong convergence phenomenon,2025 survey | Section 6.7, pp.68-69

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Under suitable nondegeneracy assumptions, establish the N^(-2/3) fluctuation scale and edge law for norms of arbitrary-degree polynomials of random matrices.

Scope

The source explicitly excludes degenerate situations and separates deterministic spikes, which may produce N^(-1/2) fluctuations.

P0187-A001 | Optimal edge-fluctuation scales for arbitrary random-matrix polynomials structural calculation

Argument or result

The nondegeneracy qualifier is necessary even for a one-variable quadratic polynomial. For a standard GUE matrix G_N let P(x)=20-(x-2)^2. With probability tending to1 its spectrum lies in a fixed neighborhood of[-2,2] where P is positive. Then norm(P(G_N))=20-min_i(lambda_i-2)^2. Since the largest GUE eigenvalue is2+O_p(N^-2/3), one has0<=20-norm(P(G_N))<=O_p(N^-4/3).

Check

The upper bound uses the largest eigenvalue as one candidate in the minimum. Spectral mapping and positivity give the norm identity. Consequently N^(2/3)*(norm(P(G_N))-20) tends to0, rather than a nondegenerate Tracy-Widom variable.

Remaining obstruction

The example is deliberately degenerate and does not contradict the stated research direction. A correct general theorem must identify regularity of the polynomial edge before asserting the universal scale.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0188 Joint strong convergence for permutation and deterministic matrices

Source: The strong convergence phenomenon,2025 survey | Section 6.8, p.69

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Develop a permutation-matrix counterpart of joint strong convergence with strongly convergent deterministic matrices.

Scope

Removing the common constant vector of permutation matrices requires an explicit convention for deterministic matrices that do not preserve its orthogonal complement.

P0188-A001 | Joint strong convergence for permutation and deterministic matrices structural calculation

Argument or result

Naively compressing arbitrary deterministic matrices creates persistent defects. Let e be the normalized all-ones vector, P=I-ee*, and B a balanced diagonal matrix with entries plus/minus1. B strongly converges to the symmetric Bernoulli law. But on e-perp, PBP has the zero eigenvector Be, while all its other eigenvalues are plus/minus1. Thus norm(I_(e-perp)-(PBP)^2)=1, whereas the corresponding Bernoulli-limit polynomial is zero.

Check

Because B^2=I and e*Be=0, PBP(Be)=PB(e)=Pe=0. More generally (PBP)^2-PB^2P=-PBee*BP, whose norm here is1. Its normalized trace tends to0 but its operator norm does not.

Remaining obstruction

This refutes the naive unchanged-deterministic-limit formulation after compression, not every possible counterpart requested by the survey. Appropriate invariant subspaces or additional deterministic hypotheses remain to be formulated and proved.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0189 Brown-measure convergence from general strong convergence

Source: The strong convergence phenomenon,2025 survey | Section 6.9, pp.69-70

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: candidate_complete_counterexample

Question

Does strong convergence of a family of random matrices imply convergence of empirical complex-eigenvalue measures of every noncommutative polynomial to its limiting Brown measure?

Scope

This is the literal general statement. It does not impose independent entries, independence of the matrix family, or a free limiting family. The narrower Ginibre-with-adjoints question is recorded as a remaining subclass, not counted again.

P0189-A001 | Brown-measure convergence from general strong convergence structural calculation

Argument or result

Complete candidate counterexample: rational clock/shift unitary pairs U_j,V_j approach an irrational rotation pair u,v strongly, while U_j^q=I and V_j^q=-I force det(zI-U_j-V_j)=z^q. Hence every finite sum is nilpotent with empirical measure delta_0. The established Brown measure of u+v is Haar measure on the unit circle. Simultaneous Haar conjugation makes the finite family random without changing any of these statements.

Check

The strong-convergence proof maps the simple irrational rotation algebra into the product of matrix algebras modulo norm-null sequences; injectivity gives every subsequential limsup polynomial norm and therefore the full limit. Normalized word traces are checked directly. The determinant calculation independently checks nilpotency. Full proof and all source dependencies are in the proof packet.

Remaining obstruction

No gap presently identified in the literal-general-scope construction, but there is no independent review and no novelty claim. It does not resolve independent Ginibre polynomials with adjoints; the June2026 holomorphic-Ginibre theorem is compatible with this example.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0190 Intrinsic freeness for non-Hermitian eigenvalues

Source: The strong convergence phenomenon,2025 survey | Section 6.9, p.70

https://arxiv.org/pdf/2507.00346

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Find a non-Hermitian complex-eigenvalue counterpart of intrinsic freeness for general nonhomogeneous random matrices, and identify the appropriate comparison objects.

Scope

This concerns nonasymptotic comparison of broad covariance models, distinct from the abstract continuity questionP0189.

P0190-A001 | Intrinsic freeness for non-Hermitian eigenvalues structural calculation

Argument or result

Hermitization exposes a missing hypothesis. Weak singular-value convergence and bounded largest singular value do not control the logarithmic potential: nu_N=(1-1/N)*delta_1+(1/N)*delta_(exp(-N)) converges weakly to delta_1 and has bounded support, but integral log(s) dnu_N=-1 rather than0. Thus a comparison based only on bounded polynomial spectral statistics cannot justify passage through the logarithm.

Check

All positive moments converge to1, while the single tiny singular value contributes(1/N)*(-N)=-1. Uniform integrability of the negative logarithm would remove this particular obstruction.

Remaining obstruction

The scalar example is an obstruction to a proof step, not itself a counterexample to empirical eigenvalue convergence. Quantitative least-singular-value control adapted to intrinsic covariance models is still missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0191 Least singular values at the polynomial constant term

Source: Yi Han: Brown measure convergence for the spectrum of polynomials in Ginibre matrices | Question 1.6

https://arxiv.org/pdf/2606.01664

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend the polynomial least-singular-value bound for a nonzero holomorphic polynomial of independent complex Ginibre matrices to the shift equal to its constant term.

Scope

The source theorem excludes that single shift. Qualitative invertibility and a quantitative polynomial lower-tail estimate are kept distinct.

P0191-A001 | Least singular values at the polynomial constant term structural calculation

Argument or result

For a monomial P=X_(i1)...X_(id), the missing shift is0 and the estimate reduces to the individual matrices. Since smin(P)>=product_k smin(X_(ik)), the event smin(P)<=epsilon implies at least one factor has smin<=epsilon^(1/d). The standard complex-Ginibre bound then gives P(smin(P)<=epsilon)<=C*d*N^2*epsilon^(2/d). Repeated factors require no independence in this union bound.

Check

For normalized complex Ginibre, N*smin has the usual bounded quadratic lower tail. Multiply the deterministic singular-value inequalities successively, and apply the union bound over the d factor positions.

Remaining obstruction

Sums of monomials can cancel, so this argument gives no least-singular-value estimate for an arbitrary noncommutative polynomial at its constant term.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0192 Brown-measure universality beyond complex entry densities

Source: Yi Han: Brown measure convergence for the spectrum of polynomials in Ginibre matrices | Question 1.7 and first part of Question 1.8

https://arxiv.org/pdf/2606.01664

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Extend Brown-measure convergence for holomorphic noncommutative polynomials to real Gaussian or discrete entry laws such as Bernoulli, allowing complex polynomial coefficients.

Scope

Real-valued and discrete entry versions are grouped as one source universality family. The source bounded-density-on-C theorem does not cover either automatically.

P0192-A001 | Brown-measure universality beyond complex entry densities structural calculation

Argument or result

Every one-matrix holomorphic polynomial is already reducible to the circular law. If X_N has a circular-law limit and P is a fixed complex polynomial, spectral mapping gives mu_(P(X_N))=P_pushforward(mu_(X_N)) exactly, including algebraic multiplicities. The continuous mapping theorem therefore yields the pushforward of the circular law. The Brown measure of P(c) is the same pushforward, by factorizing P(z)-lambda and using multiplicativity of the Fuglede-Kadison determinant.

Check

For each test function h bounded and continuous, h composed with P is bounded and continuous, so weak convergence passes through P. At the logarithmic-potential level, P(c)-lambda factors into commuting(c-root_j) terms, and determinant multiplicativity gives the same transformed Brown measure. Existing real-Gaussian and Bernoulli circular laws supply the initial one-matrix convergence.

Remaining obstruction

For several noncommuting input matrices there is no joint eigenvalue spectral mapping. Arithmetic and structural dependence in least-singular-value estimates remain unresolved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0193 Eventual log-unimodality from arbitrary compact positive initial laws

Source: Hasebe-Ueda-Wang: Log-unimodality for free positive multiplicative Brownian motion | Problem 4.7, PDF p.11; Lemma 4.1, p.8

https://arxiv.org/pdf/2009.13848

Historical status: source_says_open | Current status: unverified | Result: candidate_complete_counterexample

Question

If nu is supported on a compact interval [alpha,beta] with 0<alpha<beta, must sigma_t free-multiplied by nu be log-unimodal for all sufficiently large times?

Scope

This asks whether the support-ratio restriction in Theorem 4.4 can be removed. Log-unimodality is unimodality after taking logarithms. The known positive theorem assumes a multiplicatively symmetric log-unimodal initial law; the question here allows non-log-unimodal initial laws.

Dated literature checks

Focused primary-source searches2026-09-06 found no verified later resolution. The arXiv page has one2020version and a2022journal reference. Author publication pages and Auer2026 concern related results, not a verified resolution here. Search absence is not current-open evidence.

P0193-A001 | Four exact crossings of the multiplicative Cauchy-profile criterion

Argument or result

Candidate negative answer: nu=(delta_(1/10)+delta_10)/2 yields failure of log-unimodality for every t>=5*pi. Set a=log10 and K_R(s)=1/[2*(cosh(s)-cosR)], F_R(u)=(K_R(u-a)+K_R(u+a))/2. Choose the unique R in[pi/2,pi) such that R/sinR=t/10. Lemma4.1 then asks whether F_R(u)=1/10 has at most two roots. But F_R(0)<=10/101<1/10, F_R(+-a)>=1/8>1/10, and F_R tends to0 at both infinities. Continuity gives four distinct roots, contradicting the necessary condition for log-unimodality.

Check

Same-author second algebra route in r: at r=1 the integral is10/(101-20*cosR)<=10/101; at r=10 and1/10 the resonant atom alone contributes1/[4*(1-cosR)]>=1/8; endpoint limits are0. The four roots lie in(0,1/10),(1/10,1),(1,10),(10,infinity). Monotonicity of R/sinR follows from(sinR-R*cosR)prime=R*sinR>0.

Remaining obstruction

No gap currently identified in the full written argument. Later-literature novelty and independent review remain unverified. Check the exact necessity direction in Lemma4.1, the time normalization, and whether the published question changed from the archived version before promotion.

Verification: author_checked_complete_candidate | Independent review: none | Full question solved: False

Full proof artifact

proofs/P0193_compact_multiplicative_nonunimodality_candidate.md

P0194 Polynomial mixing with plus Ising boundary conditions

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question1, PDFp375

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is Glauber mixing on the n-by-n square with plus boundary polynomial in n at every fixed positive temperature?

Scope

The2026 phase-ordering paper still uses a quasi-polynomial plus-boundary estimate. Its biased random initialization theorem is not a worst-start plus-boundary polynomial bound. Critical and high-temperature established cases are not counted separately.

P0194-A001 | Polynomial mixing with plus Ising boundary conditions explicit structural calculation

Argument or result

The direct disagreement coupling proves the high-temperature part. Write N=n^2 and c=4*tanh(beta). A single disagreement vanishes if its site is updated, while at most four neighboring updates create disagreement with probability at most tanh(beta). Thus the expected Hamming distance contracts by1-(1-c)/N when c<1. Iteration and the coupling inequality give tmix(1/4)<=N*log(4*N)/(1-c), up to integer rounding.

Check

Changing one neighboring spin changes the heat-bath plus probability by at most tanh(beta), since the conditional probability is(1+tanh(h))/2. A path coupling along single-spin differences extends the bound to arbitrary initial pairs. Boundary sites have at most four variable neighbors, so the estimate remains valid with fixed plus boundary.

Remaining obstruction

For4*tanh(beta)>=1 the coefficient ceases to contract. This is a failure of the local metric argument, not a lower bound showing slow mixing; the low-temperature interface regime remains unresolved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0195 Monotonicity of the ferromagnetic Ising spectral gap

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question2, PDFp375

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does increasing any ferromagnetic interaction decrease the Glauber spectral gap? What stronger time-dependent total-variation monotonicity holds?

Scope

Temperature, edge addition and individual-coupling versions are grouped. The source has zero field in its general formula. Cycle and mean-field special cases are established in the cited literature.

P0195-A001 | Monotonicity of the ferromagnetic Ising spectral gap explicit structural calculation

Argument or result

The two-spin calculation supports the conjecture exactly. With interaction J>=0, let c=tanh(J) and update either spin with probability1/2. The eigenfunctions1, sigma1+sigma2, sigma1-sigma2, sigma1*sigma2-c have eigenvalues1,(1+c)/2,(1-c)/2,0. Hence the spectral gap is(1-tanh(J))/2, strictly decreasing for finite J.

Check

Conditionally E[sigma1|sigma2]=c*sigma2. Applying the update kernel to the four displayed basis functions gives the eigenvalues directly. The stationary mean of sigma1*sigma2 is c, so the last function is centered.

Remaining obstruction

For a general graph, increasing J changes both the reversible measure and the Dirichlet form. Pointwise comparison of transition probabilities alone does not order the Rayleigh quotient because its variance denominator also changes.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0196 Sharp comparison of systematic and random Ising updates

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question3, PDFpp375-376

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For ferromagnetic Ising dynamics, compare the number of updates in a systematic sweep order with random-scan mixing, with universal upper constant and lower logarithmic loss as posed in the source.

Scope

The two inequalities are one comparison problem. General Gibbs-system polynomial comparison is established by Gaitonde-Mossel2025; their counterexamples for arbitrary target laws do not by themselves refute the ferromagnetic Ising scope.

P0196-A001 | Sharp comparison of systematic and random Ising updates explicit structural calculation

Argument or result

At infinite temperature one deterministic sweep gives an exact independent fair-spin sample. Random scan has mixing time asymptotic to(N/2)*log N. The upper bound follows from Fourier eigenvalues1-|S|/N: chi-square distance is sum_(k>=1) binom(N,k)*(1-k/N)^(2t) <=(1+exp(-2t/N))^N-1. For the lower bound, magnetization from all plus has mean N*(1-1/N)^t and variance at most N, whereas stationary magnetization has variance N. Thus at t=(N/2)*(log N-C) the means are separated by order exp(C/2)*sqrt(N).

Check

For distinct spins, E[sigma_i*sigma_j]=(1-2/N)^t <=(1-1/N)^(2t), so their covariance is nonpositive and the variance bound follows. Chebyshev separates the two magnetization laws as C grows. This independently recovers the factor log N between random updates and one sweep.

Remaining obstruction

The product case shows why the logarithmic allowance is necessary. Interacting conditional-expectation projections fail to commute, so their product cannot be replaced by the power of their average.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0197 Relation between Ising relaxation and mixing on transitive graphs

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question4, PDFp376

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

On transitive graph families, is Glauber relaxation of order N equivalent to mixing of order N*log N as temperature varies?

Scope

Time counts individual random-site updates. This concerns a relation of orders, distinct from the separate question of whether there is cutoff.

P0197-A001 | Relation between Ising relaxation and mixing on transitive graphs explicit structural calculation

Argument or result

At zero interaction the heat-bath operator has eigenvalues1-k/N with multiplicity binom(N,k), so trel=N and tmix is asymptotic to(N/2)*log N. This exhibits the proposed relation, and also shows that the logarithm comes from many coordinate modes rather than a single slow eigenfunction. A gap estimate alone gives only tmix<=trel*log(4/pi_min); at beta0, pi_min=2^-N, which overestimates by a factor of order N/log N.

Check

The largest nonconstant eigenvalue is1-1/N. Summing the full spectral multiplicities yields the sharper chi-square expression(1+exp(-2t/N))^N-1 as an upper bound; retaining only the gap loses the binomial structure.

Remaining obstruction

Transitivity of vertices does not itself identify the multiplicities or nonlinear slow observables of the interacting spin configuration chain. A comparison preserving this spectral structure is missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0198 Cutoff on bounded-degree transitive expanders

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question5, PDFp376

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Must lazy random walk on every bounded-degree vertex-transitive expander sequence exhibit total-variation cutoff?

Scope

Salez2025 repeats this as a current challenge. Optimal Ramanujan expansion is an established special case. This is not the random-Cayley spectral-gap question P0181.

P0198-A001 | Cutoff on bounded-degree transitive expanders explicit structural calculation

Argument or result

The assumptions already imply mixing of order log N and the usual product condition. From any vertex, after t steps the support lies in a ball of at most(Delta+1)^t vertices, so total variation is at least1-(Delta+1)^t/N. Thus tmix(1/4)>=log(3*N/4)/log(Delta+1). A uniform positive lazy spectral gap gamma gives tmix(1/4)<=gamma^-1*log(4*N), while trel<=gamma^-1.

Check

The lower bound tests the reachable set against uniform stationarity; transitivity makes the graph regular. The upper bound is the standard reversible L2 contraction. Their ratio is bounded and trel/tmix tends to0.

Remaining obstruction

These bounds establish only a bounded transition-time ratio, not a ratio tending to1. They provide no concentration of the relevant entropic or hitting-time clock, which is the unresolved step.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0199 Cutoff for cyclic-to-random transpositions

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question6(a), PDFp376

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does the shuffle that swaps the cyclically chosen position with a uniform random position have cutoff?

Scope

This is a time-inhomogeneous deck chain or a chain sampled by full sweeps. It differs from the cyclic adjacent-transposition shuffle, whose cutoff was proved by Nam-Nestoridi.

P0199-A001 | Cutoff for cyclic-to-random transpositions explicit structural calculation

Argument or result

The one-card marginal admits an exact update that may identify candidate slow modes. If position i is chosen deterministically and the tracked-card probabilities are p_j, then p_i becomes1/N, while p_j becomes(1-1/N)*p_j+p_i/N for j different from i. Thus a centered vector z has new z_i=0 and new z_j=(1-1/N)*z_j+z_i/N. A sweep is the explicit product of these N linear maps.

Check

If the card is at i it moves uniformly; if it is at j different from i, it stays unless the random partner is j. Summing the two cases gives the formula, which preserves the uniform distribution. Adding a deterministic phase as a state coordinate would give a periodic phase marginal and cannot justify ordinary mixing of the augmented chain.

Remaining obstruction

The one-card product can supply lower-bound observables, but does not control correlations of all cards or produce a matching sharp upper bound for the full permutation.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0200 Cutoff for card-cyclic random insertion

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question6(c), PDFp376

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does the shuffle that cyclically chooses a card label, removes that card and reinserts it uniformly have cutoff?

Scope

Random-to-random insertion is a different shuffle and is resolved. Here the selected label, rather than its present position, advances cyclically.

P0200-A001 | Cutoff for card-cyclic random insertion explicit structural calculation

Argument or result

A coupon-collector strong-stationary-time argument fails even after every label has been updated. For three cards initially123, update labels1,2,3 once each. The final probabilities of123,132,312 are5/27 each, and those of213,231,321 are4/27 each, so the total-variation distance from uniform is1/18 after a complete sweep.

Check

After labels1 and2, the states213,123,132 each have probability2/9 and231,321,312 each have probability1/9. Deleting card3 leaves relative order12 with probability5/9 and21 with probability4/9. Uniform insertion of3 gives the six stated probabilities, summing to1.

Remaining obstruction

Complete update coverage is not exact stationarity because the updated card positions remain dependent. A large-N contraction or spectral mechanism with a sharp window is not obtained.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0201 Optimal Glauber mixing near the coloring degree threshold

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question7, PDFp376

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does proper-coloring Glauber dynamics mix in order N*log N whenever the color count exceeds Delta+2 on graphs of maximum degree Delta?

Scope

The book asks strictly more than Delta+2. Nearby formulations at Delta+2 are grouped. Later2026 results for high-girth graphs do not settle unrestricted graphs; vertex coloring is different from edge coloring.

P0201-A001 | Optimal Glauber mixing near the coloring degree threshold explicit structural calculation

Argument or result

The elementary Hamming path coupling stops at q>2*Delta. For configurations differing at one vertex, updating that vertex removes the disagreement. Updating one of its at most Delta neighbors creates disagreement with probability at most1/(q-Delta), by maximally coupling the two uniform available-color sets. Therefore E[H_next]<=1-1/N+Delta/[N*(q-Delta)], which contracts precisely when q>2*Delta.

Check

The two available-color sets differ by at most one exchanged color, each has at least q-Delta elements, and their uniform laws have total variation at most1/(q-Delta). Iterating the path coupling yields an N*log N bound with constant proportional to(q-Delta)/(q-2*Delta).

Remaining obstruction

The denominator degenerates before the source threshold. Exploiting local color availability or multi-vertex moves is necessary; the failed bound is not a counterexample to rapid mixing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0202 Diameter-squared upper bound for transitive mixing

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question8, PDFp377

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is lazy random-walk mixing on every finite transitive graph bounded by a universal constant times degree times diameter squared?

Scope

The2018 spectral-embedding paper still states this as an open question. Relaxation bounds of this order do not automatically prove the mixing bound.

P0202-A001 | Diameter-squared upper bound for transitive mixing explicit structural calculation

Argument or result

The cycle illustrates the missing spectral information. On the N-cycle, lazy eigenvalues are1-sin^2(pi*k/N). Since sin(pi*k/N)>=2*k/N for k<=N/2, the chi-square bound from a point is at most2*sum_(k>=1) exp(-8*t*k^2/N^2). At t=C*N^2 this sum is as small as desired for a universal C, proving the proposed diameter-squared order in this case without a log N penalty.

Check

Use symmetry k versus N-k and(1-x)^(2t)<=exp(-2tx). The Gaussian series converges and tends to0 as C increases. The cycle diameter is floor(N/2) and its degree is2.

Remaining obstruction

For a general transitive graph, the gap alone bounds all N-1 modes by the same exponential and introduces log N. A uniform bound on low-frequency multiplicities or spectral mass strong enough to remove that factor is missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0203 Mixing exponent of uniform binary row additions

Source: Levin-Peres: Markov Chains and Mixing Times, second edition | Chapter26 Question9, PDFp377; Peres current page Question2

https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the mixing order for the walk on GL_N(2) that adds one uniformly chosen distinct row to another.

Scope

The same question on the Peres author page is counted once. The book phrase saying order2^N nodes is a typo; the group size is of order2^(N^2). The asymptotic question concerns N tending to infinity.

P0203-A001 | Mixing exponent of uniform binary row additions explicit structural calculation

Argument or result

The entropy/support lower bound has an exact leading constant. After t steps at most[N*(N-1)]^t states can be reached. Since |GL_N(2)|=2^(N^2)*product_(j=1)^N(1-2^-j), and the product tends to a positive constant, total variation is at least1-[N*(N-1)]^t/|GL_N(2)|. Thus tmix(1/4)>=(N^2*log2-O(1))/log[N*(N-1)], asymptotically at least(log2/2)*N^2/log N.

Check

The group-size formula chooses successive linearly independent columns: product_(k=0)^(N-1)(2^N-2^k). The infinite product is positive because sum2^-j converges; log(1-x) is bounded below by-2x for0<=x<=1/2 after treating the first factor separately.

Remaining obstruction

Counting update words supplies no upper bound because many words cancel or represent the same matrix. The known O(N^3) upper bound leaves a logarithmic and polynomial gap.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0204 Mixing on large monotone subsets of the hypercube

Source: Yuval Peres: related open Markov-chain problems | Related open problems, Question1

https://www.yuval-peres-books.com/markov-chains-and-mixing-times/

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For an increasing subset V of the binary N-cube with at least half its vertices, is lazy induced-subgraph random-walk mixing always O(N*log N)?

Scope

The author page gives an O(N^3) theorem of Ding-Mossel and conjectures N*log N. This is the induced graph walk, whose stationary law is degree weighted, not automatically uniform.

P0204-A001 | Mixing on large monotone subsets of the hypercube explicit structural calculation

Argument or result

Every such induced graph is connected and has diameter at most2*N: from any x in V one may flip zero coordinates upwards to the all-one vertex while remaining in V. Every vertex has degree at least the number of its zero coordinates, but the degrees need not be constant. For V equal to the cube with only the all-zero vertex removed, a unit-vector vertex has degree N-1 while a vertex of Hamming weight at least2 has degree N (for N>=3). Thus uniform stationarity is already false in an almost-full monotone set.

Check

For an undirected graph, detailed balance holds with pi(x)=deg(x)/sum_y deg(y). On an edge between a unit vector and a weight-two vertex, a uniform law would have unequal transition flow1/[2*(N-1)] versus1/(2*N). The up-path connectivity argument is deterministic and uses only monotonicity.

Remaining obstruction

Applying full-cube product Fourier estimates without correcting the stationary law is invalid. Diameter and volume alone give much weaker general bounds; the monotone geometry must control bottlenecks quantitatively.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0205 Metropolis relaxation for geometric-time Cayley targets

Source: Aldous open problems: Metropolis on Cayley graphs | Questions(i)-(iii), first posted March2009

https://www.stat.berkeley.edu/~aldous/Research/OP/cayley.html

Historical status: source_says_open | Current status: unverified | Result: partial_counterexample_author_checked

Question

Control the relaxation time of Metropolis random walk targeting the distribution of a Cayley walk at an independent geometric time; in particular, test monotonic decrease in the geometric parameter.

Scope

The source defines p in[0,1] but writes tau(infinity) in its proposed bounds; that reference parameter is undefined and is retained as a source ambiguity. The monotonicity question itself is precise. Its negative answer below is a partial answer to the grouped request for general bounds.

P0205-A001 | Metropolis relaxation for geometric-time Cayley targets explicit structural calculation

Argument or result

Monotone decrease is false already on the two-element Cayley group, with relaxation defined by1/(1-lambda_2). For the nonlazy base walk that flips states, mu_p=(1/(2-p),(1-p)/(2-p)), and its Metropolis matrix is[[p,1-p],[1,0]]. Its second eigenvalue is p-1, hence tau(p)=1/(2-p), which is strictly increasing. The same conclusion survives a lazy base walk: the base matrix has all entries1/2, mu_p=((1+p)/2,(1-p)/2), the Metropolis off-diagonal rates are(1-p)/(2*(1+p)) and1/2, and tau(p)=1+p.

Check

For the flip walk, sum the geometric probabilities over even versus odd times. For the lazy base, X_0 is the identity and X_k is uniform for every k>=1. The nontrivial eigenvalue of a two-state stochastic matrix is1 minus the sum of its off-diagonal probabilities, giving both formulas directly. If laziness is imposed after forming the nonlazy Metropolis kernel, tau becomes2/(2-p), still increasing.

Remaining obstruction

The literal monotonicity direction is refuted under the standard spectral-gap relaxation convention, but this does not answer the other universal-bound questions or resolve the source undefined tau(infinity). If relaxation meant inverse absolute gap for a nonlazy chain, that convention differs; the lazy example has nonnegative spectrum and removes that ambiguity. Prior-art status remains unverified.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0208 Extremal arc-transitive walk parameters away from cycles

Source: Aldous-Fill: Reversible Markov Chains and Random Walks on Graphs, 2002 recompiled2014 | Open Problem7.27, PDFp259; definitionp254

https://www.stat.berkeley.edu/~aldous/RWG/book.pdf

Historical status: source_says_open | Current status: unverified | Result: candidate_complete_counterexample

Question

On arc-transitive n-vertex graphs other than n-cycles, are maximal commute time O(n log n), relaxation time O(n), and the maximal-commute to twice stationary-average-hitting ratio asymptotic to1?

Scope

The three linked bounds are one question. The source definition has a variable-assignment typo; the intended standard ordered-edge transitivity is used, as in the surrounding tori examples. Wreath graphs themselves are established prior art.

Dated literature checks

Focused primary searches2026-09-06 found source question and related established results, but no verified later disposition. Absence from search is not proof of current openness or novelty.

P0208-A001 | Extremal arc-transitive walk parameters away from cycles explicit counterexample

Argument or result

For odd m>=5 the established wreath graph C_m[edgeless K2] is simple, connected,4-regular and arc-transitive on n=2m vertices. Its base projection is cycle walk. Exact hitting means are j*(m-j)+m between different fibers and2m between twins. Thus tau_star=2*floor(m^2/4)+2m, tau0=(m^2-1)/6+m, and tau2=1/(1-cos(2*pi/m)). The first two scale quadratically in n, and tau_star/(2*tau0) tends to3/2, refuting all three proposed bounds.

Check

The invariant fiber-constant and fiber-antisymmetric subspaces give the full transition spectrum: cycle eigenvalues plus m zeros. Its fundamental trace agrees with the direct stationary hitting-time sum. Neighbor hitting equals2m-1=n-1 as required by the source lemma.

Remaining obstruction

Verify prior disposition of this1994 question: the graph construction is classical. The source typo and rate-one convention are explicit. No mathematical gap found by the author, but independent review and novelty remain pending.

Verification: author_checked_complete_candidate | Independent review: none | Full question solved: False

Full proof artifact

proofs/P0208_wreath_graph_counterexample_candidate.md

P0209 Uniform stationary normal approximation from gap and bounded observable

Source: Aldous-Fill: Reversible Markov Chains and Random Walks on Graphs, 2002 recompiled2014 | Open Problem4.30, PDFp137; continuous ratesp26

https://www.stat.berkeley.edu/~aldous/RWG/book.pdf

Historical status: source_says_open | Current status: unverified | Result: candidate_complete_counterexample

Question

For stationary finite reversible continuous-time chains and Epi g=0,Epi g^2=1, does a universal normal-approximation error bound psi(||g||infinity,t/tau2) tend to0 at fixed first argument?

Scope

The source uses asymptotic variance normalization, permits arbitrary continuous transition rates, and starts in stationarity. The counterexample fixes ||g||infinity exactly2 and also survives uniformizing rates to at most1.

Dated literature checks

Focused primary searches2026-09-06 found source question and related established results, but no verified later disposition. Absence from search is not proof of current openness or novelty.

P0209-A001 | Uniform stationary normal approximation from gap and bounded observable explicit counterexample

Argument or result

Let epsilon=s^-2,R=s^4, Y have rates0to1=epsilon,1to0=1-epsilon, and independently let fair sign Z flip at R/2. Gap=1. Set A=2*epsilon+sqrt((1-epsilon)*(1-4*epsilon)),B=(2-A)/(1-epsilon),g=A*Z+B*(Y-epsilon). Then mean0,variance1,supnorm2 exactly and sigma^2=2*A^2/R+2*B^2*epsilon*(1-epsilon). By time s the chance ever Y=1 is at most epsilon*(s+1); off that event the slow integral is negligible after sigma*sqrt(s) normalization. The fast normalized variance is at most A^2/[R*B^2*epsilon*(1-epsilon)] tending0. Thus the normalized integral tends in probability to0, with normal Kolmogorov distance liminf at least1/2, although s/tau2 tends to infinity.

Check

The product generator has eigenvalues0,-1,-R,-(R+1). Exact covariance integration verifies sigma and confirms finite-time variance divided by sigma^2*s tends to1 despite distributional collapse: rare events destroy uniform integrability. Scaling Q to Q/R and time to R*s leaves the standardized law and time/gap ratio unchanged, and bounds outgoing rates by1.

Remaining obstruction

No gap identified in the written quantifier and normalization checks. Current literature and independent review remain unverified; no original full-solution promotion.

Verification: author_checked_complete_candidate | Independent review: none | Full question solved: False

Full proof artifact

proofs/P0209_uniform_clt_counterexample_candidate.md

P0210 Martingales detectable by computationally practical stopping rules

Source: Aldous open problems: Martingale for practical purposes | Problem and polynomial-constraint formulation

https://www.stat.berkeley.edu/~aldous/Research/OP/fields.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Give a useful probabilistic definition of martingale for practical purposes, or a tractable class defined by polynomially many stopping-time constraints.

Scope

The two formulations are grouped as one conceptual question. A complexity model and input representation are not specified by the source; no financial claim or backtest is involved.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0210-A001 | Martingales detectable by computationally practical stopping rules structural calculation

Argument or result

Any prescribed polynomial-size family of stopping tests can miss bounded conditional drift on an exponentially large history space. Reveal m fair bits by time m, set X_t=0 through time m and X_(m+1)=f(Y). For predetermined stopping rules T_j, each optional-mean constraint is the linear equation E[f(Y)*1_{T_j=m+1}]=0. If the number of tests plus the mean-zero condition is less than2^m, a nonzero f exists in their common nullspace. Rescale it to supnorm1. Then the process passes every supplied test but E[X_(m+1)|F_m]=f(Y) is nonzero.

Check

The witness T*=m+1 on{f(Y)>0} and m otherwise is a stopping time and has E X_T*=E f_+>0. Predetermined test rules are functions of the revealed-bit history; their final-time continuation indicators do not depend on the unknown terminal value. The construction does not claim f or T* is computationally efficient.

Remaining obstruction

A polynomial number of linear tests alone cannot reproduce the full martingale property. Selecting a useful computational subclass requires an explicit representation, complexity bound and error tolerance; the nullspace construction does not supply that theory.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0211 Relaxation of linear-polymer coagulation and fragmentation

Source: Aldous open problems: Coagulation-fragmentation mixing | Problem; linked coag_frag_mix.pdf pp1-4

https://www.stat.berkeley.edu/~aldous/Research/OP/coag_frag.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the order of the relaxation time for the reversible polymer coagulation-fragmentation chain, especially fragmentation rate beta_n comparable to1/n.

Scope

Clusters merge at rate2/n per pair; each internal link splits at rate beta. Fixed-beta and small-beta regimes are grouped. The source proves bounds between asymptotic1/sqrt(beta^2+4beta) and1/beta.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0211-A001 | Relaxation of linear-polymer coagulation and fragmentation structural calculation

Argument or result

The cluster-count coordinate K has exact rates K->K+1 at beta*(n-K) and K->K-1 at K*(K-1)/n. Its deterministic equilibrium density mu solves mu^2=beta*(1-mu), and linearizing the density drift gives restoring rate beta+2*mu=sqrt(beta^2+4*beta). At beta=b/n this is order n^-1/2. The source edge coupling has discrepancy killing rate at least beta+(K-1)/n, so it also suggests this scale once K is near sqrt(b*n).

Check

For n=3 the partition states3,21,111 have rates3->21=2beta,21->3=2/3,21->111=beta,111->21=2. Direct characteristic-polynomial calculation gives gap[3beta+8/3-sqrt(beta^2+16/9)]/2. Thus finite-n small-beta behavior cannot simply be substituted for the joint n,beta_n limit.

Remaining obstruction

The required upgrade is a uniform exponential bound on the coupling survival functional E exp[-integral(beta+(K_u-1)/n)du], including atypically small K. Replacing K by its stationary mean inside the exponential has the wrong Jensen direction and is not a proof of the desired upper bound.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0212 Low-density genealogical limit of branching coalescing walks

Source: Aldous open problems: Low-density coalescing branching random walk | Problem A,2012; updateMarch2018

https://www.stat.berkeley.edu/~aldous/Research/OP/cbrw.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove the proposed low-density, time-rescaled lineage limit for transient branching-coalescing walk and the constrained-Ising analogue, after discarding short-lived branching jitter.

Scope

The two explicitly linked microscopic models are grouped; the undefined dynamic-graph research direction is not a second admitted problem. Spatial positions are forgotten but splitting and merging histories remain.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0212-A001 | Low-density genealogical limit of branching coalescing walks structural calculation

Argument or result

Immediate parent-daughter returns renormalize the branching clock. For two independent simple walks started at neighboring sites in Z^3, their relative walk is a time change of simple walk. The probability they never meet is q=1-G(e,0)/G(0,0)=1/G(0,0), using G(0,0)=1+G(e,0) by the first-step identity and lattice symmetry. Thus a rare branch at rate beta produces a macroscopically surviving split at leading rate beta*q, rather than beta.

Check

The equality of the two Green-function expressions follows from translation and reflection symmetry of all six neighbors. It concerns an isolated pair; the chance of interference by other particles has not been set to zero without justification.

Remaining obstruction

A thinning heuristic suggests time scaling1/(beta*q), but a proof needs control of third-particle encounters, dependent return episodes and joint Palm genealogies. Stationary balance of split and merge counts does not imply independent Poisson clocks.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0213 Constant-factor online scheduling with subadditive batch durations

Source: Aldous open problems: Bacon and eggs batch scheduling | Problem,postedMarch2009

https://www.stat.berkeley.edu/~aldous/Research/OP/bacon.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Find an explicit online batch-scheduling algorithm with a universal ratio of long-run weighted waiting cost to the optimal online cost, over monotone strictly subadditive service-time functions.

Scope

Arrivals form a rate1 Poisson process with uniform types. The source assumes finite optimal cost. A finite deterministic queue example below is only an obstruction to a proposed greedy step, not a steady-state counterexample.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0213-A001 | Constant-factor online scheduling with subadditive batch durations structural calculation

Argument or result

Greedy batching can be arbitrarily poor for a fixed two-job queue when holding costs differ. Let S(a)=1,S(b)=L,S({a,b})=L+epsilon with0<epsilon<1 and L>1. Give a holding rate M and b rate1. Processing both immediately costs(M+1)*(L+epsilon). Processing a first and then b costs M+1+L. For M much larger than L their ratio approaches L+epsilon, although service times are strictly monotone and subadditive.

Check

The individual completion times are L+epsilon for both in the batch, and1,L+1 in the sequential schedule. This compares exactly the same initial jobs without incoming work. For the actual stationary model, Campbell reasoning also gives the elementary lower bound W_*>=integral c(a)*S({a}) da, since each task must spend at least its singleton service time in its completing batch.

Remaining obstruction

Embedding the bad queue event in a stable Poisson-arrival system while controlling its frequency and other waiting costs remains necessary. The two-job calculation does not disprove existence of a different constant-factor algorithm.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0214 Fluctuations of shortest routes in Poisson proximity networks

Source: Aldous open problems: Shortest routes in random proximity networks | Main question;2019 simulation study discussed

https://www.stat.berkeley.edu/~aldous/Research/OP/random_proximity.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine rigorous orders of route-length variance and transversal deviation for shortest paths in random proximity networks on planar Poisson points.

Scope

Variance and transversal exponents are grouped as the source fluctuation family. A law of large numbers is established and is not counted as a solution.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0214-A001 | Fluctuations of shortest routes in Poisson proximity networks structural calculation

Argument or result

A deterministic route of length D connecting points at distance r and reaching perpendicular deviation T must satisfy D>=sqrt(r^2+4*T^2). To see this, split at a point of maximal deviation and minimize the sum of its distances to the two endpoints over the longitudinal coordinate; the minimum occurs above the midpoint. Hence T<=sqrt(D^2-r^2)/2.

Check

When D/r tends to a stretch constant mu>1, the bound gives only T/r<=sqrt(mu^2-1)/2 asymptotically. It does not force sublinear wandering. This explicitly explains why the existing route-length shape theorem does not already identify a transversal exponent.

Remaining obstruction

A KPZ-type prediction needs estimates for centered route excess and correlations of competing near-geodesics. The deterministic Euclidean inequality leaves an order-r allowance whenever the limiting stretch exceeds1.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0215 Growth exponent for common subcladograms of random trees

Source: Aldous open problems: Largest common random substructures | Original2003 Section2 Example3 PDFp5; author updates

https://www.stat.berkeley.edu/~aldous/Research/OP/substructures.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Establish existence and values of the power-law growth exponents for the largest common induced leaf-labeled subtree of two independent uniform cladograms or coalescent trees.

Scope

The two source models form one admitted family. The2024 Budzinski-Senizergues paper improves the uniform-model upper exponent below1/2; that established bound is not a new solution or proof of a limiting exponent.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0215-A001 | Growth exponent for common subcladograms of random trees structural calculation

Argument or result

For the uniform model, sampling consistency gives an exact first-moment count: the expected number of agreeing k-subsets is binom(n,k)/(2k-3)!!. Thus P(C_n>=k) is at most this ratio. Stirling gives log of the ratio=k*[log n-2log k+2-log2]+o(k) when k is of order sqrt n. Consequently the elementary bound stops at order sqrt n, with threshold constant e/sqrt2.

Check

The agreement probability is the collision probability of two uniform induced k-cladograms, exactly1/(2k-3)!!. Existence of a common larger subset implies existence of one of size k. The coalescent model has a nonuniform induced law, so its collision probability must instead be sum_T p_T^2; the uniform formula is not transferred to it.

Remaining obstruction

First moments overcount many strongly overlapping agreeing subsets. They cannot reach the stronger known exponent bound or prove convergence of log E C_n/log n. The source centroid recursion constructs particular matches and is not automatically optimal.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0216 Limit constant for common random two-dimensional partial orders

Source: Aldous open problems: Largest common random substructures | Original2003 Section2 Example4 PDFpp5-6

https://www.stat.berkeley.edu/~aldous/Research/OP/substructures.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove that the mean size of the largest common induced suborder of two independent uniform two-dimensional partial orders is asymptotic to c*n^(1/3), and determine c.

Scope

The source already gives order n^(1/3) upper and lower arguments. The calculation here makes a constant in its grid lower construction explicit; it is a rederivation of an established method, not resolution of the limit question.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0216-A001 | Limit constant for common random two-dimensional partial orders structural calculation

Argument or result

Partition each square into m-by-m cells, m=floor(n^(1/3)). Keep a label when its two points occupy the same cell, and retain it only if no other kept label shares that row or column. The retained labels have identical comparisons in both posets. For a fixed label the match probability is1/m^2; conditional on its cell, another label is a conflicting match with probability(2m-1)/m^4. Therefore the expected retained number is n/m^2*[1-(2m-1)/m^4]^(n-1), asymptotic to exp(-2)*n^(1/3).

Check

Two retained labels have distinct row and column indices, so each of their coordinate orderings is determined by cell indices in both independent samples. Independence over labels gives the exact no-conflict probability. The bound is liminf E C_n/n^(1/3)>=exp(-2).

Remaining obstruction

The isolated-cell matching discards many usable labels and supplies no matching upper constant. Optimizing a local grid construction does not establish existence of the global optimal limit.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0217 Percolation of merging planar empires

Source: Aldous open problems: Percolation and empires | Two PROBLEM paragraphs;2009 and2017 updates

https://www.stat.berkeley.edu/~aldous/Research/OP/empires.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Characterize merging-rate conditions for finite-time infinite empires, in particular whether rate1 per adjacent pair percolates.

Scope

General rate criteria and the distinguished constant-rate case are grouped. The source already proves some sufficient conditions in its2010 paper; the main constant-rate case is not inferred from bond percolation.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0217-A001 | Percolation of merging planar empires structural calculation

Argument or result

The natural area-moment Lyapunov calculation works for strongly suppressed rates. In a finite periodic partition, F=sum_A area(A)^2 changes by2ab when areas a,b merge. If r(A,B)<=1/N(A,B), where N is at least each endpoint region adjacency degree, then the generator obeys L F<=sum_edges(a^2+b^2)/N(A,B)<=F. Hence E F_t<=e^t F_0. Under an infinite-volume limit preserving this bound and translation invariance, the mean area at a typical point stays finite, excluding an infinite empire.

Check

Use2ab<=a^2+b^2 and sum over neighbors of each region of1/N<=1. This independently explains the source suppressed-rate example. For rate1 the neighbor sum becomes the region degree, which has no such uniform control.

Remaining obstruction

For rate1, knowing each original boundary survives with probability exp(-t) does not give product bounds for contours: multiple segments separating the same two current empires share one merger clock. The missing joint survival estimate prevents the naive Peierls argument.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0218 Growth exponents in the spatial city-formation model

Source: Aldous: A spatial model of city growth and formation | Section3 equations4-9 and Section3.1; author open page

https://arxiv.org/pdf/1209.5120

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove the proposed balanced/unbalanced growth regimes and city-count exponents in the Aldous-Huang spatial immigration model.

Scope

All parameter regimes, including alpha1, are grouped. A city of size N has influence c0*N^alpha*r^-beta; an arrival founds a city with probability1/(1+maximum influence). The source heuristic is not a theorem.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0218-A001 | Growth exponents in the spatial city-formation model structural calculation

Argument or result

A rigorous one-sided bound supports the balanced exponent when0<alpha<1 and beta>2alpha. Write d=beta/2-alpha>0. A union bound over influence disks gives area{I(y,t)>u}<=pi*c0^(2/beta)*u^(-2/beta)*M_t^(1-2alpha/beta)*t^(2alpha/beta), using concavity of the population power. Thus the conditional founding probability is at least1/[2*(1+K*t^alpha*M_t^d)], K=(2pi)^(beta/2)*c0. Put theta=(1-alpha)/(1+d) and m=floor(kappa*t^theta). Until M reaches m, each of the last floor(t/2) arrivals has founding chance at least p=1/[2*(1+K*t^alpha*m^d)]. Choose kappa small so this Bernoulli count has mean at least2m. A stopped coupling and Chernoff then give P(M_t<m)<=exp(-c*m) for large t.

Check

The disk estimate is valid even at the square boundary because clipping only reduces area. Sum_i N_i^(2alpha/beta)<=M^(1-2alpha/beta)*t^(2alpha/beta). On the event M_t<m the threshold has never been reached, so domination by independent Bernoulli(p) increments throughout the half-interval is legitimate. The exponent identity theta*(1+d)=1-alpha makes the mean comparable to m.

Remaining obstruction

This proves a lower city-count growth bound in the balanced regime, not the matching upper bound, comparable populations, unbalanced regime, or alpha1 logarithmic law. The upper bound requires control of spatial clustering and influence overlap that the union bound does not supply.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0219 High-density oriented percolation flow asymptotics

Source: Aldous open problems: Discrete Hammersley process and oriented percolation flows | Conjecture,formulated2004

https://www.stat.berkeley.edu/~aldous/Research/OP/hammersley_flow.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For the oriented square-lattice maximal-flow constant v(p), prove1-v(p) is asymptotic to sqrt(2*(1-p)) as p tends to1.

Scope

The p-down-to-critical regime is not counted as an invented second problem. The author heuristic contains two normalization errors, recorded explicitly below; correcting them still supports the displayed conjecture.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0219-A001 | High-density oriented percolation flow asymptotics structural calculation

Argument or result

For a continuum Poisson process of intensity delta in[0,n]^2, rescaling coordinates by sqrt(delta) gives intensity1 in[0,n*sqrt(delta)]^2. Hence its minimum increasing-chain cover has the law U*(n*sqrt(delta),1), not U*(n/sqrt(delta),1). By the poset chain-cover/antichain relation it equals the longest decreasing subsequence, whose leading Poissonized Hammersley asymptotic is2m for a unit-intensity m-square. With delta=2*(1-p), dividing2*n*sqrt(delta) by2n recovers sqrt(2*(1-p)).

Check

The expected number of Poisson points, delta*n^2, is invariant under the corrected scaling and would fail under the printed inverse scaling. Reflection converts decreasing to increasing subsequences, preserving the known leading constant2. These are checks of the continuum surrogate only.

Remaining obstruction

One still needs a uniform comparison of the lattice closed-edge path-cover problem with the continuum antichain problem as n tends to infinity and p tends to1. The two corrected constants do not justify interchanging those limits.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0220 Explicit stationary law of the leftmost Hammersley drift-jump particles

Source: Aldous open problems: One-dimensional drift-jump particle process | Problem,postedFebruary2018;1993notes Section4.1

https://www.stat.berkeley.edu/~aldous/Research/OP/left_Hamm.html

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Give a useful explicit joint stationary law for the ordered particles which drift at speed x and jump left at rate1 Poisson space-time points.

Scope

The1993 notes already contain a patience-sorting representation and small-k calculations. The1999 paper Section5.2 uses the stationary process for limit theorems. Those established representations are not counted as newly solved open work.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0220-A001 | Explicit stationary law of the leftmost Hammersley drift-jump particles structural calculation

Argument or result

The established rank representation can be expanded into a convergent combinatorial density series. Let B=(1=b1<...<bk) be the limiting first-row labels of the insertion tableau, and let q_B be its probability. The first b=bk labels form a uniform permutation, so q_B=(1/b!)*sum over partitions lambda of b with first row length k of f^lambda times the number of standard tableaux of shape lambda with first row B. Independent unit-rate Poisson spacings then give density exp(-xk)*sum_B q_B*product_(i=2..k)[(x_i-x_(i-1))^(b_i-b_(i-1)-1)/(b_i-b_(i-1)-1)!].

Check

Restriction of an insertion tableau to labels at most b is the tableau of the corresponding restricted subword. The last specified first-row entry b pins the first k labels, irrespective of later larger labels. Conditional on B, the Poisson arrival times have independent gamma increments, so every term is a normalized conditional density and nonnegative summation is valid. For k1 this gives Exp(1); known Poissonized LIS formulas give P(X2>x)=exp(-x)*I0(2sqrt(x)) and E X2=e.

Remaining obstruction

This repackages a known representation into explicit but rapidly growing finite-tableau coefficients and an infinite sum. It does not establish a new tractable closed joint law or its novelty, and should not be promoted as a full solution to the source request.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0221 Limit of the optimal online random spanning-tree cost

Source: Aldous-Angel-Berestycki: Online random weight minimal spanning trees, draft2008 | Conjecture PDFp2;Theorem3 PDFp4; author open page

https://www.stat.berkeley.edu/~aldous/Research/OP/Beres2008.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove existence and determine the limiting optimal mean cost of an online spanning tree of the complete randomly weighted graph.

Scope

Combinatorial random-order and Poisson offer models are grouped, but their finite-n costs differ. The draft explicitly lacks a complete asymptotic equivalence proof. Numerical estimates near1.9 are not certified limits.

Dated literature checks

Author-maintained page and relevant linked primary sources checked2026-09-06. Focused later-literature searches did not establish a complete resolution; current status remains unverified.

P0221-A001 | Limit of the optimal online random spanning-tree cost structural calculation

Argument or result

The Poisson dynamic-programming recursion has exact small-state solutions. For two current components of sizes x,y, F=(xy/2)*F^2 and the positive solution is2/(xy). For three singleton vertices each merged state has cost1, giving F=(3/2)*(F-1)^2. The admissible root exceeding the child cost is(4+sqrt7)/3; the other root is below1 and violates strict monotonicity.

Check

The first recursion follows by integrating the accepted-offer surplus over weights below the continuation-cost difference. For n2 the Poisson cost is2 while the combinatorial model must buy its sole Uniform(0,1) edge and has cost1/2, demonstrating why finite-n model identification is invalid.

Remaining obstruction

The recursion depends on the entire component-size partition and on n. Standard coagulation hydrodynamic limits assume a prescribed kernel, whereas these acceptance kernels are themselves defined by unknown value functions. A compactness and uniqueness argument for their joint limit is missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0222 Near-unit stretch versus length in spatial networks

Source: Aldous-Lando: The stretch-length tradeoff in geometric networks | Abstract and dense-network limit question; author open page

https://arxiv.org/pdf/1404.2653

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Establish existence and value of the exponent governing minimum network length per unit area as the allowed all-pairs stretch s decreases to1.

Scope

The Poisson average-case problem is the probability target; the source worst-case analogue is grouped as a comparison. This is different from the mean-route-length constraint for SIRSNs.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0222-A001 | Near-unit stretch versus length in spatial networks structural attempt

Argument or result

Every path of stretch at most1+epsilon between endpoints at distance r stays in an ellipse whose maximum transverse half-width is(r/2)*sqrt(2epsilon+epsilon^2). Indeed a point of deviation T forces the total path length to be at least sqrt(r^2+4T^2). Thus as epsilon decreases, each local pair needs a usable road within a tube of angular width order sqrt(epsilon). This is an explicit geometric constraint for a possible direction-packing lower bound.

Check

The ellipse inequality follows by minimizing the sum of distances from a path point to its two endpoints over the longitudinal coordinate. It is shared with the deterministic lemma in P0214, but the sourced optimization problem here has a different target.

Remaining obstruction

Counting constrained pairs is insufficient because one road segment can serve many pairs. A stochastic-geometric bound on this reuse, or a matching construction with controlled length, is needed to obtain a sharp exponent.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0223 Sharp relaxation exponent for local tree rotations and triangulation flips

Source: Aldous: Mixing times for branch rotation and triangulation walk | PROBLEM,PDFp2; historyp3

https://www.stat.berkeley.edu/~aldous/206-RWG/RWGpapers/clad-mix.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Prove an order n^(3/2) relaxation-time upper bound for the branch-rotation cladogram chain and the related convex-polygon triangulation flip chain.

Scope

The two chains are conservatively grouped as the source rotation family. The2026 Alev-Frishberg-Sarantis-Tetali theorem improves the triangulation bound to n^2 times polylogarithms and explicitly leaves the conjectured exponent unresolved.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0223-A001 | Sharp relaxation exponent for local tree rotations and triangulation flips structural attempt

Argument or result

A canonical-path construction that first rotates every tree to one fixed comb has exponential congestion, even if each route is short. Let N be the number of states and use uniform stationary mass1/N. Routing the all-pairs demand pi(x)*pi(y) through that comb sends total order1 flow through a vertex whose total stationary transition capacity is1/N. Some incident edge therefore has load-to-capacity ratio at least order N. For triangulations N is a Catalan number of order4^n/n^(3/2).

Check

The total capacity through the chosen vertex is sum_y pi(v)P(v,y)=1/N. Pairs not starting or ending there have total demand(1-1/N)^2 and must cross it in the proposed construction. Summing their edge loads gives the claimed bottleneck independently of path length.

Remaining obstruction

A proof needs routes or transport flows distributed over many intermediate shapes. The comb normal form is useful for irreducibility but cannot furnish the conjectured polynomial relaxation bound.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0224 Feasible length and stretch statistics of scale-invariant networks

Source: Aldous: Scale-invariant random spatial networks | Open Problems29 and30(i),PDFp48

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Describe the feasible triples(Delta,ell,p(1)) for SIRSNs and obtain sharp bounds for the minimum sampled-network length ell at fixed mean route stretch Delta.

Scope

The feasible-region question and its lower-boundary function are one optimization family. Source30(ii), attainment, is separately admitted because it concerns closure and existence.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0224-A001 | Feasible length and stretch statistics of scale-invariant networks structural attempt

Argument or result

The feasible set is convex under the source axioms. Choose one of two entire SIRSN laws once using an independent Bernoulli variable, and use that law for all routes. Every realization retains compatibility and all required distributional invariances. Each of Delta,ell,p(1) is an expectation or monotone limit of expectations, hence is the same convex combination of the corresponding statistics. Taking near-minimizers shows ell_*(lambda*Delta1+(1-lambda)*Delta2)<=lambda*ell_*(Delta1)+(1-lambda)*ell_*(Delta2).

Check

The random choice is global, so routes from the two networks are never spliced together. No ergodicity assumption appears in the cited definition; imposing one would invalidate this mixture argument. Finite expected statistics remain finite for a two-law mixture.

Remaining obstruction

Convexity does not locate the feasible region or give its sharp boundary. It also does not guarantee an extremal law exists; a convex set need not be closed.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0225 Attainment of the optimal SIRSN length-stretch tradeoff

Source: Aldous: Scale-invariant random spatial networks | Open Problem30(ii),PDFp48

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Does a SIRSN attain the infimum sampled-network length at a prescribed mean route stretch, and what structure can an optimizer have?

Scope

This is the existence part of the source optimization question, not a separately counted parameter choice.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0225-A001 | Attainment of the optimal SIRSN length-stretch tradeoff structural attempt

Argument or result

Uniform expected length in every compact set supplies tightness for the unlabelled length measures: Markov bounds on an exhausting sequence of compact sets control local masses, and bounded-mass sets of Radon measures are vaguely relatively compact. However the prescribed feasible-route class is not closed under even uniform convergence with bounded lengths. Inscribed polygonal approximations to a semicircle are finite unions of straight segments with uniformly bounded length, while their limit is a smooth arc and is not a jagged route made of straight segments.

Check

For the semicircle, no nontrivial subarc is a line segment; countably many straight segments lying in the arc can therefore cover only countably many points. The source itself discusses a possible extension from jagged to rectifiable routes.

Remaining obstruction

A compactness proof for length measures alone loses route labels, compatibility and the allowed-curve condition. One must choose a richer topology and prove closure or justify a relaxed model before claiming attainment.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0226 Local finiteness of scale-invariant network traffic measures

Source: Aldous: Scale-invariant random spatial networks | Open Problem31,PDFp49

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For source-destination displacement density|z|^-beta with2<beta<4, prove suitable local finiteness of induced route traffic on the major-road set E(infinity,1).

Scope

The target is traffic restricted to major roads. Finiteness of the expected traffic on the full dense route network is a different, stronger statement and fails by the calculation below.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0226-A001 | Local finiteness of scale-invariant network traffic measures structural attempt

Argument or result

For a bounded area A, translation invariance and Tonelli give expected full-network traffic through A equal to area(A)*integral_R2 |z|^-beta*E len R(0,z) dz. Scale invariance makes E len R(0,z)=Delta*|z|, yielding area(A)*2*pi*Delta*integral_0^infinity r^(2-beta) dr. This integral diverges for every beta: at infinity for beta<=3, at zero for beta>=3.

Check

Translate each route by its source and integrate the length of its intersection with A over all source locations. Each infinitesimal route segment contributes area(A). This mass-transport identity uses nonnegative integrands, so it remains valid when the integral is infinite.

Remaining obstruction

The divergent full-network expectation neither disproves almost-sure local finiteness nor answers the restricted-major-road question. A proof must exploit the major-road restriction and separately control long routes and short routes which take large excursions.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0227 Implications among SIRSN finiteness and geodesic properties

Source: Aldous: Scale-invariant random spatial networks | Open Problem32(i)-(iv),PDFp49;definitionspp10-11,44-45

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Determine the four proposed reverse implications between weak and full SIRSN finiteness, unique rays, pairwise coalescence, uniform coalescence and endpoint route continuity.

Scope

All four linked implications are grouped as one source family. Equation16 is finite ell;20 is finite p(1);49 is pairwise ray coalescence;50 is uniform small-neighborhood-to-far-target coalescence;51 is route coincidence away from nearby endpoints.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0227-A001 | Implications among SIRSN finiteness and geodesic properties structural attempt

Argument or result

The available finite-intensity estimate does not yield the first implication: p(lambda,1)<=intensity S(lambda)=ell*sqrt(lambda), which diverges with the sampling density. For the third implication, finite coalescence radii at each finite sample do not bound their increasing supremum; a concrete scalar analogue is R_k=k, finite at every k and divergent in the limit. For the second, stationary families of parallel rays show that uniqueness of a ray at each origin does not alone imply coalescence. For the fourth, continuity at each fixed destination gives no uniform estimate as the destination recedes to infinity.

Check

The scaling identity for sampled intensity follows by rescaling a rate1 Poisson process by lambda^-1/2. The parallel-ray construction is deliberately only a counterexample to the isolated logical inference; it is not claimed to satisfy the full connected-route SIRSN axioms.

Remaining obstruction

Actual counterexamples must preserve compatibility, all scale invariances and finite required moments. Alternatively a proof needs a uniform estimate absent from the displayed bounds. No member of the four-part family is settled here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0228 Uniqueness of the unbounded major-road component

Source: Aldous: Scale-invariant random spatial networks | Open Problem33,PDFp49

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Must the major-road set E(infinity,1) of every SIRSN have only one unbounded connected component almost surely?

Scope

Disconnected finite major-road pieces are allowed by definition. The question concerns unbounded components only.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0228-A001 | Uniqueness of the unbounded major-road component structural attempt

Argument or result

A precise sufficient reduction is available: if every unbounded major-road component contains a tail of a singly infinite geodesic and all such geodesics coalesce, then uniqueness follows. Two purported unbounded components would contain two tails, whose common eventual tail lies in both, forcing the components to coincide.

Check

The conclusion uses connected-component maximality, not merely that the full network is connected. Finite local total length does not by itself supply the needed geodesic tail in every unbounded component, because the route union is not assumed to be a locally finite combinatorial graph.

Remaining obstruction

Neither the universal ray-coalescence property nor the component-tail property is proved from the source axioms. A Burton-Keane style argument would require a valid local-modification principle, which arbitrary SIRSN laws need not have.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0229 Integrable supremum of routes to a bounded region

Source: Aldous: Scale-invariant random spatial networks | Open Problem34,PDFp50

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Under what additional assumptions does the expected supremum of route lengths from0 to an infinite independent uniform sample in the unit disk remain finite?

Scope

Pointwise E len R(0,x)=Delta*|x| does not control a random supremum. The sufficient increment condition below is stronger than the base axioms and is not claimed to follow from them.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0229-A001 | Integrable supremum of routes to a bounded region structural attempt

Argument or result

Let L(x)=len R(0,x). Suppose on a containing square E|L(x)-L(y)|^p<=C*|x-y|^p for a p>2, with a jointly measurable version and finite mean at one anchor. On the dyadic grid of mesh2^-k there are order2^(2k) parent-child or neighbor pairs. The expected maximum increment is at most Cprime*2^(-k*(1-2/p)). This is summable, so telescoping along grid ancestors constructs a continuous modification with finite expected supremum.

Check

Use max_j |Z_j|<=(sum_j |Z_j|^p)^(1/p) and Jensen, then sum the resulting geometric series. The increment bound implies stochastic continuity, identifying the modification at each deterministic point; Fubini identifies it at each independent uniform sample point, and countability handles the entire sampled sequence.

Remaining obstruction

The source only gives first moments of individual route lengths, not p-moment increments of L. Its compatibility condition does not make Euclidean route length an automatic metric with a triangle inequality, so a metric-chaining shortcut is unjustified.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0230 Asymptotic total length of finite SIRSN samples

Source: Aldous: Scale-invariant random spatial networks | Open Problem35,PDFp50

https://arxiv.org/pdf/1204.0817

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

For k uniform points in a square of area k, prove that the mean total length of their spanning route network is asymptotic to ell*k.

Scope

Length includes route pieces outside the sampling square. This global excursion contribution is the main source obstruction.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0230-A001 | Asymptotic total length of finite SIRSN samples structural attempt

Argument or result

A local truncation argument gives the lower bound liminf E length/k>=ell. Tile the bulk of the large square by unit cells. For fixed R and M, retain in each cell only the length, truncated at M, of routes between sampled endpoints within distance R of that cell. Its local binomial point configuration converges in total variation to the corresponding rate1 Poisson configuration. The retained length is bounded by M, so its expectation converges. Summing disjoint unit-cell contributions and then taking k to infinity gives the Poisson local truncated expectation as a lower bound. Let M and then R increase: monotone convergence recovers the full sampled-network intensity ell.

Check

Only a bounded endpoint region is used before taking k to infinity; its binomial count tends to a Poisson count and conditional positions have the same uniform law. Truncation avoids any unproved uniform-integrability step. Boundary cells are an asymptotically negligible fraction for fixed R.

Remaining obstruction

This proves no upper bound on total length. Rare paths can leave the sampling square by a large distance, and their exterior contribution is absent from the local lower-bound argument. Controlling those excursions remains necessary for equality.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0231 A zero-one law for Poisson-coloring frontier dimension

Source: Basdevant-Blanc-Curien-Singh: Fractal properties of the frontier in Poissonian coloring,2023 | Discussion before Open Question1.2,PDFp3

https://alea.impa.br/articles/v20/20-59.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is the Hausdorff dimension of the limiting Poisson-coloring frontier almost surely deterministic?

Scope

The strict inequalities d-1<dimension<d are already proved in Theorem1.1. This entry concerns the explicitly remaining zero-one question, not that resolved conjecture.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0231-A001 | A zero-one law for Poisson-coloring frontier dimension structural attempt

Argument or result

The dimension event is not visibly a tail event of the independent arriving points. Changing an early point can change the color of infinitely many later descendants, so Kolmogorov zero-one cannot be invoked without a finite-modification invariance proof. The potential route is a regeneration principle producing, inside every surviving interface neighborhood, copies with the full essential supremum of the dimension law; countable-union stability of dimension would then force the maximal value.

Check

The mere facts that the frontier is closed and has dimension strictly between1 and2 do not imply determinism: a Bernoulli mixture of two fixed compact fractal curves with different dimensions satisfies those facts and has random dimension. This is a logical control, not a Poisson-coloring counterexample.

Remaining obstruction

Independent arrivals do not make recursively inherited colors independent across subregions. A regeneration or local-universality statement strong enough to remove early-history effects is missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0232 Continuous curves in the Poisson-coloring frontier

Source: Basdevant-Blanc-Curien-Singh: Fractal properties of the frontier in Poissonian coloring,2023 | Open Question1.2,PDFpp3-4

https://alea.impa.br/articles/v20/20-59.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is the planar limiting frontier a countable union of continuous curves?

Scope

A curve means an image of a continuous map from[0,1]. It is not required to be rectifiable. Point and segment variants are grouped.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0232-A001 | Continuous curves in the Poisson-coloring frontier structural attempt

Argument or result

The established dimension lower bound already rules out a countable union of rectifiable curves: every rectifiable curve has Hausdorff dimension at most1, and the dimension of a countable union is the supremum of the component dimensions. Thus any affirmative construction must use nonrectifiable curves. Compact connectedness alone also cannot supply path connectedness, as shown by the topologist sine-curve closure.

Check

The sine-curve closure of{(x,sin(1/x)):0<x<=1} together with the limiting vertical segment is compact and connected but not path connected. This does not by itself refute a countable-curve covering; it specifically blocks the inference from connectedness to a single continuous parametrization.

Remaining obstruction

A scale-uniform crossing/tightness estimate is needed to construct continuous interface paths. The source lacks the correlation inequality used to establish such estimates in standard planar models.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0233 Absence of double points on Poisson-coloring interface curves

Source: Basdevant-Blanc-Curien-Singh: Fractal properties of the frontier in Poissonian coloring,2023 | Open Question1.3,PDFp4

https://alea.impa.br/articles/v20/20-59.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If the limiting frontier is a union of curves, are its component curves almost surely simple?

Scope

This conditional question is distinct from existence of curves. The segment-model single-curve consequence is grouped.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0233-A001 | Absence of double points on Poisson-coloring interface curves structural attempt

Argument or result

A quantitative four-arm estimate would settle the proposed double-point obstruction. In a bounded region cover by order epsilon^-2 boxes. If the probability that a box supports four alternating color arms from scale epsilon to a fixed macroscopic radius is at most C*epsilon^(2+eta), a union bound makes the chance of any such box at most Cprime*epsilon^eta. Taking epsilon down to0 and then countably many macroscopic radii excludes points with four nontrivial alternating arms.

Check

A double point with four arms reaching a positive radius must lie in one of the grid boxes and induce a slightly enlarged box arm event. Changing fixed geometric constants does not alter the positive exponent margin.

Remaining obstruction

No four-arm exponent exceeding2 is established for this dependent coloring process. Excluding four-arm points would still have to be connected carefully to the exact parametrization and component statement, rather than silently equated with every form of self-touching.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0234 A safe margin around extended initial color seeds

Source: Basdevant-Blanc-Curien-Singh: Fractal properties of the frontier in Poissonian coloring,2023 | Open Question1.4,PDFp4

https://alea.impa.br/articles/v20/20-59.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

If the initial red seed is a segment or ball disjoint from the blue seed, does the limiting blue region almost surely avoid the entire initial red seed?

Scope

Pointwise protection of each fixed initial point is established in related arguments and does not imply simultaneous protection of a continuum. Disjoint initial colors are part of the meaningful scope.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0234-A001 | A safe margin around extended initial color seeds structural attempt

Argument or result

A sufficient uniform estimate can be stated precisely. Let the compact seed K have covering number at most C*delta^-d0. If every x in K satisfies P(dist(x,B_infinity)<=2delta)<=Cprime*delta^alpha uniformly, with alpha>d0, a delta-net and union bound give P(K intersects B_infinity)<=Cdouble*delta^(alpha-d0), hence zero. Compactness converts zero separation into intersection for the closed limiting blue set.

Check

Pointwise zero intersection probability is inadequate: for a uniformly random U in a segment, every fixed x differs from U almost surely, but the random singleton{U} intersects the segment surely. The net argument explicitly identifies the missing uniform rate.

Remaining obstruction

The available shielding results give positive protection probability or fixed-point statements, not a uniform small-distance exponent exceeding the dimension of the seed. The full safe-margin question remains unresolved here.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0235 A Feller topology for limits of reversible Markov chains

Source: Aldous: Limits for processes over general networks,2018 | Conjecture,slide5;author current compact page

https://www.stat.berkeley.edu/~aldous/Research/OP/talk_towsner_2018.pdf

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Can the measurable limit process of finite reversible chains be given a natural complete separable state metric with the Feller property?

Scope

Towsner2015 already supplies the measure-theoretic existence and refined uniqueness results. This entry concerns only the remaining topological upgrade.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0235-A001 | A Feller topology for limits of reversible Markov chains structural attempt

Argument or result

The suggested unweighted heat-kernel metric can be infinite in a basic admissible example. For Brownian motion on the two-torus, the squared row-distance integrated against exp(-t)dt equals sum_(k in Z^2) |1-exp(2pi*i*k.h)|^2/[1+8pi^2*|k|^2]. At h=(1/2,0), every odd k1 has numerator4. Each dyadic annulus contains order N^2 such terms of order N^-2, so the series diverges. A repair is to use w(t)=exp(-t)/(1+G(2t)), where G is the integrated diagonal heat trace: Fubini gives finite weighted row norm for almost every state.

Check

Parseval computes the row norm exactly; integrating the nonnegative Fourier series is justified by Tonelli. For the repaired weight, the average row squared norm is integral w(t)*G(2t)dt<=1. This produces an almost-everywhere finite Hilbert-space pseudometric after identifying indistinguishable rows.

Remaining obstruction

Brownian motion already has a good usual topology, so the divergence refutes only the illustrative formula, not the existence conjecture. Completing a heat-kernel embedding can add states; proving valid transition kernels and the Feller property on that completion is still missing.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

P0236 Infinite mean route length at each large fixed Euclidean separation

Source: Aldous: Route Lengths in Invariant Spatial Tree Networks | Introduction conjecture,PDFp2;Section3.3

https://arxiv.org/pdf/2103.00669

Historical status: source_says_open | Current status: unverified | Result: attempted_unsolved

Question

Is there a universal r0 such that every invariant tree network on planar rate1 Poisson points has infinite mean within-tree distance at almost every fixed separation r>=r0?

Scope

Theorem2 proves the averaged statement for pairs at distance at most r. The stronger fixed-separation conjecture and the authors possible thin-annulus counterconstruction remain distinct from that theorem.

Dated literature checks

Primary source and available author updates read2026-09-06; focused later searches recorded in source registry. No current-open certification or novelty claim.

P0236-A001 | Infinite mean route length at each large fixed Euclidean separation structural attempt

Argument or result

Divergence of the averaged mean cannot alone imply pointwise infinite means. A scalar control is rho(s)=1+1/|s-a| for s different from a, with any finite value assigned at a. Then rho is finite at every s, but integral_0^r s*rho(s)ds is infinite for every r>a. Thus even all sufficiently large cumulative pair averages can diverge while no fixed-radius value is infinite.

Check

The planar radial averaging density is proportional to s, so a singularity at a positive a survives the weighting. This example is only a function-level obstruction; it is not asserted to arise from any invariant tree network.

Remaining obstruction

One needs regularity or geometric constraints on the actual Palm mean function rho, or a completed invariant-tree counterconstruction. The source thin-annulus idea still requires cycle removal and connectivity without destroying the desired finite conditional means.

Verification: author_checked_limited_argument | Independent review: none | Full question solved: False

Excluded and resolved source candidates

P0096 Four-edge-minimum lower bound for the FPP time constant

current status

resolved_in_literature

result status

literature_counterexample_rederived

statement

For iid nonnegative edge times on Z^2 with F(0)=0, is the axial time constant at least the expected minimum of four independent edge times?

statement caveat

E_4 is expectation of a minimum, not a fourth moment. The target is the universal survey formulation; the motivating simulation paper reports selected families.

resolution evidence

Daniel Ahlberg, Tertiles and the time constant, arXiv1911.09949 (2019;v2May2020), Journal of Applied Probability57(2),407-408,2020. The full two-page paper was read. Theorem1 gives mu<=2t_(2/3); final paragraph explicitly refutes Alm-Deijfen and surveyQuestion12. Published DOI10.1017/jpr.2020.9.

resolution url

https://arxiv.org/pdf/1911.09949

reason

Already resolved before this campaign; own construction is retained as a rederivation, excluded from1000 current-open target and original-solution counts.

Retained derivation or attempt

Take tau=eta+epsilon U with P(eta=0)=3/4 and U uniform[0,1]. For epsilon>0 this is atomless and bounded with F(0)=0. The four-edge minimum has expectation at least1/256. The eta-model time constant is0. For each n, passage times converge to their eta-model values and are bounded by2n; dominated convergence and mu=inf_n E T(0,ne1)/n imply mu_epsilon tends to0. All sufficiently small epsilon reverse the proposed inequality.

Status limit

No mathematical gap identified in the written candidate; later-literature and erratum search remain incomplete. No explicit numerical epsilon threshold or independent review is claimed.

Retained derivation or attempt

Ahlberg2019/2020 already refuted the question. Applying his mu<=2t_(2/3) to the independently drafted tau=eta+epsilon U family gives t_(2/3)=8epsilon/9, hence mu<=16epsilon/9. With epsilon=1/1024, mu<=1/576<1/256<=E minfour. This supplies an explicit atomless bounded counterexample using an established theorem.

Status limit

No current-open problem remains for this universal inequality. This is literature-established, not an original solution.

P0140 Random critical drift in the annealed thinned Levy process

current status

resolved_in_literature

result status

literature_result_rederived_with_scope_correction

statement

Explain directly how mixing the heavy-tailed degree weights yields the independent-increment iid-degree exploration limit.

statement caveat

The deterministic drift in the short2017 formulation must be replaced by the random critical shift Lambda_0. Dhara et al2019 equation2.22 and Remark11 already identify the corrected law indirectly.

later resolution

{"finding": "Corrected law is indirectly identified; direct marking proof supplied here is not counted as a new theorem", "locator": "Equation2.22 and Remark11, PDF pp.9-10", "url": "https://arxiv.org/pdf/1612.00650"}

Retained derivation or attempt

For PPP weights with intensity(tau-1)c*x^(-tau)dx and exponential marks of rate x/mu, the random drift Z/mu cancels the empirical quadratic compensator and leaves independent increments from a deterministic marked-Poisson intensity. With a deterministic drift, a mixed log-Laplace difference is positive for small a, so increments are dependent.

Status limit

No unproved theorem is claimed: the corrected law was already identified indirectly in2019. Exact source normalization and later availability of the direct proof are retained as scope/novelty limits.

P0206 Random-to-random insertion cutoff

current status

resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Bernstein-Nestoridi prove cutoff at(3/4)*N*log N in Annals of Probability2019.

resolution url

https://arxiv.org/abs/1703.06210

P0207 Mixing of the supercritical hypercube giant

current status

claimed_resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Anastos-Diskin-Lichev-Zhukovskii arXiv2510.13348v4May2026 Theorem1(b) proves order d^2 for fixed c>1; abstract and full theorem read.

resolution url

https://arxiv.org/abs/2510.13348

P0237 Strictly fractal dimension of the Poisson-coloring frontier

current status

resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Basdevant-Blanc-Curien-Singh2023 Theorem1.1 proves d-1<dim_H F<d and explicitly says it answers this Aldous conjecture.

resolution url

https://alea.impa.br/articles/v20/20-59.pdf

P0238 Zero area of the Poisson-coloring frontier

current status

resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Preater2009 Theorem2 proves zero Lebesgue measure; the2023 primary paper explicitly identifies this as the earlier answer to Aldous Conjecture3.

resolution url

https://www.stat.berkeley.edu/users/aldous/Research/OP/preater.pdf

P0239 Exclusion of tree subnetworks in SIRSNs

current status

resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Aldous2021 Theorem2 forces infinite average route length for large bounded separations in every invariant Poisson tree network. A SIRSN has E length at separation s equal to Delta*s with finite Delta, giving finite radial average2*Delta*r/3, a contradiction. This is an application of the established theorem, not a new open-problem solution.

resolution url

https://arxiv.org/abs/2103.00669

P0240 Poisson-line construction of a SIRSN

current status

resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Kahn2016 proves that the marked Poisson-line metric is a SIRSN in every dimension.

resolution url

https://arxiv.org/abs/1503.03976

P0241 Measure-theoretic compactification of finite reversible chains

current status

resolved_in_literature

result status

excluded_prior_resolution

resolution evidence

Towsner2015 proves existence and refined uniqueness of the measurable limit construction. The topological upgrade remains separately recorded as P0235.

resolution url

https://arxiv.org/abs/1404.3815

R0001 Complete bunkbed inequality

current status

resolved_in_literature

result status

literature_counterexample

resolution url

https://www.math.ucla.edu/~pak/papers/BunkBed4.pdf

R0002 One-dimensional Brownian snails extinction

current status

resolved_in_literature

result status

literature_solution

resolution url

https://www.normalesup.org/~hartarsky/files/research/publi/snails.pdf

R0003 Schramm locality for critical percolation

current status

resolved_in_literature

result status

literature_resolution

resolution evidence

Theorem 1.1; abstract checked; 2023 preprint

resolution url

https://arxiv.org/abs/2310.10983

R0004 Generating-set invariance of free uniform-forest connectivity

current status

resolved_in_literature

result status

literature_resolution

resolution evidence

Abstract and author institutional version; Annals of Probability 2022

resolution url

https://arxiv.org/abs/2006.06387

R0005 Analytic biased-walk speed on leafless Galton-Watson trees

current status

resolved_in_literature

result status

literature_resolution

resolution evidence

Question 17.34 immediately following sentence; original theorem conditions still to verify

resolution url

https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf

R0006 Harmonic dimension lower bound by expected log offspring

current status

resolved_in_literature

result status

literature_resolution

resolution evidence

Theorem 1; primary theorem statement checked

resolution url

https://perso.lpsm.paris/~slin/papers/harmonic_bias.pdf

R0007 Exact Gaussian ellipsoid-fitting quarter threshold

current status

claimed_resolution_in_recent_primary_preprints

result status

literature_claim_excluded_pending_review

resolution evidence

Misiakiewicz-Wen, Theorem1.2, PDF p.2, both exact feasibility below1/4 and unrestricted infeasibility above1/4; statement read, full47-page proof not reviewed

resolution url

https://arxiv.org/pdf/2608.10184

additional resolution claims

["https://arxiv.org/abs/2608.12415", "https://arxiv.org/abs/2608.27372"]

Source registry

Ahlberg: Tertiles and the time constant (2019;published2020)

ahlberg_tertiles_2020 | https://arxiv.org/pdf/1911.09949

access evidence

Public primary arXiv manuscript; both pages including complete proof and counterexample read

reading status

unread

statement validation

not_started

sha256

202f4455544198fc1a7cd832f4264ab0f8184d23f00e706a058ecaf5e327538b

acquired at

2026-09-06T19:17:14Z

AIM braid groups, clusters and free probability problems

aim_braid_free_probability | https://aimath.org/WWN/braidgroups/braidgroups.pdf

access evidence

Public primary author or workshop PDF; individual open-question statements and later status require separate validation

reading status

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statement validation

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88f54825edb119532be13ece404988f8ed82dfef4716d24397e110743c19c1e9

acquired at

2026-09-06T19:34:20Z

AIM free analysis problems, August2006

aim_free_analysis_2006 | https://aimath.org/WWN/freeanalysis/freeanalysis.pdf

access evidence

Public primary author or workshop PDF; individual open-question statements and later status require separate validation

reading status

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statement validation

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a44e98be34192d107d730b5b3f4b892dd4640563123b50972d5b58ada54d5b47

acquired at

2026-09-06T19:33:56Z

AIM stochastic methods for non-equilibrium dynamical systems, Ben Webb notes

aim_nonequilibrium | https://aimath.org/pastworkshops/noneqdynsysproblems.pdf

access evidence

Public primary author or workshop PDF; individual open-question statements and later status require separate validation

reading status

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statement validation

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acquired at

2026-09-06T19:32:44Z

AIM phase transitions workshop problems, compiled by Charles Radin

aim_phase_transitions | https://aimath.org/WWN/phasetransition/phasetransition.pdf

access evidence

Public primary author or workshop PDF; individual open-question statements and later status require separate validation

reading status

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statement validation

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sha256

5a2f7236eb67c095fbc95c88213c28aacf07d6fe434ba026abd9597c0669cea4

acquired at

2026-09-06T19:33:28Z

AIM random analytic functions workshop open problems, April2006

aim_random_analytic_2006 | https://aimath.org/WWN/randomzeros/arcc1.pdf

access evidence

Public primary author or workshop PDF; individual open-question statements and later status require separate validation

reading status

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statement validation

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sha256

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acquired at

2026-09-06T19:31:50Z

AIM random matrices workshop problem list

aim_random_matrices | https://aimath.org/WWN/randommatrices/randommatrices.pdf

access evidence

Public primary author or workshop PDF; exact statements and later status separately checked

reading status

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statement validation

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5f3f0e00106140d5401fe671da093bbc2e23ab4f8faa4e2db034ebc5b50e080d

acquired at

2026-09-06T19:50:04Z

Aldous open problems: Bacon and eggs batch scheduling

aldous_bacon_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/bacon.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

unread

statement validation

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ac9b73e56230e8f4150301f86bd9f583301476b435070f560262538be3a22d67

acquired at

2026-09-06T22:04:58Z

A survey of max-type recursive distributional equations

aldous_bandyopadhyay_2005 | https://arxiv.org/pdf/math/0401388

access evidence

Public primary author website or arXiv manuscript

reading status

unread

statement validation

not_started

sha256

a54121c0a0fe066392edf3dba6652ef2fd22210b610c65aa9416aa710f09631a

acquired at

2026-09-06T16:43:17Z

Aldous open problems: Metropolis on Cayley graphs

aldous_cayley_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/cayley.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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acquired at

2026-09-06T22:05:06Z

Aldous open problems: Low-density coalescing branching random walk

aldous_cbrw_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/cbrw.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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69f398a7ef7fc65e190b7382479bb1084f948067f93a3c25e734085baf817600

acquired at

2026-09-06T22:04:49Z

Aldous: A spatial model of city growth and formation

aldous_cities_2012 | https://arxiv.org/pdf/1209.5120

access evidence

Public primary source; precise open questions and later resolutions checked separately

reading status

unread

statement validation

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sha256

eeecb371b6e6c2860c01e1b73d812468b619ddb7e3c4cd1d0e3affc056e8a077

acquired at

2026-09-06T22:16:15Z

Aldous open problems: Spatial city growth and formation

aldous_cities_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/cities.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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acquired at

2026-09-06T22:05:59Z

Aldous open problems: Branch-rotation chain mixing

aldous_clad_mix_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/clad-mix.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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eb77b2a7bde90c9afda28c7b8ed738a2f6db0c5d2c52813d586d6d75c2b21208

acquired at

2026-09-06T22:05:22Z

Aldous: Mixing times for branch rotation and triangulation walk

aldous_clad_rotation_source | https://www.stat.berkeley.edu/~aldous/206-RWG/RWGpapers/clad-mix.pdf

access evidence

Primary source statement and later-literature scope verification2026-09-06

reading status

unread

statement validation

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sha256

005420add4edf92cd7ad46df7799ef082db1414a07f46d80818df10939316654

acquired at

2026-09-06T22:48:36Z

Aldous open problems: Coagulation-fragmentation mixing

aldous_coag_frag_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/coag_frag.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

unread

statement validation

not_started

sha256

4205247b098318c0b262384df06c2efc93ad8b673efaf8216e4f54c24b7c9d47

acquired at

2026-09-06T22:04:42Z

Aldous-Fill unposted notes: coagulation fragmentation mixing

aldous_coag_frag_notes | https://www.stat.berkeley.edu/~aldous/Research/OP/coag_frag_mix.pdf

access evidence

Public primary source; precise open questions and later resolutions checked separately

reading status

unread

statement validation

not_started

sha256

1ff70d6c2dba03cc80738b723f22e476509f3255131343b655726d0a6e86b7d7

acquired at

2026-09-06T22:16:12Z

Aldous open problems: Compactification of finite reversible chains

aldous_compact_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/compact.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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sha256

96289780c33904b0f61c155e378d5d2508684845a355d77db2494881719086e7

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2026-09-06T22:10:45Z

Aldous: Compactification notes for reversible Markov chains

aldous_compact_notes | https://www.stat.berkeley.edu/~aldous/Talks/MCcompact.pdf

access evidence

Public primary source; precise open questions and later resolutions checked separately

reading status

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statement validation

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sha256

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acquired at

2026-09-06T22:16:10Z

Aldous open problems: Congestion and percolation-fragmentation

aldous_congestion_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/congestion.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

unread

statement validation

not_started

sha256

341bf2e5f1e024c9e654a450cc0849518d1c3ad52fcfd086702a7e4fd28baed8

acquired at

2026-09-06T22:04:19Z

Aldous open problems: Percolation and empires

aldous_empires_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/empires.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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acquired at

2026-09-06T22:05:50Z

Aldous open problems: Martingale for practical purposes

aldous_fields_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/fields.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

unread

statement validation

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a6e69e62ec5ca5c77750245b07d1182dd44488eab863cdc83d8ca1a6122694e5

acquired at

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Aldous-Fill: Reversible Markov Chains and Random Walks on Graphs, 2002 recompiled2014

aldous_fill_2014 | https://www.stat.berkeley.edu/~aldous/RWG/book.pdf

access evidence

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reading status

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8f9b934111c30b3f8fd29d94a18794d4b86f824b19f7a3c27b1153cdfac08e79

acquired at

2026-09-06T22:03:09Z

Aldous open problems: Bayesian graph Laplacian spectral gap

aldous_gap_ion_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/gap_ion.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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acquired at

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Aldous open problems: Compression of sparse labelled graphs

aldous_graph_compression_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/graph_compression.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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d6aa44f23bd408ec6a45d20ef59d0ff25fa442f63303ac3f164fd5c612d3ff02

acquired at

2026-09-06T22:04:34Z

Aldous open problems: Discrete Hammersley process and oriented percolation flows

aldous_hammersley_flow_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/hammersley_flow.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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1c3078bb05b25554b7ab7bc12e6d8567b777407b6c6ad24fecfc89715088a301

acquired at

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Aldous open problems: One-dimensional drift-jump particle process

aldous_left_hamm_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/left_Hamm.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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1a208ce3851d65022ecc998bedcfe97fd8afe5240e52cec3172aef0d9a89b2b6

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Bowen-Chapman-Lubotzky-Vidick: The Aldous-Lyons conjecture I, subgroup tests

aldous_lyons_negative_2024 | https://arxiv.org/pdf/2408.00110

access evidence

Public primary source; precise open questions and later resolutions checked separately

reading status

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754f587aa5592c147ded596422b5c9350a1fc8767bc1c0517f054f2f19ccd2f5

acquired at

2026-09-06T22:16:18Z

Aldous: Limits for processes over general networks,2018

aldous_markov_topology_2018 | https://www.stat.berkeley.edu/~aldous/Research/OP/talk_towsner_2018.pdf

access evidence

Primary URL discovered through focused literature search2026-09-06; substantive title and theorem verification follows acquisition.

reading status

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statement validation

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fdf0d2bea05dc981662d33622fb6a104650efc9261d3d5bdcd118ad3694d6b8c

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Aldous-Angel-Berestycki: Online random weight minimal spanning trees, draft2008

aldous_online_mst_2008 | https://www.stat.berkeley.edu/~aldous/Research/OP/Beres2008.pdf

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Primary URL discovered through focused literature search2026-09-06; substantive title and theorem verification follows acquisition.

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2026-09-06T22:39:04Z

Aldous open problems: Online minimum spanning tree

aldous_online_zeta3_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/online_zeta3.html

access evidence

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David Aldous: Some Open Problems, May2003

aldous_open_2003 | https://www.stat.berkeley.edu/~aldous/Research/OP/problems_old.pdf

access evidence

Primary author publicly accessible source; numbered open questions require later-status verification

reading status

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statement validation

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3ddbd2d6b95c7bcfccfd9767f3e9d4ecf221ddd6fb036d4e36e9700ec1e53418

acquired at

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David Aldous: current open-problem index and resolved-question registry

aldous_open_index_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/index.html

access evidence

Primary author publicly accessible source; numbered open questions require later-status verification

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statement validation

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69de0a61207bac04b9452263afbc75321409adb75e673d0f6dfb29a2441adb40

acquired at

2026-09-06T21:44:46Z

Aldous: Random partitions of the plane via Poissonian coloring and coalescing planar partitions

aldous_partition_2017 | https://arxiv.org/pdf/1701.00131

access evidence

Public primary source; precise open questions and later resolutions checked separately

reading status

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statement validation

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c8b365c639a8611db5c5941bd92a458a17baa4e974699157e112fbdf73a2433d

acquired at

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Aldous open problems: Random plane partition topology

aldous_partition_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/partition.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

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8cce7766af016f82b00a40fe03ccfb20da692d124fd92eebee48815d8b6928ff

acquired at

2026-09-06T22:06:12Z

Aldous-Diaconis: unpublished1993 patience-sorting notes

aldous_patience_1993 | https://www.stat.berkeley.edu/~aldous/Research/OP/patience.pdf

access evidence

Public primary source; precise open questions and later resolutions checked separately

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7d7457424b2f1422e30900692e1626b9fe07d172e8b1cfe12c3470f418ac717c

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Aldous-Diaconis: Longest increasing subsequences, from patience sorting to the Baik-Deift-Johansson theorem,1999

aldous_patience_1999 | https://www.stat.berkeley.edu/~aldous/Papers/me86.pdf

access evidence

Public primary source; precise open questions and later resolutions checked separately

reading status

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statement validation

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19350af5be8b46aca281c32a9dc063e46bf31c6f8e97ae919ff9e8eb547e7317

acquired at

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Aldous open problems: Shortest routes in random proximity networks

aldous_random_proximity_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/random_proximity.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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Aldous open problems: Scale invariant random spatial networks

aldous_sirsn_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/sirsn.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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0f2d77bc8c702dc50c02f41fbc2cf1be1898d86ed82f174be22f8701535ec748

acquired at

2026-09-06T22:06:08Z

Aldous: Open problems in spatial networks, Paris2022

aldous_spatial_talk_2022 | https://www.stat.berkeley.edu/~aldous/Talks/Paris_2022_talk.pdf

access evidence

Primary URL discovered through focused literature search2026-09-06; substantive title and theorem verification follows acquisition.

reading status

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statement validation

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891dc17aa4273ba87b6458762c94603d761309fd4d831197a074c4087ef09ac4

acquired at

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Aldous-Lando: The stretch-length tradeoff in geometric networks

aldous_stretch_2015 | https://arxiv.org/pdf/1404.2653

access evidence

Primary source statement and later-literature scope verification2026-09-06

reading status

unread

statement validation

not_started

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92ae1ef8a49209a937bf12b5e7dcbc8c709545b2230490c2638e8719f5b11738

acquired at

2026-09-06T22:48:41Z

Aldous open problems: Spatial network stretch-length tradeoff

aldous_stretch_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/stretch.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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statement validation

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Aldous open problems: Largest common random substructures

aldous_substructures_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/substructures.html

access evidence

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reading status

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statement validation

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26952c614e9736843f48855eb5e9fe79fcf627c5d54b2af3ed213614a3c93b5e

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Aldous: Route Lengths in Invariant Spatial Tree Networks

aldous_tree_lengths_2021 | https://arxiv.org/pdf/2103.00669

access evidence

Primary URL discovered through focused literature search2026-09-06; substantive title and theorem verification follows acquisition.

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942fb21291dba57e7214dcea915e6338b7907669d9d2f872940bfdec5b397bd7

acquired at

2026-09-06T22:38:07Z

Aldous open problems: Local weak limits and unimodularity

aldous_unimodular_20260906 | https://www.stat.berkeley.edu/~aldous/Research/OP/unimodular.html

access evidence

Public primary author source, accessed2026-09-06; mathematical content and current status checked separately

reading status

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Basdevant-Blanc-Curien-Singh: Fractal properties of the frontier in Poissonian coloring,2023

basdevant_coloring_2023 | https://alea.impa.br/articles/v20/20-59.pdf

access evidence

Primary URL discovered through focused literature search2026-09-06; substantive title and theorem verification follows acquisition.

reading status

unread

statement validation

title_and_relevant_theorem_read

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6757c4979446f51b3f155d4aa40ecd625b7e0a559a255fdf8f9345e2c13839f0

acquired at

2026-09-06T22:39:35Z

Yi Han: Brown measure convergence for the spectrum of polynomials in Ginibre matrices

brown_ginibre_2026 | https://arxiv.org/pdf/2606.01664

access evidence

Primary author or publisher public manuscript; no license or current-status inference from access

reading status

unread

statement validation

not_started

sha256

cbce9cea23d33efc19032bba7a35d79cfd080b24222d85befe79000f66b2f656

acquired at

2026-09-06T21:27:13Z

Berestycki and Sauzedde: Brownian paths as loop-decorated SLEs,2026

brownian_loop_sle_2026 | https://arxiv.org/pdf/2602.04673

access evidence

Public primary author or workshop PDF; exact statements and later status separately checked

reading status

unread

statement validation

not_started

sha256

992bf2231f5a814a8e45ff022df405ef47e01ff37743875f785265cc7d77f2f0

acquired at

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Krzysztof Burdzy: My favorite open problems

burdzy_open_20260906 | https://sites.math.washington.edu/~burdzy/open_mathjax.php

source date

undated; official page includes 2024 update

access evidence

Public primary author website or arXiv manuscript

reading status

all_problem_statements_read

statement validation

author_read_exact_locators

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1620000be15308b6996ed3c215b10c36828e91328bc783146c789e02a58ecd86

acquired at

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Forward Brownian motion

burdzy_scheutzow_2014 | https://math.washington.edu/sites/math/files/documents/research/1302.6958.pdf

source date

2014; arXiv:1302.6958

access evidence

Public primary author website or arXiv manuscript

reading status

all_problem_statements_read

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8c18daa54bac6ccc5da96a3a0d3d614c41cc41678df671d117b3a86168920b9e

acquired at

2026-09-06T16:41:55Z

Deift: Some open problems in random matrix theory and integrable systems

deift_problems_2007 | https://arxiv.org/pdf/0712.0849

access evidence

Public primary author or workshop PDF; exact statements and later status separately checked

reading status

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statement validation

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40d0f6564a81d3222e30430940d8935804778e5c5d350ce7e411a91bd3e2d947

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Deift: Some open problems in random matrix theory and integrable systems II

deift_problems_2017 | https://arxiv.org/pdf/1703.04931

access evidence

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675b61a50b2a22a20b2c3d29b32044ccdd9714b9516262202888060845c7d5a1

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Alekseev et al: Generalised Linial-Nisan Conjecture Is False for DNFs

dnf_almost_independence_2025 | https://drops.dagstuhl.de/storage/00lipics/lipics-vol339-ccc2025/LIPIcs.CCC.2025.29/LIPIcs.CCC.2025.29.pdf

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aac710a13425798b1fa0feb3b36472e60ad6f988cb74dcd0f5b75734a050905a

acquired at

2026-09-06T20:47:59Z

Duminil-Copin thesis: Chapter17 Many questions and few answers

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access evidence

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203a2abbe1237d63ac6018db908db1d7bf925754aa061404191bf465283018c4

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Gonzalez,MacManus,Pereyra: Further evidence towards the Fourier Entropy-Influence conjecture

fei_evidence_2026 | https://arxiv.org/pdf/2606.00246

access evidence

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reading status

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0cbdddd0c99e9aa6c525b60cddfdf292b4abccd34aafbf7e669d097fdf11a8f4

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Auffinger, Damron and Hanson: 50 years of first-passage percolation

fpp_survey_2015 | https://arxiv.org/pdf/1511.03262

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89ab8416bc42814c3d529475ad7e5608a7c67ba368fcf7c6192c54eb82bf424d

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2026-09-06T18:34:26Z

Free probability and model theory of tracial W-star algebras

free_model_theory_2022 | https://arxiv.org/pdf/2208.13867

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reading status

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aa9031b2bb359626b9670abe01fc1e9301baa925adc0c62dba3d041db003c7f0

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An Study of The Role of Software Project Manger in the Outcome of the Project

free_positive_unimodality_2020 | https://arxiv.org/pdf/2009.13869

access evidence

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reading status

excluded_off_domain

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wrong_arxiv_identifier_caught_before_problem_admission

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d522f76fcaca29c50e686a4e6d304b362f75cccc8fa51ec9f4ff02317567d21d

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Hasebe-Ueda-Wang: Log-unimodality for free positive multiplicative Brownian motion

free_positive_unimodality_2020_correct | https://arxiv.org/pdf/2009.13848

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cf993cc0fbd26af6cc88b15f78db007401f61832d1cd68f63847440daec5842a

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Barbieri-Jezernik: Strong convergence of random representations of free products of finite groups

free_products_strong_2026 | https://arxiv.org/pdf/2602.11905

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362c34faffdb6dcbed4f493fd2a90235bd4409eb963f949d26895020aa3e96bc

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Hasebe,Ueda: Large time unimodality for classical and free Brownian motions

free_unimodality_2018 | https://arxiv.org/pdf/1710.08240

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Hasebe-Ueda: Unimodality for free multiplicative convolution on the unit circle

free_unitary_unimodality_2019 | https://arxiv.org/pdf/1903.05327

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Roberts-Rosenthal: General state space Markov chains and MCMC algorithms

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Delgosha-Anantharam: A universal low complexity compression algorithm for sparse marked graphs

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1f9cd89e8aab3f23e8aa6ce92a92c911ffbccb69e253443a00395f741b93ddb7

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Ben Green:100open problems; select probability questions only

green_100 | https://people.maths.ox.ac.uk/greenbj/papers/open-problems.pdf

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Selected problems in probability theory

grimmett_2022_selected | https://arxiv.org/pdf/2205.07318

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21d15bcfb892c8b19b4ca593f87ba8dfc58504de3c02aaad896ec65551aee167

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Jorritsma,Maillard,Moerters: The critical percolation window in growing random graphs

growing_critical_2025 | https://arxiv.org/pdf/2512.18937

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Can,Giardina,Giberti,van der Hofstad: Annealed Ising model on configuration models

ising_cm_2019 | https://arxiv.org/pdf/1904.03664

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57d84e129c70d875078c9274f075aca636fc0b2e7325966d41feee5c9b3962ef

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Rapid phase ordering of Ising dynamics on Z2,2026

ising_phase_order_2026 | https://arxiv.org/pdf/2605.08052

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9d039f66d423000d5bcb33e14851a272489e215e3afb6c31dcdf604147c87470

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Kahn: Improper Poisson line process as SIRSN in any dimension

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Proof appendix: P0001_cactus_forests

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0001: negative dependence for random forests

Target remains unsolved here. This note proves a special case and refutes an overly broad proof route. The special case is author checked; novelty is not claimed. No independent reviewer or proof assistant has checked this note.

The sourced target is Grimmett, *Selected problems in probability theory* (2022), Conjecture 4.2, PDF page 8: for a finite simple graph, uniformly choose an acyclic edge subset F. Are distinct edge indicators always negatively correlated? Tang and Zhang, arXiv:2603.10738v1 (March 2026), Conjecture 1.3, still states this general question. Their complete-graph results do not settle arbitrary graphs.

A special case with an exact formula

Let G be a finite cactus graph: each edge belongs to at most one simple cycle. Give every edge a finite positive weight w_e. Choose an acyclic edge set F with probability proportional to the product of its selected edge weights. This includes the uniform forest measure when every w_e=1.

Proposition. This measure is negatively associated: if A and B are disjoint edge sets and f and g are increasing real functions of the indicators in A and B respectively, then Cov(f,g) <= 0.

Proof. Begin with independent Bernoulli edge indicators X_e with p_e=w_e/(1+w_e). Conditioning on acyclicity gives exactly the desired measure, because the product probability of an edge set is a constant times its product weight.

The cactus cycles have disjoint edge sets. Acyclicity is therefore the intersection, over cycles C, of the events that not every edge of C is selected. These events depend on disjoint coordinate sets. Conditioning preserves independence between the cycle blocks and the remaining bridge edges.

Consider one cycle block. Write H for the event that every coordinate equals one and q=P(H)=product(p_e). For disjoint coordinate sets A and B in that block, set a=E f, b=E g, f_*=f(1_A), and g_*=g(1_B). Independence gives E(fg)=ab. On H, f=f_* and g=g_*. Consequently

E(f | H^c) = (a-q f_*)/(1-q),
E(g | H^c) = (b-q g_*)/(1-q),
E(fg | H^c) = (ab-q f_* g_*)/(1-q).

Subtracting the product of the first two equations from the third yields

Cov(f,g | H^c)
= -q (f_*-a)(g_*-b)/(1-q)^2 <= 0.

The inequality follows from monotonicity: f_* >= a and g_* >= b. The denominator is positive because every p_e<1.

For completeness, independence products preserve negative association. Suppose X and Y are independent negatively associated blocks. Condition first on Y. The conditional covariance of f(X_A,Y_C) and g(X_B,Y_D) is nonpositive whenever A,B are disjoint and C,D are disjoint. The two conditional expectations are increasing functions of the disjoint Y-coordinate sets C,D; their covariance is also nonpositive. The total covariance identity proves the claim for (X,Y). Induct over the finitely many blocks, including independent bridge coordinates. This proves the proposition.

For two distinct edges e,f in the same cycle, the formula becomes

Cov(X_e,X_f | H^c)
= -q (1-p_e)(1-p_f)/(1-q)^2.

For a uniform cycle with m edges this is

-2^(m-2)/(2^m-1)^2.

Edges in different blocks have covariance zero.

Second derivation and adversarial check

For the uniform cycle, count subsets directly. There are 2^m-1 forests, 2^(m-1)-1 containing a given edge, and 2^(m-2)-1 containing a specified pair. Thus the covariance numerator is

(2^(m-2)-1)(2^m-1) - (2^(m-1)-1)^2 = -2^(m-2),

which agrees with the conditional-product calculation. For a triangle it gives -2/49. For a tree, all edge subsets are forests, and independent Bernoulli(1/2) indicators give zero covariance. These checks test both the sign and the degenerate no-cycle case.

The strongest reason this does not solve the target is that general cycles share edges. Their exclusion events no longer factor into independent blocks. One cannot simply say that conditioning independent bits on a decreasing event preserves negative association.

Here is a minimal counterexample to that proposed general principle. Take three fair independent bits (X1,X2,X3), conditioned on neither X1 X2 nor X2 X3 equaling one. The five allowed states are

000, 100, 010, 001, 101.

Then E X1=E X3=2/5 and E(X1 X3)=1/5, so Cov(X1,X3)=1/25>0. The conditioning event is decreasing, but creates positive correlation. This is a counterexample to a proof route, not to the forest conjecture: the forbidden pairs do not form the circuit family of a graphic matroid. Circuit elimination would require another circuit, which is absent.

Exact unresolved obligation

To extend this argument one needs a valid inequality controlling the interaction of overlapping cycle constraints, using graphic-matroid structure. Arbitrary decreasing-event conditioning cannot provide it. The target is not promoted beyond a partial attempt.

Sources

- https://arxiv.org/abs/2205.07318 (Conjecture 4.2, page 8).
- https://arxiv.org/html/2603.10738v1 (Conjecture 1.3 and the discussion immediately following).

The calculations and proof above were derived during this campaign. They are not attributed to the cited papers.

Proof appendix: P0004_quenched_limsup

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0004: what can be proved about the limsup

Status: author-checked elementary partial result. No novelty claim. The source's question about existence of a limit remains unsolved here.

Target source: Geoffrey Grimmett, *Selected problems in probability theory*, arXiv:2205.07318, Question 2.3, PDF pages 3-4. The target is the exponential growth limit of self-avoiding walks on the infinite supercritical bond-percolation cluster of the square lattice.

Deterministic root-independence lemma

Let G be an infinite connected graph of maximum degree Delta < infinity. Let c_n(v) count n-step self-avoiding walks starting at v, with c_0(v)=1. Then the quantity

M(v) = limsup as n tends to infinity of c_n(v)^(1/n)

does not depend on v.

Proof. First let v and w be adjacent. Partition the walks from v according to whether they visit w. A walk which avoids w becomes an (n+1)-step self-avoiding walk from w after prepending the edge wv. If a walk visits w, let k in {1,...,n} be its unique visiting time. Reverse its prefix from v to w and retain its suffix from w. These are respectively k-step and (n-k)-step self-avoiding walks from w. Forgetting their disjointness only enlarges the number of pairs. Thus

c_n(v) <= c_(n+1)(w) + sum_(k=1)^n c_k(w)c_(n-k)(w).

An infinite connected locally finite graph contains a ray from every vertex, by the elementary finite-branching tree argument applied to finite self-avoiding walks. Hence 1 <= M(w) <= Delta. For every epsilon>0 there is a finite C>=1 such that c_m(w)<=C(M(w)+epsilon)^m for every m>=0, by the definition of limsup and enlarging C for the finitely many exceptions. The displayed inequality gives

c_n(v) <= [C(M(w)+epsilon) + n C^2](M(w)+epsilon)^n.

Take nth roots and limsup, then let epsilon decrease to zero. This gives M(v)<=M(w). Interchanging the roots gives equality. A finite path connects any two roots. QED.

Independent check: apply the argument to a tree. The visiting decomposition still holds although the exact counts can vary dramatically by root at finite n; the polynomial factor disappears at the exponential scale. The proof never uses transitivity or stationarity and therefore cannot, by itself, prove convergence of c_n(v)^(1/n).

Consequences in percolation and an annealed upper bound

Using uniqueness of the infinite cluster and translation ergodicity of independent supercritical bond percolation, the common limsup is translation-invariant and therefore deterministic. These two standard percolation theorems are dependencies, not proved here. This conclusion is already in the literature: Hubert Lacoin, *Non-coincidence of quenched and annealed connective constants on the supercritical planar percolation cluster*, arXiv:1203.6051.

There is also an elementary upper bound. For a fixed lattice root, let c_n^omega count open n-step self-avoiding walks, including environments where the root has a finite cluster. Every lattice self-avoiding walk uses n distinct edges, so

E c_n^omega = p^n c_n^lattice.

Write mu for the lattice connective constant. For any epsilon>0, eventually c_n^lattice<=(mu+epsilon)^n. Markov's inequality yields

P(c_n^omega >= [p(mu+2epsilon)]^n)
<= ((mu+epsilon)/(mu+2epsilon))^n.

The right side is summable. Borel-Cantelli, followed by a countable epsilon sequence tending to zero, proves limsup(c_n^omega)^(1/n)<=p mu almost surely. It remains valid after conditioning the root to belong to the infinite cluster. Lacoin proves a strictly smaller bound in the planar supercritical regime p<1; that strict improvement is not derived here.

Why the obvious subadditivity route stops

Cutting a walk at time m only gives

c_(m+n)^omega(v) <= c_m^omega(v) max_(u reachable in m steps) c_n^omega(u).

The endpoint maximum depends on the same environment and is not c_n^omega(v), nor a single predetermined stationary translate. Fekete's lemma therefore does not apply to log c_n^omega(v). Kingman's theorem needs a specified stationary subadditive process; no such process has been constructed by this inequality. Replacing the maximum with a typical-root value would require a uniform estimate over the growing set of endpoints and cannot be justified by an almost-sure statement at each fixed root alone.

Exact obstruction: prove that the liminf has the same exponential rate as the deterministic limsup.
Root-independence, annealed expectations, and subexponential prefactors in the displayed comparison do not resolve it.

Proof appendix: P0032_brownian_prefix_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0032: a Brownian-prefix embedding argument

Written 2026-09-06. Status: complete candidate proof, undergoing author checks and a prior-literature search. No independent review and no novelty claim. Do not promote this file merely because a ledger validator accepts it.

Source problem: Burdzy and Scheutzow, *Forward Brownian motion*, arXiv:1302.6958, Problem 7.5, PDF page 26 (published 2014). The source asks whether a process on nonnegative time whose law is equivalent to Wiener measure can be preceded by a random finite path segment so that the concatenated process has Wiener law.

The interpretation here allows an extension of the probability space and dependence between prefix and suffix. The joining time is an arbitrary finite random time, not a stopping time for the ordinary Brownian filtration. Requiring a Brownian stopping time would immediately force the suffix itself to have Wiener law by the strong Markov property, so that different requirement cannot hold for a general equivalent law.

A general lemma

Let (E,F,P) be a standard Borel probability space, and let theta:E->E be an invertible, measurable, measure-preserving, ergodic transformation with measurable inverse. Let Q have density f>=0 with respect to P, with integral f dP=1. On an extension supporting independent uniform random variables, there exists an almost surely finite integer N>=0 such that, for Z with law P, theta^N Z has law Q.

This is a filling construction related to classical work of Chacon-Ornstein, Rost, and Thorisson. The following proof is supplied in full; claiming its application as a new research result requires a separate literature check.

Construction of remaining densities

For an integrable density h let Lh=h composed with theta^(-1). Thus pushing hP forward under theta gives (Lh)P. Set

a_0=1, b_0=f,
m_n=min(a_n,b_n),
a_(n+1)=L(a_n-m_n),
b_(n+1)=b_n-m_n.

All these functions are measurable and nonnegative. Inductively 0<=a_n<=1. The b_n decrease and are bounded above by f, so they have an integrable pointwise limit b_infinity. Since L preserves integrals, both remaining densities have the same mass r_n:

integral a_n dP = integral b_n dP = r_n,
r_n decreases to r = integral b_infinity dP,
sum_(n>=0) integral m_n dP = 1-r.

The remaining mass must be zero

Iteration gives the exact identity

a_n = 1 - sum_(k=0)^(n-1) L^(n-k)m_k.

For each fixed k, the ergodic theorem for the measure-preserving transformation theta^(-1) gives

(1/M) sum_(n=k+1)^(M-1) L^(n-k)m_k
-> integral m_k dP

in L1(P). The L1 assertion follows from the usual almost-sure ergodic theorem and equality of the integrals for nonnegative averages, or by approximation from bounded functions. Initial omitted terms do not affect the limit.

Truncate the sum in k at K. Its tail has L1 norm at most sum_(k>=K) integral m_k dP, uniformly in M. That tail tends to zero as K grows. Therefore

(1/M) sum_(n=0)^(M-1) a_n -> r

in L1(P), where r on the right is the constant function.

Suppose r>0. There is a delta>0 such that B={b_infinity>=delta} has positive P-measure. On B,

m_n=min(a_n,b_n) >= min(a_n,delta)
>= min(1,delta) a_n,

where the last step uses a_n<=1. Averaging and integrating over B gives

liminf_M (1/M) sum_(n<M) integral_B m_n dP
>= min(1,delta) r P(B) > 0.

But the entire sum of the integrals of m_n is at most one, so its Cesaro average tends to zero. This contradiction proves r=0. In particular sum_n m_n=f in L1 and almost everywhere.

Realizing the matching as a random shift

Sample Z with law P and independent U_0,U_1,... uniform on [0,1]. At stage n, if not previously accepted, inspect theta^n Z and accept when

U_n <= h_n(theta^n Z),
h_n=m_n/a_n on {a_n>0}, and h_n=0 elsewhere.

By induction, the subprobability distribution of theta^n Z before this test is a_n P. The mass accepted at stage n has distribution m_n P, and the surviving mass pushed forward has distribution a_(n+1)P. Hence P(N>=n)=r_n tends to zero, while the law of theta^N Z is sum_n m_n P=fP=Q. Every U_n is independent of Z and of preceding U's, so the induction is valid even though h_n can inspect an entire trajectory. This proves the lemma.

Application to Brownian motion

Let E=C_0(R) carry two-sided Wiener measure P. Define

(theta w)(t)=w(t+1)-w(1), t in R.

This is an invertible measurable measure-preserving transformation. It is ergodic: the path is measurably equivalent to its bi-infinite sequence of unit-length Brownian increment segments, and theta shifts this independent identically distributed sequence by one. Ergodicity of that shift follows by approximating an invariant event by finite-coordinate cylinder events and using independence of sufficiently separated cylinders.

Let Q_plus be the desired law of X on C_0([0,infinity)), and let f_plus=dQ_plus/dW_plus. Define f(w)=f_plus(w restricted to [0,infinity)). Its integral under P is one. Apply the lemma to Q=fP. The forward part of theta^N W,

Y_t=W_(N+t)-W_N, t>=0,

has exactly law Q_plus, and N is finite and nonnegative. The path obtained by taking W on [0,N] and continuing with W_N+Y_(t-N) is identically the original path W on nonnegative time. It therefore has Wiener law.

If the requested suffix X already lives on a specified probability space, use the regular conditional distribution of (W,N) given Y in this coupling to extend that space. Standard Borel path spaces ensure this disintegration exists. In the resulting coupling Y=X almost surely, so the prescribed X receives a prefix without changing its law.

Thus the argument gives the requested embedding under the stated interpretation, and even uses only absolute continuity Q_plus<<W_plus rather than equivalence. It does not require a finite expectation for N and does not give a computable finite-lookahead rule.

Checks and strongest objections

1. Pushforward direction: Lh=h composed with theta^(-1), so removing m_n and then moving forward really gives the stated a_(n+1). Using h composed with theta would reverse the density transport.
2. The bound a_n<=1 is indispensable in the contradiction. It follows from stationarity of the starting probability P; an arbitrary initial distribution would need a different argument.
3. No independence of the sampled future blocks is asserted. The density recursion replaces the invalid idea that infinitely long Brownian futures can be used as disjoint rejection trials.
4. The proof preserves the unconditional Wiener marginal of W. Selecting a suffix changes its law but does not condition the original W on the success of an event; acceptance occurs almost surely.
5. A two-state cyclic transformation with uniform P and target density (2,0) matches half the mass at n=0 and the other half after one shift. The density recursion yields Q exactly and checks its direction and normalization.
6. The mathematical and interpretive risk is that Problem 7.5 imposes an additional implicit restriction on the prefix. The displayed source wording contains no independence or stopping-time restriction. Check the surrounding paper and subsequent literature before labeling the problem solved.
7. A pre-existing embedding/filling theorem may already imply this conclusion. The argument would remain a valid explanation, but novelty and open-at-start status would then need to be withdrawn.

Prior-literature leads being checked: Pitman-Tang (2015), *The Slepian zeros and Brownian bridge embedded in Brownian motion*, Theorem 5.7 (a two-sided random shift), and Chacon-Ornstein/Rost filling schemes. Theorem 5.7 alone allows a shift of either sign; the nonnegative direction is established directly above.

A determinism obstruction and why the auxiliary uniforms matter

A counter-check communicated by the parent agent identifies a real boundary of the lemma. If an integer shift n(z) is a deterministic measurable function of z alone, the full shifted law has density

g(y) = sum_(k in Z) 1_{n(theta^(-k)y)=k}

with respect to P. Indeed, for every measurable A, partitioning by n=k and applying invariance gives the integral of 1_A times this sum. Tonelli applies because the summands are nonnegative. Thus g is integer-valued almost everywhere; its integral is one, so the sum is finite almost everywhere. Arbitrary noninteger f cannot arise this way. This is a counterexample to a stronger, deterministic version of the general lemma, not to the displayed randomized construction.

The present argument explicitly uses independent uniforms. Averaging their conditional acceptance probabilities yields real-valued densities m_n, so the integer-valued obstruction does not apply. Alternatively, a future marginal may hide randomness in the unobserved past; a statement about that projection must be distinguished from a statement about the full two-sided law. No deterministic full-path transport is claimed here.

This check was independently suggested but does not constitute an independent review of the complete candidate proof. The author has re-derived the density formula above. Holroyd and Peres, *Extra heads and invariant allocations* (2005), arXiv:math/0306402, Theorem 17 and the following remark, provide another prior-literature lead: their greedy transport can restrict displacements on Z to nonnegative integers. Their displayed setting uses binary target capacities; the density-matching proof here supplies its own general-density argument.

Proof appendix: P0033_stationary_flow_counterexample_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0033: a stationary Galton-Watson flow that is not ergodic

Written 2026-09-06. Complete candidate counterexample; author checking and prior-literature search are in progress. No independent review or novelty claim. This file is not counted as a solved open problem merely because it exists.

Source: Lyons and Peres, *Probability on Trees and Networks*, version 19 August 2026, Question 17.18, printed page 575 (PDF page 592). The question asks whether a flow rule admitting a stationary measure equivalent to Galton-Watson measure must induce an ergodic Markov chain. Proposition 17.17 proves this under strict positivity of all child weights. Exercise 17.22 gives a nonergodic example with a merely absolutely continuous stationary measure; equivalence is the important extra requirement here.

Construction and claim

Let GW be the law of a rooted Galton-Watson tree with independent offspring numbers D satisfying

P(D=2)=3/5, P(D=3)=2/5.

The tree has no leaves and mean offspring 12/5. We construct an isomorphism-invariant measurable color c(T) in {0,1}, each with probability 1/2, such that almost surely:

* at a vertex with two children, its color is the AND of its child colors;
* at a vertex with three children, its color is the OR of its child colors.

Both Boolean rules preserve unanimity. Consequently every vertex has at least one child of its own color. At every vertex, the flow chooses uniformly among those children. The resulting flow is consistent under restriction to a descendant subtree. GW itself is stationary for the Markov chain that follows this flow; nevertheless the root color is constant along the chain and has probability 1/2 for each value. Thus the chain is not ergodic, despite a stationary measure equal to GW.

The color is a function of the uncolored tree

For a finite truncation of T at level n, give the vertices at that level independent Bernoulli(1/2) colors and propagate colors upward using AND at two-child vertices and OR at three-child vertices. Let q_n(T) be the probability, over these temporary boundary colors, that the root color is one. Thus q_0=1/2. It is a measurable, isomorphism-invariant function of the uncolored tree through level n, with values in [0,1].

The mean output probability when the children independently have probability p of color one is

F(p)=(3/5)p^2+(2/5)[1-(1-p)^3].

Direct substitution gives F(1/2)=1/2. Let F_n denote the sigma-field of the tree through level n, meaning that the offspring numbers at depths below n have been exposed. Conditional on F_n, the offspring numbers of the level-n vertices are independent with the displayed law. The expected probability of a one at each such vertex after one additional level is 1/2. The root Boolean formula is a multilinear polynomial in the probabilities at its boundary vertices, because it is a read-once formula on a tree. Taking conditional expectations therefore gives

E[q_(n+1) | F_n]=q_n.

Hence q_n is a bounded martingale. It converges almost surely and in L2 to a measurable q_infinity, with E q_infinity=1/2.

We must show that q_infinity is {0,1}-valued; otherwise a random boundary color would remain and would not define the required flow on uncolored trees. Put r_n=E[q_n^2]. Equivalently, use two independent boundary colorings of

the same uncolored truncation: r_n is the probability that both root outputs are one. Each output has marginal probability 1/2. For a two-child AND, the probability of two one outputs is r_n^2. For a three-child OR, inclusion-exclusion gives

1-2(1/2)^3+r_n^3=3/4+r_n^3,

since for a pair with both marginals 1/2 the probabilities of 00 and 11 agree. Different child subtrees are independent. It follows that

r_0=1/4,
r_(n+1)=3/10+(3/5)r_n^2+(2/5)r_n^3.

For 1/4 <= r < 1/2, the increment factors as

3/10+(3/5)r^2+(2/5)r^3-r
=(r-1/2)[(2/5)r^2+(4/5)r-3/5] > 0.

The quadratic factor is negative throughout [1/4,1/2], and the recursion maps this interval into itself because it is increasing and fixes 1/2. The only fixed point in this interval is 1/2: the other positive root is (sqrt(10)-2)/2 > 1/2. Thus r_n increases to 1/2. By L2 convergence,

E[q_infinity(1-q_infinity)] = 1/2-lim r_n = 0.

Nonnegativity forces q_infinity in {0,1} almost surely. Define c(T)=q_infinity(T) on this probability-one set. The recursion for q_n at the root is a continuous polynomial in the q_(n-1) values of its child subtrees, so passage to the limit proves the AND/OR color recursion. Apply this simultaneously to all vertices, a countable set. For a global Borel definition on leafless locally finite rooted trees, set c(T)=0 outside the Borel set where the displayed limit exists and is binary (and outside the offspring-2-or-3 tree space). At each vertex apply the same local same-color rule; if there are no same-color children, use equal weights on all children. Defining ray probabilities by products of these local weights gives consistency everywhere, including the null exceptional set, and leaves all GW claims unchanged.

Stationarity with no density change

Write A_c={T:c(T)=c} and GW_c=GW(. | A_c), for c=0,1. Both events have probability 1/2. Conditional on the root offspring number D=d and the vector of its child colors (c_1,...,c_d), the child subtrees are independent, and child i has law GW_(c_i). This follows directly from independent Galton-Watson child subtrees and the fact that each color is a measurable function of its own subtree.

The color of the parent is a deterministic function of d and this vector. Let I_c={i:c_i=c} when the parent color is c. This set is nonempty by the AND/OR rules. Choosing uniformly from I_c uses no further information about the internal shape of the chosen subtree. For every bounded measurable function h,

E[sum_(i in I_c) h(T_i)/|I_c| | D=d, (c_i)]
= integral h dGW_c.

Averaging conditional on the parent color c proves GW_c K=GW_c for the child-selection kernel K. Finally,

GW K = (1/2) GW_0 K + (1/2) GW_1 K
= (1/2) GW_0 + (1/2) GW_1 = GW.

Thus the stationary measure is exactly GW, not merely an absolutely continuous measure supported on a smaller set of trees.

Consistency and failure of ergodicity

The local rule assigns probability 1/|I_c| to each same-color child and zero to the other children. Products of these transition weights assign mass to all finite initial ray segments. Their consistency defines a Borel probability measure on the infinite boundary by the usual cylinder extension. At a descendant receiving positive mass, the conditional continuation is the identical rule applied to that descendant subtree. This is a flow rule in the sense of the source.

The event A_1 has GW measure 1/2, and K(T,A_1)=1_(A_1)(T) almost surely. Therefore it is a nontrivial invariant event for the stationary Markov chain. Equivalently, in its stationary path space the root-color event is invariant under the time shift and has probability 1/2. The chain is not ergodic.

Adversarial checks and remaining status work

1. A recursive distributional fixed point alone is insufficient: it might require extra randomness after the full tree is known. The second-moment calculation above proves this does not happen here.
2. The two boundary trials are independent conditional on the tree, so their joint probability really is E q_n^2. Their unconditional correlation is retained in r_n; treating them as independent would destroy the proof.
3. Merely conditioning GW on the existence of an n-ray does not give equivalence to GW, as the source's Exercise 17.22 illustrates. Here the two invariant classes partition a GW-full set and GW itself is stationary.
4. The selected child may depend on all child colors, but these are exactly the conditions used in the disintegration argument. No unconditional independence after selection is asserted.
5. Rooted isomorphisms permute children; both Boolean rules, q_n, its limit, and uniform selection of eligible children are invariant under those permutations. No hidden labels are used.
6. Strict positivity of every child weight fails on some configurations, as it must in view of Proposition 17.17. At a two-child vertex with colors 0,1 only the zero child is chosen; at a three-child vertex with both colors only the one children are chosen.
7. All arithmetic is exact rational algebra. Independent symbolic or finite-depth checks are diagnostics only; the argument proves all depths and the infinite-tree limit.
8. Search prior work on endogeny and interpretable tree automata before calling this a new resolution. Johnson, Podder, and Skerman, *Random tree recursions: which fixed points correspond to tangible sets of trees?*, arXiv:1808.03019, is a relevant primary source. Its general endogeny criteria are not assumed in this self-contained argument.

Exact scope check: the definition in Section 17.5, printed pages 573-574, explicitly permits nonnegative child probabilities and imposes conditional-flow consistency only at children with positive mass. Equivariance is required and was checked above. Thus zeros in this construction are permitted by the source definition; they are not a silent relaxation of it.

Proof appendix: P0096_FPP_minimum_bound_counterexample_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0096: continuous-law counterexample to the four-edge lower bound

Status update19:18Z2026-09-06: ALREADY REFUTED IN THE LITERATURE. Ahlberg1911.09949 (2019;published2020) explicitly settles this question. The proof below is retained as an author-checked rederivation; no novelty or original solved-open credit. The two-page primary paper was read in full; no independent review of this document occurred.

Source and exact target

Auffinger, Damron and Hanson, *50 years of first-passage percolation*, arXiv:1511.03262, manuscript26September2016, Question12 (PDF p.33; repeated p.146), asks whether iid nonnegative edge passage times on Z^2 with F(0)=0 always satisfy

\[\mu(e_1)\geq \mathbb E\min(\tau_1,\tau_2,\tau_3,\tau_4).\]

E_4 denotes expectation of a minimum of four independent copies, not a fourth moment. The survey motivates this by Alm-Deijfen's simulation study, arXiv:1412.5924 (December2014). That paper describes an observed relation for selected distribution families. The target here is the survey's universal formulation, not a universal theorem attributed to the simulation authors.

Primary sources: https://arxiv.org/pdf/1511.03262 and https://arxiv.org/pdf/1412.5924. The two standard time-constant inputs below are stated in the latter paper, equations(2),(3), PDF p.3.

Construction and lower bound

Independently for each lattice edge e, take eta_e in {0,1} with P(eta_e=0)=3/4 and an independent U_e uniform on [0,1]. For 0<epsilon<=1 put

\[\tau_e^{(\epsilon)}=\eta_e+\epsilon U_e.\]

The resulting iid law is bounded and absolutely continuous: its density is 3/(4epsilon) on (0,epsilon) and 1/(4epsilon) on (1,1+epsilon). It has no atoms, F_epsilon(0)=0, and infimum of support zero. Every moment exists.

For four independent copies, the event that all eta values equal1 has probability (1/4)^4=1/256. Their minimum is at least1 on this event, so

\[\mathbb E\min_{1\le i\le4}\tau_i^{(\epsilon)}\geq1/256.\tag{1}\]

Time constants tend to zero

For bounded iid nonnegative edge weights the established subadditive formula is

\[\mu(e_1)=\inf_{n\ge1}\mathbb E T(0,ne_1)/n.\tag{2}\]

The established positivity criterion says mu(e_1)>0 iff F(0)<p_c(Z^2)=1/2. Hence the eta-only model has mu_0=0 because its zero mass is3/4. Each positive-epsilon model has mu_epsilon>0.

Fix n and couple all epsilon values with the same eta,U. Trivially T_epsilon(0,ne_1)>=T_0(0,ne_1). Given delta>0, choose a finite path gamma whose eta-cost is at most T_0(0,ne_1)+delta. Such a path exists by the definition of an infimum; a minimizing path is not assumed. Since U_e<=1,

\[T_\epsilon(0,ne_1)\le T_0(0,ne_1)+\delta+\epsilon|\gamma|.\]

First send epsilon to zero and then delta to zero. This proves T_epsilon(0,ne_1) tends almost surely to T_0(0,ne_1). The straight coordinate path gives 0<=T_epsilon(0,ne_1)<=2n. Dominated convergence therefore gives

convergence of expectations.

For each fixed n, (2) now implies

\[\limsup_{\epsilon\downarrow0}\mu_\epsilon(e_1)
\leq \mathbb E T_0(0,ne_1)/n.\]

Take the infimum over n and use mu_0=0. Nonnegativity yields

\[\lim_{\epsilon\downarrow0}\mu_\epsilon(e_1)=0.\tag{3}\]

By(1),(3), every sufficiently small positive epsilon satisfies

\[0<\mu_\epsilon(e_1)<1/256\leq\mathbb E\min_{i\le4}\tau_i^{(\epsilon)}.\]

Thus this explicit family eventually reverses the proposed universal inequality. The proof gives existence of a small-enough parameter, not a numerical cutoff. All sufficiently large integers k give examples with epsilon=1/k.

Adversarial checks and limits

The argument never interchanges an infinite-path minimization with a limit. It uses a finite approximate minimizer for each fixed n, then an upper bound valid for every n. No uniform bound on geodesic length is needed.

The limiting distribution has an atom at zero; the actual counterexample distributions do not. Their support has a gap, which the survey's stated hypotheses do not prohibit. Thus this is stronger than an atomic-law loophole.

A local entrance cost remains positive even when long travel can exploit a supercritical cheap-edge network; dividing a single entrance cost by distance explains the failure of the proposed bound. As a second literature lane, Cox-Kesten continuity of the time constant, cited in Alm-Deijfen, gives the same limiting conclusion for this bounded weakly convergent family. The direct proof above does not need that general continuity theorem.

Searches for Question12, Alm-Deijfen lower bounds, and minimum-of-four counterexamples did not locate a matching prior construction in the inspected results. This does not establish novelty. Before promotion, recheck source notation and standard theorem hypotheses, compare the original simulation discussion and later errata, and obtain independent review. No mathematical gap has been identified in the current argument, but neither independent review nor current open status is claimed.

Later-literature correction and an explicit parameter

Ahlberg, Tertiles and the time constant, https://arxiv.org/pdf/1911.09949, Theorem1 and final paragraph, already gives counterexamples. The publication is Journal of Applied Probability57(2),407-408(2020), DOI10.1017/jpr.2020.9. The earlier search statement above records the preliminary search only and is superseded by this finding.

For our family, its two-thirds quantile is8epsilon/9. Ahlberg proves mu<=2t_(2/3), so epsilon=1/1024 gives mu<=1/576<1/256<=E minfour. This is an explicit atomless bounded example derived from an established theorem. The continuity argument above remains a separate qualitative derivation. No original solution is claimed.

Proof appendix: P0140_thinned_levy_random_drift_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0140: Poisson marking explains the random critical drift

Status: author-checked derivation, no independent review, no novelty claim. Excluded from the current-open target and original-solution counts. The filename preserves the initial candidate workflow. The later primary paper already states an indirect identification with Joseph's independent-increment limit. What is supplied here is a direct calculation and an explicit obstruction to holding the critical drift deterministic.

Sources: van der Hofstad, *Stochastic processes on random graphs*, July12,2017, equations4.5.5 and4.5.20, Open Problem4.1, PDF pp.206,210; https://www.win.tue.nl/~rhofstad/SaintFlour_SPoRG_fin_Saint_Flour.pdf . Dhara,van der Hofstad,van Leeuwaarden,Sen, *Heavy-tailed configuration models at criticality*, December2019 version, equation2.22 and Remark11, PDF pp.9-10; https://arxiv.org/pdf/1612.00650 . Joseph's original independent-increment limit: https://arxiv.org/abs/1012.2352 . Exact comparison with every convention in Joseph's characteristic function remains a separate check; the independence statement below is direct.

Setup and the distinction in the question

Fix 3<tau<4, c>0, and mu>0. Let Gamma_i be the arrival times of a rate-one Poisson process and set theta_i=(c/Gamma_i)^(1/(tau-1)). Thus the theta_i form a Poisson point process on(0,infinity) with intensity

rho(dx)=(tau-1)c x^(-tau) dx.

Conditionally on these points, give each theta_i an independent exponential clock T_i with rate theta_i/mu. The source's process with a deterministic real parameter lambda is

S_lambda(t)=sum_i theta_i[1{T_i<=t}-(theta_i/mu)t]+lambda*t.

The ordered series is well defined. For small theta_i, its conditional means are O(theta_i^3) and its conditional variances are O(theta_i^3); sum theta_i^3 is finite almost surely. There are finitely many points above1, each finite. A process-level construction also follows from the compensated Poisson representation below.

The important parameter for iid degrees is the RANDOM variable Lambda_0 in equation2.22 of the later paper, not a fixed lambda. Define

Z=lim_(epsilon down0)[sum_(theta_i>epsilon) theta_i^2 - integral_(epsilon,infinity) x^2 rho(dx)].

The small-point part of this compensated Poisson integral converges in L2 and almost surely because integral_(0,1) x^4 rho(dx)<infinity. The large-point part is finite almost surely and has finite first absolute moment because integral_(1,infinity) x^2 rho(dx)<infinity. Its second moment need not be finite. Set Lambda_0=Z/mu.

At the random cutoff epsilon=theta_n, including the nth point changes the sum only by theta_n^2, which tends to0. Since

integral_(theta_n,infinity) x^2 rho(dx)=c^(2/(tau-1))*Gamma_n^(1-2/(tau-1))/(1-2/(tau-1)),

the definition of Z/mu is exactly equation2.22 after telescoping the integrals over[Gamma_(i-1),Gamma_i]. This checks the random-drift convention against the primary source, not merely by analogy.

Direct independent-increment representation

For every epsilon>0 the cancellation is an identity:

S_0,epsilon(t)+(t/mu)Z_epsilon
= sum_(theta_i>epsilon) theta_i 1{T_i<=t} -(t/mu) integral_(epsilon,infinity) x^2 rho(dx).

By independent Poisson marking, the pairs(theta_i,T_i) form a Poisson random measure with deterministic intensity

nu(dx,ds)=rho(dx)*(x/mu)*exp(-xs/mu) ds.

The jump sum in the last display has independent increments in t, since disjoint time intervals select disjoint sets of this Poisson measure. Its centering is deterministic. Passing to finite-dimensional limits preserves independence, so

L(t):=S_0(t)+(t/mu)Z=S_(Lambda_0)(t)

has independent increments. They are generally NONSTATIONARY: the jump intensity contains exp(-xs/mu).

For completeness, this passage is not based on a formally divergent cancellation. Write the truncated expression as M_epsilon(t)+D_epsilon(t), where

M_epsilon(t)=sum_(theta_i>epsilon) theta_i1{T_i<=t}
-integral_(epsilon,infinity) x(1-exp(-xt/mu))rho(dx),

D_epsilon(t)=integral_(epsilon,infinity) x[1-exp(-xt/mu)-xt/mu]rho(dx).

On bounded time intervals, the centered contribution from points x<=1 is an L2 martingale in t. The variance contributed by0<x<=epsilon is bounded by

(T/mu) integral_(0,epsilon) x^3 rho(dx),

which tends to0 because tau<4. Doob's inequality gives uniform-in-time L2 convergence of this small-point martingale. The deterministic drift converges uniformly on bounded time intervals because its integrand near0 is O(x^3), uniformly for bounded t. The finitely many large points and their finite deterministic compensator cause no problem. Hence L has a cadlag version obtained by this construction, and S_0=L-(t/mu)Z does too.

The deterministic drift can also be evaluated:

D(t)= - (tau-1)c * Gamma(4-tau)/[(tau-2)(tau-3)] * (t/mu)^(tau-2).

To check the sign and constant, substitute u=xt/mu and integrate twice by parts in the positive integral of u^(1-tau)(exp(-u)-1+u). All boundary terms vanish for3<tau<4. The remaining integral is Gamma(4-tau)/[(tau-2)(tau-3)]. In particular D(t)<0, consistent with depletion of previously discovered vertices.

A deterministic lambda does not have independent increments

This part checks the strongest way the preceding conclusion could be misreported. Consider increments A=S_0(mu) and B=S_0(2mu)-S_0(mu). A marked point of size x contributes

F1=x1{T<=mu}-x^2,
F2=x1{mu<T<=2mu}-x^2.

Let K(a,b)=log E exp(aA+bB), for a,b>=0. These Laplace transforms are finite. Indeed, large points have nonpositive F1,F2 once x>=1; for small points the linear terms have integrable expectation O(x^3) and the quadratic remainder has integrable expectation O(x^3). The compensated Poisson formula, or its finite-cutoff version followed by these bounds, gives

K(a,a)-K(a,0)-K(0,a)
=integral rho(dx) E_T[(exp(aF1)-1)(exp(aF2)-1)].

For x>=1 both F1 and F2 are nonpositive, so the integrand is nonnegative. If0<a<=1/4 and x lies in[a^(-1/2),2a^(-1/2)], then x>=2 and F1,F2<=-x^2/2. Consequently the integrand is at least(1-exp(-1/2))^2 on this interval. Its rho-mass is

c*(1-2^(1-tau))*a^((tau-1)/2).

The potentially negative part is confined to0<x<1. The two indicators in F1,F2 are disjoint, and

E|F1F2| <= x^3 P(T<=2mu)+x^4 <=3x^4.

Since |exp(au)-1|<=a|u|exp(a|u|) and |Fj|<=2 on this range, the absolute contribution of this whole region is at most

3a^2 exp(4a) integral_(0,1) x^4 rho(dx)=C*a^2 exp(4a).

Thus the mixed log-Laplace difference is bounded below by a positive constant times a^((tau-1)/2) minus C*a^2 exp(4a). Since (tau-1)/2 is strictly less than2, it is positive for all sufficiently small a>0. Independence would make this difference identically0. Therefore A and B are dependent. Adding any deterministic lambda changes only their deterministic means and cannot fix this dependence.

This is a direct counterexample to interpreting the 2017 question as a claim about deterministic lambda after mixing the weights. It does not refute the corrected iid-degree statement, whose random drift is explicitly supplied in the later paper.

Checks and remaining status limits

The argument has two complementary checks: Poisson marking proves independence after the exact random-slope cancellation; the mixed Laplace estimate disproves independence without it. The latter avoids covariance calculations, which would be invalid because the random slope can have infinite variance. The PPP intensity follows independently from the tail count E number{theta_i>x}=c*x^(-(tau-1)). The source's Gamma-coordinate expression telescopes to exactly the same compensated slope.

No simulation is used as proof. No independent referee has checked this text. The 2019 paper already identifies the corrected law indirectly, so this derivation does not establish an unproved theorem for the corpus and is not counted as an original solution. Later literature may also contain this direct marking calculation; that novelty search remains incomplete.

Proof appendix: P0174_free_brownian_unimodality_constant_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0174: candidate sharp diameter threshold for free Brownian unimodality

Status: complete candidate proof, author-checked in two analytic formulations; no independent review and no claim of established novelty. Current-literature search is incomplete. Do not count as a promoted original solution.

Source: Hasebe and Ueda, *Large time unimodality for classical and free Brownian motions with initial distributions*, ALEA 15 (2018), 353-374, Problem 3.3; archived arXiv1710.08240v2, PDF p.6. The PDF carries the arXiv revision date14 September2018 and a manuscript header4 November2021; retain that discrepancy. Primary links: https://arxiv.org/abs/1710.08240 and https://alea.math.cnrs.fr/articles/v15/15-15.pdf . The source defines S(0,t) to have variance t, and D_mu to be support diameter.

The source proves the sufficient constant4 and asks for the optimal universal C in t>=C D_mu^2. Its paragraph says the symmetric Bernoulli law on{-1,1} has threshold t=4 and diameter2, then prints2<=C<=4. Those preceding numbers imply the lower bound C>=1. The candidate upper bound below is a separate argument, not a conclusion drawn from that arithmetic inconsistency.

Candidate theorem

If mu is a compactly supported probability measure on R with support diameter D, then mu free-convolved with the centered semicircle law of variance t is unimodal whenever t>=D^2. The optimal universal constant in the source's normalization is therefore C=1.

When D=0 the measure is a point mass and the assertion is immediate. Henceforth D>0. No symmetry, density, or non-atomicity of mu is assumed.

An elementary moment inequality

Let Q be a probability law of a centered real variable X supported in an interval of length D. Write a=E_Q X^2 and b=E_Q X^4. Then

\[
a\le D^2/4,\qquad b+3a^2\le D^2a.
\]

For the first inequality, if the supporting interval is[l,h], then E[(X-l)(h-X)]>=0 gives a<=-lh, and -lh<=D^2/4 because l+h has a real square. For the second, let X' be an independent copy. Pointwise, (X-X')^4<=D^2(X-X')^2. Taking expectations and using E X=0 gives2b+6a^2<=2D^2a. This derivation checks both the coefficient3 and the use of the diameter rather than radius.

Equality a=D^2/4 forces l=-D/2,h=D/2 and Q to put equal mass on the two endpoints: equality in E[(X-l)(h-X)]>=0 forces endpoint support, and centering fixes the weights.

Cauchy-profile lemma

For R>0 define

\[
F_R(u)=\int\frac{d\mu(x)}{(x-u)^2+R^2}.
\]

Claim: F_R has no local minimum with value at most1/D^2.

Take any stationary point u of F_R with F_R''(u)>=0. Put y=R^2>0 and

\[
Z=\int ((x-u)^2+y)^{-2}\,d\mu(x),\qquad
dQ(x)=Z^{-1}((x-u)^2+y)^{-2}\,d\mu(x).
\]

Differentiation under the integral is justified by R>0 and compact support. Stationarity says E_Q(X-u)=0. With a=E_Q(X-u)^2 and b=E_Q(X-u)^4, the preceding moment inequalities apply. The second derivative is

\[
\frac{F_R''(u)}{2Z}
=E_Q\frac{3(X-u)^2-y}{(X-u)^2+y}
=3-4y E_Q\frac1{(X-u)^2+y}.
\]

Jensen's inequality and F_R''(u)>=0 imply

\[
\frac1{a+y}\le E_Q\frac1{(X-u)^2+y}\le\frac3{4y},
\qquad y\le3a.
\]

The normalization of Q gives an exact reciprocal-profile identity:

\[
\frac1{F_R(u)}
=\frac{E_Q[((X-u)^2+y)^2]}{E_Q[(X-u)^2+y]}
=\frac{b+2ay+y^2}{a+y}.
\]

Consequently

\[
\begin{aligned}
\frac1{F_R(u)}-D^2
&\le\frac{y^2+2ay-3a^2-D^2y}{a+y}\\
&=\frac{(y-3a)(y+a)+(4a-D^2)y}{a+y}\le0.
\end{aligned}
\]

The first inequality uses b<=D^2a-3a^2; the final two summands are nonpositive because y<=3a and4a<=D^2. In particular, F_R(u)>=1/D^2.

The equality case matters for the endpoint t=D^2. If F_R(u)=1/D^2 and F_R''(u)>=0, then the two final nonpositive terms both vanish: y=3a and4a=D^2, since y>0. Equality in the variance bound forces Q to be symmetric Bernoulli on u plus/minus D/2. On these two points the reweighting denominator is equal, so mu is the same symmetric Bernoulli law. Writing c=D/2 and s for displacement from its midpoint, with R^2=3c^2, gives

\[
F_R(u+s)=\frac{s^2+4c^2}{s^4+4c^2s^2+16c^4},
\qquad
F_R(u+s)-\frac1{4c^2}
=-\frac{s^4}{4c^2(s^4+4c^2s^2+16c^4)}<0\quad(s\ne0).
\]

Thus the only possible equality point is a strict maximum, not a local minimum. Every local minimum therefore has value strictly greater than1/D^2, proving the claim including equality.

Unimodality via Biane's density parametrization

For fixed t>=D^2 and fixed R>0, the equation F_R(u)=1/t has at most two solutions. Indeed, if there were three solutions u1<u2<u3, then either F_R dips below1/t somewhere in[u1,u3], producing an interior local minimum below that level, or F_R>=1/t throughout that interval, making u2 a local minimum at that level. Both alternatives contradict the Cauchy-profile lemma because1/t<=1/D^2. The analytic function cannot be identically constant on an interval, since F_R tends to0 at infinity.

Now use exactly the density framework stated in source Section2.2 and Lemma3.1. Define v_t(u)>=0 by

$$
\int\frac{d\mu(x)}{(x-u)^2+v_t(u)^2}=\frac1t
$$

where its positive solution exists, setting v_t=0 elsewhere. Biane's real homeomorphism psi_t satisfies

$$
p_t(\psi_t(u))=\frac{v_t(u)}{\pi t},
$$

where p_t is the continuous density of mu free-convolved with S(0,t), analytic wherever positive. Each positive density level p_t=a corresponds to F_(pi*a*t)(u)=1/t, and therefore has at most two points. Source Lemma2.2, equivalently Lemma3.1, now gives unimodality. All source assumptions match: mu is compactly supported, t>0, and the semicircle parameter is variance.

Sharpness

Let mu be symmetric Bernoulli on{-c,c}, c>0, so D=2c. For c^2<t<4c^2, the positive solution at the symmetry center has v_t(0)^2=t-c^2. Implicit differentiation at0 yields

$$
v_t''(0)=\frac{3c^2-v_t(0)^2}{v_t(0)(c^2+v_t(0)^2)}
=\frac{4c^2-t}{t\sqrt{t-c^2}}>0.
$$

Thus v_t has a positive strict local minimum at0, and the density under the homeomorphism psi_t does as well. It is not unimodal. For any proposed C<1, choose t strictly between max(C,1/4)D^2 and D^2. Then t>=CD^2 but unimodality fails. This proves the lower bound C>=1 independently of the source's printed inequality.

Second analytic check and remaining review obligations

The same moment computation can be applied directly at critical points of v_t. At such a point, the profile's first derivative vanishes and the sign of v_t'' equals the sign of F_v-fixed''. The calculation above says a nonnegative second derivative would force t<=D^2. Hence for t>D^2 every critical point of v_t is a strict maximum, giving the same threshold. The equality analysis in the Cauchy-profile proof closes the boundary without appealing to weak closure of unimodal laws.

The load-bearing points to challenge are the Q-normalization identity, the centered fourth-moment inequality, the direction of Jensen, and the equality case at t=D^2. Each is written explicitly; neither a numerical validator nor brain retrieval supplies independent mathematical review. Later literature might already contain this sharp threshold or the same argument. The statement remains a candidate until further proof checking and literature comparison are complete.

Proof appendix: P0189_brown_measure_strong_convergence_counterexample_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0189: candidate counterexample to Brown-measure continuity under strong convergence

Status: complete candidate construction, checked by the author; no independent review and no novelty claim. This addresses the literal general conjecture in van Handel, *The strong convergence phenomenon*, arXiv2507.00346v2 (11 February2026), Section6.9, pp.69-70. It does not address the narrower conjectures for independent Ginibre matrices or free limiting families.

Primary source for the question: https://arxiv.org/abs/2507.00346 . Primary source for the established limiting Brown measure: Fang, Jiang, Lin and Xu, *On generalized universal irrational rotation algebras and the operator u+v*, https://arxiv.org/abs/1210.4771 , Theorem12.2. Its introduction also states simplicity and uniqueness of trace of the irrational rotation algebra. These literature theorems are dependencies, not results proved here.

Finite matrices

Fix an irrational theta in(0,1). Choose reduced rational approximants p_j/q_j tending to theta, with q_j tending to infinity. Let omega_j=exp(2*pi*i*p_j/q_j). On the standard basis indexed by k modulo q_j define

\[
U_j e_k=\omega_j^k e_k,\qquad
S_j e_k=e_{k-1},\qquad
V_j=e^{\pi i/q_j}S_j.
\]

Then U_j and V_j are unitary and satisfy

\[
V_jU_j=\omega_jU_jV_j,\qquad U_j^{q_j}=I,\qquad V_j^{q_j}=-I.
\]

Put A_j=U_j+V_j. Its characteristic polynomial is exactly z^{q_j}. Indeed, in the determinant of zI-U_j-V_j only the all-diagonal permutation and the full cyclic permutation contribute (for q_j>=2). The former contributes

\[
\prod_{k=0}^{q_j-1}(z-\omega_j^k)=z^{q_j}-1,
\]

because omega_j is primitive. The latter contributes minus the product of the cyclic weights, namely -exp(pi*i)=+1. Thus det(zI-A_j)=z^{q_j}. Cayley-Hamilton gives A_j^{q_j}=0, and every empirical eigenvalue distribution is delta_0.

An equivalent algebraic check is the root-of-unity binomial identity (U_j+V_j)^{q_j}=U_j^{q_j}+V_j^{q_j}=0. The determinant calculation supplies the conclusion without relying on that identity.

Strong convergence of the pair

Let A_theta be the universal unital irrational rotation C*-algebra generated by unitaries u,v satisfying vu=exp(2*pi*i*theta)uv. This algebra is simple and has a unique faithful tracial state tau, as recalled in the cited primary paper.

Consider the norm quotient

\[
\mathcal Q=\prod_j M_{q_j}(\mathbb C)\big/\bigoplus_j M_{q_j}(\mathbb C),
\]

\]

where the direct sum consists of sequences whose operator norms tend to zero. The classes of (U_j) and (V_j) are unitaries and satisfy the irrational relation exactly, since omega_j tends to exp(2*pi*i*theta). Universality therefore gives a unital *-homomorphism A_theta to Q. Simplicity makes it injective, hence isometric. Since the quotient norm of a bounded sequence is its limsup norm, for every noncommutative *-polynomial P,

\[
\limsup_j\|P(U_j,V_j,U_j^*,V_j^*)\|
=\|P(u,v,u^*,v^*)\|.
\]

The same argument applies to every subsequence. If the norms failed to converge to the displayed value, a subsequence uniformly separated from that value would contradict the same limsup identity. Consequently the pair converges strongly.

The normalized traces also converge to tau. Using the commutation relation, every fixed word is a phase times U_j^a V_j^b. When b is a fixed nonzero integer and j is large, the shift has no diagonal entries, so its trace is zero. If b=0 and a is fixed nonzero, the geometric sum of omega_j^{ak} is zero for all q_j>|a|. The identity word has trace one, and phases converge. These are exactly the canonical irrational-rotation trace values. Thus both norm and tracial versions of strong convergence are satisfied, with a faithful limiting trace.

In particular A_j converges strongly, including all *-polynomials in A_j alone, to a=u+v.

The survey Definition 1.1 also allows fixed finite-dimensional matrix coefficients. For each fixed D, amplify the quotient homomorphism to M_D(A_theta). An injective *-homomorphism is completely isometric, and M_D(Q) identifies with the quotient of the product of M_D(M_qj) by norm-null sequences. Therefore the same limsup identity and subsequence argument apply to every D-by-D matrix-valued *-polynomial. The identification follows entrywise in finite D: a matrix sequence tends to zero in norm exactly when all its entries do, and the quotient norm is the limsup of the block-matrix norms. Thus the precise matrix-coefficient definition is covered.

A separate norm-convergence proof uses the trace computation above for the lower bound and needs no appeal to simplicity. Write P_j for a fixed matrix-valued polynomial and P for its limit. For every positive integer m, with normalized traces on the matrix factors,
\[
\|P_j\|\ge\big[(\mathrm{tr}_D\otimes\mathrm{tr}_{q_j})((P_j^*P_j)^m)\big]^{1/(2m)}.
\]
Taking the lower limit in j and then m to infinity gives liminf ||P_j|| >= ||P||, because the limiting trace tr_D tensor tau is faithful. Complete contractivity of the universal quotient homomorphism gives limsup ||P_j|| <= ||P||. This independently proves the required full norm convergence. The parent suggested these two explicit extensions during its first bounded review on 6 September 2026; the author checked and incorporated them. The remaining review and novelty status is pending.

The Brown-measure mismatch

The cited Theorem12.2 states that the Brown measure of u+v, in the tracial von Neumann algebra generated by the irrational rotation pair, is Haar probability measure on the unit circle. It is therefore different from delta_0. The finite empirical measures are all delta_0 and do not converge to the Brown measure of a, despite strong convergence.

The same example also prevents a universal lower spectral-radius conclusion: every A_j has spectral radius zero, whereas the source Theorem11.1 gives spectrum of u+v equal to the closed unit disk and hence spectral radius one.

To make the matrices random rather than deterministic, take Haar unitary Q_j and replace the pair by (Q_j U_j Q_j*,Q_j V_j Q_j*). For every outcome, norms, normalized traces, nilpotency, and eigenvalue distributions are unchanged. This supplies genuinely random families satisfying the same convergence. The two members are dependent; the construction does not claim otherwise. Equivalently, the single random matrix Q_j A_j Q_j* is

already a counterexample for a one-variable polynomial.

Optional realization in every integer dimension

The dimensions q_j suffice for a counterexample along a sequence tending to infinity. If an all-integer-dimension formulation is required, take consecutive continued-fraction denominators, which are coprime. For each sufficiently large N, choose the largest j with q_j*q_{j+1}<=N. The elementary coin theorem gives N=a*q_j+b*q_{j+1} for nonnegative integers a,b. Define the N-dimensional pair as the block direct sum of a copies of the j-th pair and b copies of the(j+1)-st pair. Both block dimensions tend to infinity as N does. Polynomial norms are the maximum of their block norms, traces are convex combinations of block traces, and the sum remains nilpotent. Thus the same counterexample exists with dimension exactly N for every sufficiently large integer N.

Scope and remaining checks

This is a candidate refutation of the survey's literal implication from arbitrary strong convergence to empirical Brown-measure convergence. It is not a refutation for independent-entry ensembles, free limits, or holomorphic polynomials in independent Ginibre matrices. Yi Han's June2026 paper, arXiv2606.01664, claims the latter case and is compatible with this construction.

Load-bearing obligations are the precise survey scope, the primitive-root determinant sign, isometry of the quotient homomorphism, faithful-trace interpretation, and the cited Brown-measure theorem. All are explicit here. It remains possible that the construction or the general failure of continuity is already known; current-literature novelty is unverified. No validator, numerical eigenvalue computation, or brain ingestion is treated as proof review. In fact numerical eigenvalues of these nilpotent matrices would be an especially unreliable check because of their extreme nonnormality.

Proof appendix: P0193_compact_multiplicative_nonunimodality_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0193: a compactly supported initial law with persistent failure of log-unimodality

Status: complete author-checked candidate counterexample, 2026-09-06. No independent review and no novelty certification. This is a proposed negative answer to Hasebe, Ueda and Wang, *Log-unimodality for free positive multiplicative Brownian motion*, arXiv:2009.13848v1 (29 September 2020), Problem 4.7, PDF page 11. The published reference is Colloquium Mathematicum 169 (2022), 209-226; the archived version used here is the arXiv version. Source: https://arxiv.org/abs/2009.13848.

Claim and normalization

Let $\nu=(\delta_{1/10}+\delta_{10})/2$. In the source normalization
$$\Sigma_{\sigma_t}(z)=\exp\!\left(\frac t2\frac{z+1}{z-1}\right),$$
the law $\sigma_t\boxtimes\nu$ is **not log-unimodal for every $t\ge5\pi$**. Its initial law has support in the compact interval $[1/10,10]$, so this disproves eventual log-unimodality for arbitrary compactly supported positive initial laws, if the argument and source match survive review.

Log-unimodality means unimodality of the image measure under the logarithm; ordinary unimodality on the positive line is a different claim. The initial law is multiplicatively symmetric but is not log-unimodal. Thus the source theorem for symmetric log-unimodal initial laws does not apply.

The exact source criterion

Lemma 4.1(3), PDF page 8, says that $\sigma_t\boxtimes\nu$ is log-unimodal if and only if, for every $R\in(0,\pi)$, the equation
$$\frac{\sin R}{R}\int_0^\infty\frac{r\xi}{1+r^2\xi^2-2r\xi\cos R}\,d\nu(\xi)=\frac1t\tag{1}$$
has at most two solutions $r>0$. We use its contrapositive by finding four distinct solutions for one value of R, for each specified t.

Put $a=\log10$, $u=\log r$, and
$$K_R(s)=\frac{1}{2(\cosh s-\cos R)},\qquad F_R(u)=\frac{K_R(u-a)+K_R(u+a)}2.$$
The integral in (1) equals $F_R(u)$, because dividing its integrand's numerator and denominator by $r\xi$ gives
$$\frac{r\xi}{1+r^2\xi^2-2r\xi\cos R}=\frac1{r\xi+(r\xi)^{-1}-2\cos R}=K_R(\log r+\log\xi).$$
Hence (1) is equivalent to
$$F_R(u)=\frac{R}{t\sin R}.\tag{2}$$
For fixed $R\in(0,\pi)$, the function F_R is continuous, positive and even, and tends to zero at both infinities.

Selecting the level and time

The function $g(R)=R/\sin R$ is strictly increasing from 1 to infinity on $(0,\pi)$. Indeed
$$g'(R)=\frac{\sin R-R\cos R}{\sin^2R}>0,$$
because the numerator vanishes at zero and has derivative $R\sin R>0$. Also $g(\pi/2)=\pi/2$.

For each $t\ge5\pi$, there is therefore a unique $R_t\in[\pi/2,\pi)$ with
$$g(R_t)=\frac t{10}.$$
For this choice the right side of (2) is exactly $1/10$, and $\cos R_t\le0$.

Four strict crossings

Since $\cosh(\log10)=101/20$,
$$F_{R_t}(0)=K_{R_t}(a)=\frac1{2(101/20-\cos R_t)}\le\frac{10}{101}<\frac1{10}.\tag{3}$$
At $u=a$, the contribution from the centered kernel alone gives
$$F_{R_t}(a)=\frac{K_{R_t}(0)+K_{R_t}(2a)}2\ge\frac1{4(1-\cos R_t)}\ge\frac18>\frac1{10}.\tag{4}$$
The same inequality holds at $-a$ by evenness. Together with the limits zero at both infinities, (3)-(4) and the intermediate value theorem give a solution of $F_{R_t}(u)=1/10$ in each of the four disjoint intervals
$$(-\infty,-a),\quad(-a,0),\quad(0,a),\quad(a,\infty).$$
Exponentiation is injective, so these give four distinct positive solutions of (1). Lemma 4.1 proves the claim.

Separate algebra check in the original variable

The same signs can be checked without hyperbolic functions. At $r=1$, both atom integrands equal
$$\frac{10}{101-20\cos R}\le\frac{10}{101}<\frac1{10}.$$
At $r=10$, the atom $\xi=1/10$ contributes, including its mass one half,
$$\frac1{2(2-2\cos R)}=\frac1{4(1-\cos R)}\ge\frac18.$$
At $r=1/10$, the atom $\xi=10$ gives the same lower bound. The integral tends to zero as $r\downarrow0$ and as $r\uparrow\infty$. Thus there are crossings in $(0,1/10),(1/10,1),(1,10),(10,\infty)$. These strict rational inequalities include the boundary time 5π, for which $R_t=\pi/2$.

Adversarial checks and limits

The crucial direction is the equivalence in Lemma 4.1, rather than a merely sufficient unimodality condition. Its full statement and proof must remain pinned to the source. The four crossings concern a positive density level, not just multiple extrema in an auxiliary transform. The time is selected separately for each t, as the criterion permits. The upper bound on the central value is uniform for $R\ge\pi/2$; the lower bound at the two flanks is uniform for all $R\in(0,\pi)$. No numerical root finder or plot is used as proof. The argument only needs failure for arbitrarily large times; it proves failure for every time in the displayed half-line. It makes no assertion about smaller times or optimality of 5π.

The large-time mechanism can also be understood from $K_R(s)\to[4\cosh^2(s/2)]^{-1}$ as $R\uparrow\pi$: its width does not diverge. A separated two-point mixture of this limiting kernel remains bimodal. This observation motivated the construction; the finite-time rational inequalities above are the proof.

Focused primary-source searches on 6 September 2026 found the original paper, the authors' current publication pages, and Auer's 2026 paper on the law started at one. No matching later resolution of this exact compact-initial-law question was verified. This is not a certificate of current openness or originality. Author review and source checking are separate from independent review, which is pending.

Proof appendix: P0208_wreath_graph_counterexample_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0208: the wreath graph and Aldous-Fill Problem 7.27

Status: complete counterexample candidate, checked by the author through hitting-time and spectral calculations. No independent review or novelty certificate. The graph itself is an established construction, not an invention here. This is a response to a historically sourced question; present-day open status remains unverified.

Primary question: Aldous and Fill, *Reversible Markov Chains and Random Walks on Graphs*, 2002 unfinished monograph, 2014 recompilation, Open Problem 7.27, PDF p.259, https://www.stat.berkeley.edu/~aldous/RWG/book.pdf . Section 7.2 is internally dated January 31, 1994. It asks, for arc-transitive graphs other than cycles on n vertices, whether maximal mean commute time is O(n log n), relaxation time is O(n), and the ratio of maximal commute time to twice stationary-average hitting time tends to one. The three assertions are counted as one related question.

Source scope and a printed definition error

Page 254 prints an inconsistent variable assignment in the definition of arc-transitivity: the images of two possible identical starting vertices would be allowed to be distinct. Read literally, that formula excludes any graph with a vertex having two neighbors. The surrounding discussion, examples of tori and Cayley graphs, and Lemma 7.19 unmistakably use the usual definition: the automorphism group is transitive on ordered adjacent pairs. Our result concerns that intended standard definition, and does not silently claim to satisfy the misprinted formula.

Conder's 2013 Rogla summer-school notes, p.8, independently describe the same established wreath/necklace graph and its arc-transitivity: https://www.famnit.upr.si/sl/resources/files/konference/rogla2013/conderlecturenotes.pdf . This is evidence that the construction is prior art. It does not establish when the probability question was answered.

Construction and symmetry

Let m be an odd integer at least five, and put

$$
V_m=(\mathbb Z/m\mathbb Z)\times\{0,1\},\qquad
(i,b)\sim(i+1,c),\ (i,b)\sim(i-1,c)\quad(c=0,1).
$$

This is the lexicographic product of a cycle with an edgeless two-vertex graph. It is connected, simple, 4-regular, and has n=2m vertices, so it is not an n-cycle. Dihedral symmetries act on the first coordinate. Independently swapping the two vertices in any fiber preserves every edge. Given two directed edges, a dihedral symmetry sends their ordered base edges to each other, after which independent swaps in the two distinct endpoint fibers match both second coordinates. Thus the graph is arc-transitive.

Use discrete-time simple random walk, or its rate-one continuization. The book states on p.115 that the parameters at issue are unchanged by this continuization. The first-coordinate projection is simple random walk on the m-cycle, and the second coordinate is replaced at each step by a fresh fair bit independent of the base step.

Exact hitting times

Write h(v,w) for the mean first hitting time, with h(w,w)=0. From a base position at cyclic distance j in {1,...,m-1}, the mean first arrival at base fiber zero is

$$
H_j=j(m-j).
$$

\]

Indeed H_0=H_m=0 and H_j=1+(H_{j-1}+H_{j+1})/2. At the first fiber arrival, the fair bit hits a specified vertex with probability one half.

Let h be the mean hitting time from its twin to the specified vertex. Starting in that fiber, the first positive return of the base walk has mean m. The arrival bit succeeds with probability one half; on failure the same twin state recurs. Hence h=m+h/2, giving h=2m. Therefore

\[
h((j,b),(0,0))=j(m-j)+m\quad(j\ne0),\qquad
h((0,1),(0,0))=2m.
\]

The formula is symmetric in source and target and gives 2m-1=n-1 for neighboring vertices, exactly as Lemma 7.19 requires. Its maximum is floor(m²/4)+m for m at least five. Thus

\[
\tau^*=2\lfloor m^2/4\rfloor+2m.
\]

The stationary law is uniform. Summing the same hitting-time formula gives

\[
\tau_0=\frac1{2m}\left[2m+2\sum_{j=1}^{m-1}
\{j(m-j)+m\}\right]
=\frac{m^2-1}{6}+m.
\]

On a vertex-transitive graph the stationary mean to a fixed target equals the source's stationary-average target parameter, so either equivalent convention yields this value.

Independent spectral route

For functions constant on each two-vertex fiber, the transition operator is the cycle operator, with eigenvalues cos(2πk/m), k=0,...,m-1. For functions antisymmetric within every fiber it is zero. These orthogonal invariant subspaces both have dimension m and exhaust the state space. Consequently

\[
\tau_2=\frac1{1-\cos(2\pi/m)}\sim\frac{m^2}{2\pi^2}.
\]

The fundamental-matrix trace is

\[
\sum_{k=1}^{m-1}\frac1{1-\cos(2\pi k/m)}+m
=\frac{m^2-1}{6}+m,
\]

which agrees with the direct hitting-time sum. The trigonometric sum can also be recovered from the already derived cycle hitting-time formula by stationary averaging; it is not needed as an unproved numerical identity for the counterexample.

Consequences and limitations

Since n=2m, both τ* and τ2 are of order n². They violate the proposed n log n and n upper bounds. Moreover

\[
\frac{\tau^*}{2\tau_0}

$$=\frac{\lfloor m^2/4\rfloor+m}{(m^2-1)/6+m}
\longrightarrow\frac32,$$

which violates the third assertion. These are exact formulas, not simulation estimates.

The strongest reasons for caution are the old unfinished source, its definition typo, and the fact that wreath graphs have long been standard. Focused searches on September 6, 2026 located the original question and primary graph-theory descriptions but no verified later disposition of Problem 7.27. Absence from that search is not evidence of novelty. A future reviewer should check the intended arc-transitivity convention, rate-one normalization, and prior probability literature before any promotion.

Proof appendix: P0209_uniform_clt_counterexample_candidate

Candidate and author-check labels inside this appendix govern its status. Mathematical source notation is preserved.

P0209: a four-state obstruction to the proposed uniform normal approximation

Status: complete negative-answer candidate, author checked by spectral covariance and a direct rare-event argument. No independent review or novelty certificate. The construction responds to the literal historical question, whose current status is unverified.

Source: Aldous and Fill, *Reversible Markov Chains and Random Walks on Graphs*, 2002 unfinished monograph, 2014 recompilation, Open Problem 4.30, PDF p.137, https://www.stat.berkeley.edu/~aldous/RWG/book.pdf . The chapter is internally dated October 11, 1994. With a stationary finite reversible continuous-time chain and a function g satisfying Eπg=0 and Eπg²=1, the source asks for a universal bound

$$
\sup_x\left|\mathbb P_\pi\left\{\frac{\int_0^t g(X_u)\,du}
{\sigma\sqrt t}\le x\right\}-\Phi(x)\right|
\le\psi(\|g\|_\infty,t/\tau_2),
\qquad \psi(b,s)\longrightarrow0\quad(s\longrightarrow\infty).
$$

Here σ² is the asymptotic variance rate and τ2 is the relaxation time. We keep the first argument exactly equal to two, so there is no hidden assumption of uniformity or continuity in b.

Four-state reversible family

For an integer s at least three, set ε=s^{-2} and R=s^4. Let Y be a stationary two-state chain on {0,1}, with rates 0→1 equal to ε and 1→0 equal to 1-ε. Its stationary success probability is ε and its nonzero generator eigenvalue is -1. Independently let Z be a fair stationary sign chain, flipping at rate R/2, with nonzero generator eigenvalue -R. The product X=(Y,Z) has four states, is irreducible and reversible, and has generator eigenvalues 0,-1,-R,-(R+1). Thus τ2=1.

Define

$$
A=2\varepsilon+\sqrt{(1-\varepsilon)(1-4\varepsilon)},\qquad
B=\frac{2-A}{1-\varepsilon},\qquad
g(Y,Z)=AZ+B(Y-\varepsilon).
$$

For 0<ε≤1/9, A and B are positive. The positive root A solves

$$
A^2-4\varepsilon A+5\varepsilon-1=0,
$$

which is equivalent to A²+B²ε(1-ε)=1. Independence and centering therefore give Eπg=0 and Eπg²=1. Also A+B(1-ε)=2. Every possible absolute value of g is at most max{A+Bε,A+B(1-ε)}=2 because ε<1/2, and the value 2 occurs when Y=1,Z=1. Hence ||g||∞=2 exactly for every s. Finally A→1 and B→1.

Exact variance rate

The stationary autocovariance is

$$
\mathbb E[g(X_0)g(X_u)]
$$

=A^2e^{-Ru}+B^2\varepsilon(1-\varepsilon)e^{-u}.
\]

This follows directly from the two nonconstant coordinate eigenfunctions. Integrating twice over the covariance triangle gives

\[
\sigma^2=2\int_0^\infty\mathbb E[g(X_0)g(X_u)]\,du
=\frac{2A^2}{R}+2B^2\varepsilon(1-\varepsilon)>0.
\]

Each fixed member of the family has the ordinary asymptotic normal limit. The issue is uniformity over chains at a time depending on the member.

Collapse at time t=s

Observe the chain until t=s. Write F=A∫□ˢZ_u du and H=B∫□ˢ(Y_u-ε)du. The probability that Y is ever one on this interval is at most

\[
\mathbb P(Y_0=1)+\mathbb E[\#\{0\to1\hbox{ transitions in }[0,s]\}]
\le\varepsilon+s\varepsilon=\frac1{s^2}+\frac1s.
\]

On the complementary event, H=-Bεs, and

\[
\left|\frac{H}{\sigma\sqrt s}\right|
\le\sqrt{\frac{\varepsilon s}{2(1-\varepsilon)}}\longrightarrow0.
\]

The fast component is negligible in L²:

\[
\mathbb E F^2
=2A^2\int_0^s(s-u)e^{-Ru}\,du
\le\frac{2A^2s}{R},
\qquad
\mathbb E\left(\frac F{\sigma\sqrt s}\right)^2
\le\frac{A^2}{RB^2\varepsilon(1-\varepsilon)}\longrightarrow0.
\]

Here Rε=s². Combining the rare-event bound with Chebyshev yields

\[
\frac{\int_0^s g(X_u)\,du}{\sigma\sqrt s}
\longrightarrow0\quad\hbox{in probability}.
\]

For any fixed x>0 its distribution function thus tends to one. The limiting inferior of its Kolmogorov distance from a standard normal is at least 1-Φ(x). Letting x decrease to zero shows that this limiting inferior is at least one half. Yet ||g||∞=2 and t/τ2=s→∞. No function ψ of the stated kind can therefore exist.

Second checks and rate normalization

The apparent conflict with variance convergence is illusory. For a coordinate with covariance c e^{-λu}, the exact finite-time variance is 2c[t/λ-(1-e^{-λt})/λ²]. Substitution shows Var(∫g)/(σ²s)→1. Rare visits carry the variance while their probability tends to zero; the normalized squares are not uniformly integrable. Variance convergence alone cannot force a central limit theorem.

Page 26 explicitly permits nonnegative transition rates with no constraint on their sums. Even imposing total jump rate at most one would not rescue the proposed bound. Replace the generator Q by Q/R and observe for time Rs. Every outgoing rate is at most (R/2+1)/R≤1. The relaxation time becomes R, the variance rate becomes Rσ², and the time integral has the same law as R times the old integral. Both the standardized law and the ratio t/τ2=s are unchanged.

The independent coordinate construction also makes clear the missing quantity: an asymptotic variance scale relative to the function and the slow rare-event mode. A gap and a bounded instantaneous variance do not prevent the asymptotic variance from concentrating in rare excursions. This observation is explanatory, not a claim that any particular modified bound has been proved here.

Focused primary-source searches on September 6, 2026 found the original statement and the author's chapter notes, which discuss fixed-chain Berry-Esseen constants. No later resolution of this precise uniform question was verified. This is not a novelty certificate. A future reviewer should recheck the exact fixed L∞ normalization, stationary initialization, spectral gap, covariance integral, and uniform-limit quantifiers.